

Subdiagonal and superdiagonal partitions (OEIS A238875 and A238873)

Two conjectures of Archibald, Blecher, Elizalde and Knopfmacher proved;
the subdiagonal ratio to all orders; the superdiagonal rate, Airy term and limit shape;
the full superdiagonal equivalent, conditional on a preprint

Five research reports of one external research session
(bundle Reports 156, 159, 158, 160 and 162), merged into one ProveIt report

All Parts: 3 October 2026 merged: 6 October 2026

Abstract

Write partitions in weakly increasing order $\lambda_1 \leq \dots \leq \lambda_k$. Let $s_r(n)$ count those of n with $\lambda_i \leq i+r$ for every i , so that $s_0(n)$ is OEIS A238875 (subdiagonal partitions), and let $A(n)$ count those with $\lambda_i \geq i$, OEIS A238873 (superdiagonal partitions). Archibald, Blecher, Elizalde and Knopfmacher (Afrika Matematika 36:77, 2025) conjectured $q(n)/A(n) \rightarrow 0$, with $q(n)$ the number of partitions into distinct parts (their Conjecture 5.2), and $s_0(n)/p(n) \rightarrow \frac{1}{2}$ (Conjecture 5.3). This report prints five manuscripts as Parts I–V. Part I proves both conjectures: $0 \leq \mathbb{P}_n(R \leq r) - s_r(n)/p(n) = O(1/\log n)$ uniformly in $r \geq 0$, with R Dyson’s rank, a logistic crossover in $r/\sqrt{n}$, a Lambert- W inverse, and $\liminf \log A(n)/\sqrt{n} \geq B_* = \sqrt{\pi^2/3 + 2\log^2 2} > \pi/\sqrt{3}$. Part II proves $s_0(n)/p(n) = \sum_{j \leq R} c_j \beta^j + O(\beta^{R+1})$ for every R , $\beta = \pi/\sqrt{6n}$, $c_0 = \frac{1}{2}$, $c_1 = -\frac{1}{8}$, with integer inverse windows: a second proof of Conjecture 5.3, at $r = 0$ only. Part III proves $\log A(n) = B\sqrt{n} + O(n^{1/3})$ with $B = \sqrt{\pi^2/3 + 4\log^2 2} = 2.2829\dots$, which replaces Part I’s lower rate. Part IV proves $\log A(n) = B\sqrt{n} - Dn^{1/6} + o(n^{1/6})$, $D = \zeta \log 2 (\pi^2/12 + \log^2 2)^{-1/6}$ with $-\zeta$ the first zero of Ai, using Mallein’s published killed-walk estimates, and the limit shape of a uniform superdiagonal partition. Part V proves $A(n) \sim Kn^{-11/12} \exp(B\sqrt{n} - Dn^{1/6})$ with an explicit K , *conditionally* on the unrefereed preprint arXiv:2607.27504v1 of Yu-Tian Li; the unconditional superdiagonal statements are those of Parts III and IV. The write adds a proof that partitions with choosable initial intervals are exactly the superdiagonal ones, which settles Gus Wiseman’s conjecture in A238873 (the same argument was posted in A387112 by Clément Garrot on 31 August 2026), and the defect $\mathbb{P}_n(R \leq 0) - s_0(n)/p(n) = \beta/4 + (\frac{7}{16} - \frac{3}{2\pi^2})\beta^2 + O(\beta^3)$, from Part II. Effective constants, fluctuations, all orders for $A(n)$ and an unconditional amplitude remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The five Parts are five manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 3988bf5c4 (batch 107, cluster 107-DIAGP). All five write partitions in weakly increasing order $\lambda_1 \leq \dots \leq \lambda_k$ and study the classes of Archibald, Blecher, Elizalde and Knopfmacher [1]:

$$s_r(n) = \#\{\lambda \vdash n : \lambda_i \leq i + r \ \forall i\}, \quad A(n) = \#\{\lambda \vdash n : \lambda_i \geq i \ \forall i\},$$

with $s(n) = s_0(n) = A238875(n)$ and $A(n) = A238873(n)$, the empty partition counted at $n = 0$.

Part	Bundle report	Delivered title	Labels
I	156 (base)	Proofs of two conjectures on subdiagonal and superdiagonal partitions: Quantitative rank coupling and exponential separation	dgp:conj:
II	159	All orders asymptotics for subdiagonal partitions: Endpoint reductions and integer inverse localization	dgp:sub:
III	158	Exact exponential growth of increasing superdiagonal partitions: An elementary proof and quantitative threshold inversion	dgp:rate:
IV	160	Airy correction and limit shape for superdiagonal partitions: Pointwise enumeration and macroscopic structure	dgp:airy:
V	162	Full prefactor and refined inverse for superdiagonal partitions: Exact coefficient asymptotics for A238873 (conditional; see below)	dgp:pre:

Arrangement. The base of the merge is Report 156: it is the only manuscript that treats both classes, every shift $r \geq 0$, and both conjectures of [1, Section 5]; its `Report156.tex` was staged as this `article.tex`, and it is printed first, as Part I. The other four follow in dependency order, grouped by class. Part II (subdiagonal, $r = 0$ only) cites Part I and sharpens it at $r = 0$. Parts III–V are superdiagonal: Part III replaces Part I’s lower rate by the exact rate; Part IV adds the Airy term and uses Part III’s coarse lower bound for its limit-shape theorem; Part V, whose proof is independent of Parts I–IV but refines their statements under an extra, unrefereed input, comes last. The bundle numbers do not follow this order (159 is Part II).

Part V is conditional. Part V’s Theorem 39.1 and Corollary 39.2 use Theorem 1 and Corollary 2 of Yu-Tian Li’s preprint arXiv:2607.27504v1 (submitted 29 July 2026) [13], which is unrefereed and whose proof nobody at intake or write verified. They are *conditional* on that preprint, and are labelled so in their statements. The unconditional superdiagonal statements of this report are those of Parts III and IV (whose external inputs are elementary or published).

Numbering. Sections are numbered continuously, and statements and equations within sections. Part I keeps Report 156’s Sections 1–9 and all its numbers. Report 159’s Section k is Section $k + 9$ here, Report 158’s is $k + 19$, Report 160’s is $k + 29$ and Report 162’s is $k + 38$; a statement or equation $k.j$ of a manuscript becomes $(k + \text{offset}).j$. For example, Report 159’s Theorem 1.1 is Theorem 10.1, Report 158’s Theorem 1.1 is Theorem 20.1, Report 160’s Theorems 1.1 and 1.3 are Theorems 30.1 and 30.3, and Report 162’s Theorem 1.1 is Theorem 39.1. Subsection 1.1 (with Proposition 1.3) and Section 47 were added in the write; no delivered number moved.

Notes added in the write are set in small type and tagged [**write**], with their date (6 October 2026). They give cross-references between the Parts, mark questions that a later Part answers and

statements that a later Part replaces, mark restated material, and record what the write checked. No delivered statement was changed. The write’s interventions in delivered text are listed under “Provenance and merge decisions”: six restated proofs replaced by pointers (each disclosed in place), a “conditional” tag in two statement headings of Part V, unified citation keys, and two L^AT_EX macro names. Each Part ends with its manuscript’s own limits or questions; Section 47 gathers every open question and non-claim for the whole report under Vladimir’s standing rule of 4 October 2026.

What is proved, and where

Here $p(n)$ and $q(n)$ count all partitions and partitions into distinct parts; R is Dyson’s rank (largest part minus number of parts) of a uniform partition of n ; $c = \pi/\sqrt{6}$ and $\beta = c/\sqrt{n} = \pi/\sqrt{6n}$; $b = \log 2$, $C = \pi^2/12 + b^2$, $B = 2\sqrt{C} = \sqrt{\pi^2/3 + 4\log^2 2} = 2.2829\dots$; $a_1 = -2.3381\dots$ is the largest zero of Ai , $\zeta = -a_1$, and $D = \zeta b C^{-1/6} = 1.5507\dots$

Statement	Where
<i>Subdiagonal partitions (unconditional)</i>	
$0 \leq \mathbb{P}_n(R \leq r) - s_r(n)/p(n) \leq C'/\log n$ uniformly in $r \geq 0$; hence $s(n)/p(n) \rightarrow \frac{1}{2}$: ABEK Conjecture 5.3 proved	Part I, Theorem 1.1
Uniform logistic crossover $s_r(n)/p(n) = (1 + e^{-\pi r/\sqrt{6n}})^{-1} + O(1/\log n)$ (uses Dousse–Mertens)	Part I, Corollary 4.1
Lambert- W inverse of s with window $O(\log y/\log \log y)$	Part I, Theorem 5.1
$s(n)/p(n) = \sum_{j \leq R} c_j \beta^j + O_R(\beta^{R+1})$ for every R , $c_j \in \mathbb{Q}[\pi^{-2}]$, $c_0..c_3$ explicit ($c_1 = -\frac{1}{8}$): a second, sharper proof of Conjecture 5.3, <i>at $r = 0$ only</i> (uses Liu–Zhou and Zhou)	Part II, Theorem 10.1
Integer inverse windows $\lceil X_J \mp C_J x_0^{-J/2} \rceil$	Part II, Theorem 10.2
$\mathbb{P}_n(R \leq 0) - s(n)/p(n) = \frac{\beta}{4} + (\frac{7}{16} - \frac{3}{2\pi^2})\beta^2 + O(\beta^3)$	[write] note at Part I’s Section 9, (9.1)
<i>Superdiagonal partitions (unconditional)</i>	
$\liminf \log A(n)/\sqrt{n} \geq B_* = \sqrt{\pi^2/3 + 2\log^2 2} = 2.0617\dots >$ $\pi/\sqrt{3}$, so $q(n)/A(n) \rightarrow 0$: ABEK Conjecture 5.2 proved	Part I, Theorem 1.2
$C/t - O(t^{-1/3}) \leq \log \sum_n A(n)e^{-tn} \leq C/t + O(1)$; $B\sqrt{n} - O(n^{1/3}) \leq \log A(n) \leq B\sqrt{n} + O(1)$: the exact rate B , replacing Part I’s lower rate B_* ; inverse to $O((\log y)^{5/3})$	Part III, Theorems 20.1 and 20.2
$\log A(n) = B\sqrt{n} - Dn^{1/6} + o(n^{1/6})$ and its radial analogue (uses Mallein’s published estimates); inverse $T(y) = x + \zeta b C^{-2/3} x^{2/3} + o(x^{2/3})$, $x = (\log y/B)^2$	Part IV, Theorem 30.1, Corollary 30.2
Limit shape: $tK \rightarrow 2b$, $K/\sqrt{n} \rightarrow 2\log 2/\sqrt{C} = 1.2145\dots$; cumulative and row profiles g , f ; half the limiting rows on the diagonal (macroscopically)	Part IV, Theorem 30.3, Figure 1
<i>Superdiagonal partitions, conditional on arXiv:2607.27504v1</i>	
$A(n) \sim K n^{-11/12} \exp(B\sqrt{n} - Dn^{1/6})$, $K = \sqrt{2} C^{5/12} / (\sqrt{\pi} \text{Ai}'(a_1)) = 1.2705\dots$; the radial amplitude K_F and first radial correction α ; inverse to the $\sqrt{x}$ term	Part V, Theorem 39.1, Corollary 39.2
<i>OEIS</i>	

Statement	Where
A partition has choosable initial intervals iff it is superdiagonal, so their number is $A(n)$: Wiseman’s A238873 conjecture (same argument published by Clément Garrot in A387112)	[write] Proposition 1.3
<i>Open</i> The defect for fixed $r \geq 1$ and growing r ; $r/\sqrt{n}$ -sensitive errors; effective onsets everywhere; convergence and higher coefficients of Part II’s series; superdiagonal fluctuations, discrete diagonal contacts and extreme rows; a pointwise relative correction and all orders for $A(n)$; an $x^{1/3}$ inverse term; an <i>unconditional</i> proof of the amplitude K ; literature. See Section 47.	

The five companions check finite mechanisms in exact arithmetic (counts, injections, identities, finite inequalities, coefficient algebra); no asymptotic statement rests on them.

The two journal conjectures, and the OEIS entries

The conjectures. M. Archibald, A. Blecher, S. Elizalde and A. Knopfmacher, *Subdiagonal and superdiagonal partitions*, Afrika Matematika **36**, article 77 (2025), received 12 November 2024, accepted 27 February 2025, published online 7 April 2025, open access under CC BY 4.0 [1]. In their Section 5 (“Asymptotics”, p. 15), P_n is the set of weakly increasing partitions of n , $A_n \subseteq P_n$ and $S_n \subseteq P_n$ are the superdiagonal ($\lambda_i \geq i$) and subdiagonal ($\lambda_i \leq i$) ones, and $Q_n \subseteq P_n$ is the subset of partitions with distinct parts, with the remark $Q_n \subseteq A_n$. The two conjectures read, verbatim (the write read the open-access text on 6 October 2026):

Conjecture 5.2 $\lim_{n \rightarrow \infty} \frac{|Q_n|}{|A_n|} = 0.$

“We conjecture that, asymptotically, half of the partitions are subdiagonal.”

Conjecture 5.3 $\lim_{n \rightarrow \infty} \frac{|S_n|}{|P_n|} = \frac{1}{2}.$

After each, the paper says that similar problems were studied in its references [4] and [5]. It proves $|A_n|/|P_n| \rightarrow 0$ (Proposition 5.1), that balanced partitions (largest part equal to the number of parts) have density zero (Lemma 5.4, by the no-ones injection that Part I’s Lemma 2.1 extends to every rank), and $\limsup |S_n|/|P_n| \leq \frac{1}{2}$ (Proposition 5.5). Its closing paragraph says that it “would be interesting” to describe the asymptotic behaviour of $|A_n|$ similarly. These are conjectures of a refereed journal article, not OEIS conjectures; neither OEIS entry states them.

What this report proves about them. Part I proves both: Conjecture 5.3 by Theorem 1.1 (with $|S_n| = s_0(n)$, $|P_n| = p(n)$, and the rate $O(1/\log n)$), and Conjecture 5.2 by Theorem 1.2 ($|Q_n| = q(n)$, $|A_n| = A(n)$; Theorem 1.2 gives $q(n)/A(n) \leq \exp(-(B_* - \pi/\sqrt{3} - \eta)\sqrt{n})$ eventually, for every $\eta > 0$) (wording corrected after the independent check of 6 October 2026; the first wording read “the ratio decays like $\exp(-(B_* - \pi/\sqrt{3} - \eta)\sqrt{n})$ ”, but the theorem gives an upper bound only). The write read both proofs in full and found no gap; the intake re-derived their key steps (the interior lemma, the fixed-rank injection, the prefix–tail injection and the tail comparison). Part II proves Conjecture 5.3 a second time, at $r = 0$ only, with the full expansion. Part III answers the paper’s closing question at the logarithmic scale (the limit of $\log A(n)/\sqrt{n}$ exists and equals B); Part IV adds the next term, unconditionally; Part V gives the full equivalent, conditionally. With Part III’s rate, Conjecture 5.2 holds with the exponent $B - \pi/\sqrt{3} = 0.4691\dots$ in place of Part I’s $B_* - \pi/\sqrt{3} = 0.2479\dots$

The OEIS entries, as read by the write on 6 October 2026 (A238873 revision #36 of Oct 07 2025; A238875 revision #33 of Apr 14 2025, the revisions Part I's bibliography records):

- **A238875** (Joerg Arndt, Mar 24 2014), “Subdiagonal partitions: number of partitions (p1, p2, p3, ...) of n with $p_i \leq i$,” with the comment that the partitions are weakly increasing lists, a g.f. via A129176, and the link to [1] (“See p. 9”). No conjecture.
- **A238873** (Joerg Arndt, Mar 23 2014), “Number of superdiagonal partitions: partitions (p1, p2, p3, ...) of n such that $p_i \geq i$,” with a comment that the weakly decreasing convention gives A003114 (Gus Wiseman, Oct 06 2025), the link to [1] (“See p. 5”), and one conjecture, unrelated to those of [1], which reads verbatim:
 Conjecture: Also the number of integer partitions y of n with choosable initial intervals, meaning it is possible to choose distinct positive integers (z_1, z_2, ...) such that $z_i \leq y_i$ for all i.
 – _Gus Wiseman_, Sep 26 2025

No manuscript addresses Wiseman's conjecture. It is true: Proposition 1.3 (added by the write, with a complete proof) shows that a partition has choosable initial intervals if and only if its increasing arrangement is superdiagonal. The argument is not new: on 31 August 2026 Clément Garrot added to OEIS A387112 (revision #19 of Sep 07 2026) a comment with the same proof [7], stating that it also proves the corresponding conjectures in A238873, A387118 and A388711. The A238873 entry still displays the line as a conjecture. The intake record proposed the argument without knowing of Garrot's comment; the write found it. Nothing was submitted to the OEIS.

The inverses (relation to the transseries volume [17]). Statement by statement:

- Part I, Theorem 5.1: the centre $x_0(y)$ of (5.2) is an *instance* of the volume's Lambert core `p0:thm:lambert-core` after the volume's monomial reduction `p0:lem:alpha-reduction` with $\alpha = \frac{1}{2}$: the model $M(x) = e^{2c\sqrt{x}}/(8\sqrt{3}x)$ is of exponential-power type (`p0:def:model`) with data $(1/(8\sqrt{3}), 2c, \frac{1}{2}, -1, \cdot, 0)$; in $\xi = \sqrt{x}$ the equation $M = y$ is $2c\xi - 2\log \xi = \log y + \log(8\sqrt{3})$, the core with $a = 2c$, $b = -2 < 0$, whose large solution `p0:eq:core-lambert-minus` is $\xi = c^{-1}(-W_{-1}(-c/\sqrt{8\sqrt{3}y}))$, i.e. $x_0 = \xi^2$. The window $O(\sqrt{x_0}/\log x_0)$, the bracketing and the nested-log initializer (5.4) are Part I's own.
- Part II, Theorem 10.2: the same centre x_0 (an instance as above); the centres X_J are, after the same reduction, the reversion of `p0:thm:lambert-centered` for the model M_J of (17.2), whose data in ξ are $a = 2c$, $b = -2$ and $Q(\tau) = \log(1 + \sum_j a_j c^j \tau^j)$, squared back by `p0:eq:alpha-inverse-transport`: an *instance* of the coefficient formula. The write checked the first two coefficients against the volume's recurrence `p0:eq:c-bell` ($n = 1, 2$): it gives $b_0 = h$ and $b_1 = ht - a_2 + h^2/2$, Part II's values. The ceiling windows and Lemma 17.1 are Part II's own, and they are *not* an instance of `p0:thm:staircase`: that theorem presupposes an admissible interpolation of the sequence, while Part II brackets the root of a smooth model of $s(n)$ directly at the integers.
- Parts III, IV and V, Theorem 20.2 and Corollaries 30.2 and 39.2: not instances of anything in the volume. Part III has only an $O(n^{1/3})$ error, and the logarithms of Parts IV and V contain the growing fractional power $n^{1/6}$, outside `p0:def:model` (whose corrections are $b \log x$ and a series in $x^{-\delta}$). Each inverse is proved directly by monotonicity and integer bracketing. The power-and-logarithm balance of Part V's inverse is at most analogous to the volume's monomial-logarithmic template `plt:thm:lw-template`.

None of the manuscripts claims novelty for the inversion mechanics.

Notation across the five Parts

Common to all Parts. Increasing order; $A(n)$ the superdiagonal count with $A(0) = 1$; Z_j (Part I: Z_1) the multiplicity of the part j ; $T(y) = \min\{n \geq 0 : \cdot \geq y\}$ a first threshold; $p(n)$ the partition number. *Fixed for this guide and Section 47:* s_r , $s = s_0$, $c = \pi/\sqrt{6}$, $\beta = c/\sqrt{n}$, $b = \log 2$, $C = \pi^2/12 + b^2$, $B = 2\sqrt{C}$, $a_1 < 0$ the largest zero of Ai , $\zeta = -a_1 > 0$, $D = \zeta b C^{-1/6}$. No symbol of a manuscript was renamed by the write (only two macro names, which do not change the printed text). Part V itself renames Li's $A_q(z)$ to $H_q(y)$ because $A(n)$ is taken; Li's scaled argument ζ is Part V's s . The symbols below change meaning between the Parts or inside one; each Part fixes its own meaning, and opens with a short table of its readings.

Symbol	Meanings
b	Part I: $b = \pi/\sqrt{3}$, the rate of $q(n)$ (1.6). Part II: Zhou's variable $b = \pi/\sqrt{6(n-1/24)}$ (Section 11). Parts III–V: $b = \log 2$. <i>Tempting false reading:</i> Part I's b is not $\log 2$.
$B, \mathcal{B}$	Part I: lower rates B_* , B_0 , B_h ((1.6), (6.8), (7.8)). Part II: $B(r)$ a cubic rank coefficient (11.9); B a number of top rows and B_{src} , B_{tgt} complements (Lemma 13.2); $\mathcal{B}_n$, $\mathcal{B}(x; n)$ bottom events. Parts III and V: $B = 2\sqrt{C}$, the exact rate (Part IV writes $2\sqrt{C}$, and B in its inverse); Part III also B_k tail heights; Part V $\mathcal{B}_t$ an index block. <i>False reading:</i> Part I's B_* is a lower bound, not the rate.
$C, \mathcal{C}$	Part I: C a generic constant (Theorem 1.1); C_M Catalan numbers, $\mathcal{C}_M$ prefixes. Part II: C_i cumulative multiplicities; C_J , C inverse constants. Parts III–V: $C = \pi^2/12 + \log^2 2$ fixed; C_j cumulative multiplicities. Part IV reserves C and writes generic constants c_* , K_* .
t, β	Parts I, III, IV, V: t a Boltzmann (radial) variable, e^{-t} the activity (Part I in Section 3). Part II: $t = c^{-2} = 6/\pi^2$, and its radial variable is $\beta = c/\sqrt{n}$ (also β_x , β_0). Part I: $\beta = c/\sqrt{n}$ (proof of Corollary 4.1). Parts IV, V: $\beta = d - 2b - \log(1 - e^{-d}) > 0$ a buffer constant. <i>False reading:</i> Part II's t is a number, not a variable.
ζ, a	Part IV: $\zeta = -a_1 \approx 2.3381$. Part V: $a = a_1 < 0$, a_2 ; Li's preprint: ζ the scaled argument (Part V's s). Part I: $a_h = \log(\rho_h/2)$. Part II: $a = \frac{3}{16} - \frac{t}{4}$ (11.9), a_j model coefficients. Part IV: a , d_0 interval ends, $a(t) = t^{-19/12}$ (34.3). <i>False reading:</i> Part IV's $-\zeta b$ and Part V's ba are the same number.
D	Part I: $D(n) = p(n) - p(n-1)$; $D_h(M)$ Dyck counts; D a constant in the proof of Theorem 5.1. Part II: $D_k(n, a)$ a counting table; D_J constants. Part III: $D_h(M)$; D a constant in the inverse. Part IV: $D_M(t) = e^{2bM - tM(M+1)/2}$; $D = \zeta b C^{-1/6}$ (Section 35). Part V: $D = -abC^{-1/6}$, the same number; $D_M = 4^M e^{-tM(M+1)/2}$, equal to Part IV's $D_M(t)$; d a cutoff $> b$ and d an inverse coefficient (44.3).
K	Parts I, II: K the length cutoff ((3.2), (12.4)). Part III: constants $K_0, \dots, K_3$. Part IV: K the number of parts; constants $K_0, \dots, K_8$. Part V: the amplitudes K , K_F ; K_N a length cap; $\mathcal{K}$ a compact set. <i>False reading:</i> Part IV's K is a length, Part V's an amplitude.
L	Parts I, II: the largest part; Part II also $L_{x,y,r}(n)$. Parts III, IV: $L(x) = x \log 4 - x^2/2 + \text{Li}_2(e^{-x})$ (the same function in both; Part IV also L a grid bound). Part V: $L = 2b = \log 4$, as in Li's preprint.
A, H	$A(n)$ throughout; $[1]$'s $A(q, t)$ its generating function; Part II: Zhou's normalizer A ; Part I: $H = \log y$; Part II: $H(x)$ the Rademacher term (11.1); Parts IV, V: heights $H_j = j - C_j$; Part V: $H_q(y)$ Ramanujan's entire function (Li's A_q); $\mathcal{A}_M$ an area (Parts IV, V).
N	Parts I, II: $N(r, n)$ rank counts. Part IV: $N = J - M$; $\mathcal{N}$ the size. Part V: $N = \sum j Z_j$ the size; $N_t(s)$ Li's normalization (40.6); $\mathcal{N}(0, 2C)$ a normal law.
s, q	Part I: $s_r(n)$; $q(n)$ distinct partitions. Part II: $s(n) = s_0(n)$; $q = e^{-\beta}$; s an index. Part III: q rational. Part IV: s a tilt; q Mallein's potential. Part V: s the scaled Airy argument, s_t the first-zero coordinate; $q = e^{-t}$.

Symbol	Meanings
R, ρ	Parts I, II: R Dyson’s rank; Part II also R a truncation order. Parts I, III: $\rho_h = 4 \cos^2(\pi/(h+2))$. Part IV: ρ a ramp parameter. Part V: $\rho = \rho_1 < \rho_2 < \dots$ the zeros of H_q ; $R = y_t(s)$ a radius.
M, m	Part I: M a prefix length, $M(x)$ a model; m the interior scale. Part II: M the endpoint cutoff, $M_J(x)$ models; m as in Part I. Parts III–V: $M \approx b/t$ the cutoff. Part IV: $m(x)$ the limiting mean profile; m a walk length. Part V: m_t a centre; $m = \lfloor 1/t \rfloor$; m a target in Proposition 43.2.
f, g, h	Part I: $f_M(N)$ distinct tails; $f(x) = -\log(1 - e^{-tx})$; $g = \frac{1}{4} \operatorname{sech}^2(x/2)$; $h(x) = \log M(x)$; h a height. Part II: $f(z) = 1/(1 + e^z)$; $h = \frac{7}{24} + \frac{t}{2}$. Part III: $f(x) = -\log(1 - e^{-x})$; h a height. Part IV: g and $f = g^{-1}$ the limit profiles (30.6)–(30.7); $h(x) = -\log(1 - e^{-x})$. Part V: $f(x) = -\log(1 - e^{-x})$; g_j, G_q product coefficients. <i>False reading</i> : Part IV’s f is the row profile, not Part III’s f (Part IV’s h).
W, G, Q	Parts I, III: $W_h(M)$ a weight bound; Part I also W_{-1} (Lambert). Part IV: $W_P, W_T, \mathcal{W}_t$. Part II: G_b gaps; Q_β the Boltzmann law. Part III: $G_M(t)$ a tail sum. Part IV: G a buffer; $Q_{t,M}(h)$ survival; Q_t a product law. Part V: G a buffer, G_q ; Q_1, Q_2 Airy densities.
$p, P, \mathbb{P}$	$p(n)$ partitions (Parts I, II, and in the guide); p a geometric parameter (Parts III–V), p_* ; Part V also $p = 11/12$. Part II: $P_r(n)$ cumulative rank counts; Parts I, II: $P(q)$ the partition product; Parts IV, V: P_M a tail product. $\mathbb{P}_n$: uniform law on partitions of n (Parts I, II) or on superdiagonal partitions of n (Part IV); $\mathbb{P}_t$ Boltzmann laws (Parts III–V).
E, α	Part II: $E_n, E_n^{(M)}$ failure counts. Part III: $E(t)$ an error. Part IV: E_t, E_H events. Part V: $E_0 = \pi^2/12 - b^2$; $\alpha = a^2(\frac{1}{4} - \frac{b}{30})$. Part I: $\alpha = \lim r_n/\sqrt{n}$.

Provenance and merge decisions

Five manuscripts of the session bundle of Reports 1–243. Their author lines read “Report 156”, “Report 159”, “Report 158”, “Report 160” and “Report 162”; none names an author, a tool or an addressee, each sets an empty PDF author field, and none carries “prepared for private review” wording. All are dated 3 October 2026, pin no ProveIt commit and contain no repository path. They arrived in commit 60f54ea06 and were placed by 3988bf5c4 (batch 107, cluster 107-DIAGP).

- **Part I**, Report 156, archive `Two_Partition_Conjectures_Proofs_and_Quantitative_Extensions_Source.zip` (`Report156.tex`, 861 lines, 14 pages); the base. Contributes both conjecture proofs, the uniform shifted coupling and fixed-rank atom bound, the logistic crossover, the Lambert inverse, the Catalan and bounded-height prefix lower bounds.
- **Part II**, Report 159, archive `Subdiagonal_Partitions_All_Orders_Expansion_and_Integer_Inverses_Source.zip` (`Report159.tex`, 1019 lines, 17 pages). Contributes the all-orders expansion of $s(n)/p(n)$ with $c_0, \dots, c_3$, the mixed row–column bijection, and the integer inverse windows.
- **Part III**, Report 158, archive `Superdiagonal_Partitions_Exact_Exponential_Growth_and_Inverse_Source.zip` (`Report158.tex`, 749 lines, 13 pages). Contributes the exact rate B with the tilted upper bound, the survival lemma, the uniform spectral bound, the elementary coefficient transfer and the inverse.
- **Part IV**, Report 160, archive `Superdiagonal_Partitions_Airy_Correction_and_Limit_Shape_Source.zip` (`Report160.tex`, 1308 lines, 17 pages). Contributes the Airy term (radial and pointwise), the exact-weight tail lemma, the second-order inverse, and the length and shape theorem.

- **Part V**, Report 162, archive `Superdiagonal_Partitions_Full_Prefactor_and_Refined_Inverse_Source.zip` (`Report162.tex`, 933 lines, 15 pages). Contributes, conditionally, the radial amplitude and first correction, the pointwise equivalent, the refined inverse; unconditionally, the first-pole transfer lemma, the cutoff lemma and the local limit theorem, and the two source-display repairs.

No package embeds another (no byte-identical file across the five). Where the merge had to choose:

1. *Order and base*: as in the Guide.
2. *Restatements printed once*. The barrier lemma, the survival lemma, the monotonicity injection and the dilogarithm evaluation $L(b) = C$ are first printed in Part III, and the Dyck representation in Part I. Six later *proofs* of them are replaced by pointers, each with a dated note saying what is not reprinted: Part III’s proof of Lemma 23.1 (Part I, Sections 6–7); Part IV’s proof of the barrier equivalence (Section 30), of the survival bound (Section 33) and of the dilogarithm evaluation (Section 31), and Part IV’s and Part V’s monotonicity injections (Sections 35 and 44). Every statement and every labelled display stays. Restatements woven into a longer proof stay, with a note: Part II’s interior step (Section 12; Part I’s Lemma 3.1), Part V’s barrier sentence, its summation by parts (43.6) (Part IV’s (31.2)) and its ruin bound (Part III’s Lemma 24.1, Part IV’s (34.16)). The Fourier and local-limit arguments of Part IV (Section 34.2) and Part V (Section 43) use the same buffer, ruin and three-arc cover but are assembled differently (a tilted local limit theorem with exact weights, against a moment-generating-function limit with total-variation constraint removal); both are printed, as two routes.
3. *Conditional status*. The headings of Part V’s Theorem 39.1 and Corollary 39.2 carry the added tag “[**write**] conditional on arXiv:2607.27504v1”; nothing else in them changed.
4. *Front matter and bibliography*. The five title blocks, abstracts and scope paragraphs are printed at the heads of their Parts; one table of contents replaces five. The five bibliographies are merged into one: an entry cited by several Parts is printed once, with every Part’s annotation kept verbatim; citation keys were unified (for example Part I’s `oeissuper` and Part III’s `oeis` are both `oeisA238873`; Part I’s `dlnmf` is DLMF §27.14, Part IV’s is DLMF Table 9.9.1, now two keys). The entries “Report 156” and “Report 158” point to Parts I and III. The write added the entries for the transseries volume and for OEIS A387112, and dated annotations to the entries of the papers it read.
5. *Macros*. Parts I–II define `\PP` as $\mathbb{P}_n$ and Parts III–V as $\mathbb{P}$; Part III defines `\Li` as Li and Part V as Li_2 . One definition each is kept and the uses were rewritten (`\PP` as `\PP_n` in Parts I–II, `\Li` as `\Li_2` in Part V); the printed text is unchanged.
6. *Write additions*. This front matter, the reading-conventions tables of the Parts, Subsection 1.1 with Proposition 1.3, the defect formula (9.1), Section 47, the dated notes, and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 107, 5 October 2026) the five manuscripts were read in full and no mathematical claim was found false; only two source-display repairs (Part V’s) and no wrong statement. The intake re-derived the key steps of both

conjecture proofs by hand; recomputed $A(n)$ to $n = 16000$ and $s(n)$ to $n = 1500$ by its own dynamic programs (equal to the companions' exact tables for $n \leq 400$ and $n \leq 300$); recomputed every constant (B_* , B_0 , B , C , a_1 , $\text{Ai}'(a_1)$, D , K_F , K , α , $2D/B = \zeta b C^{-2/3} = 1.3585577\dots$, $2 \log 2 / \sqrt{C}$) to 30 digits; checked that $(s/p - \sum_{j \leq 2} c_j \beta^j) / \beta^3$ drifts towards $c_3 = -0.3162$ (-0.544 at $n = 100$, -0.388 at $n = 1500$); and checked the normalization of Li's Theorem 1 against Part V. It fitted Part V's constants by least squares (float dynamic program to $n = 16000$): the pointwise residual $\log A(n) - B\sqrt{n} + Dn^{1/6} + \frac{11}{12} \log n - \log K$ on $n = 2000, 2250, \dots, 16000$ in the basis $1, n^{-1/6}, n^{-1/3}, n^{-1/2}$ gives constant terms -0.038 and -0.026 ; the radial residual on 23 equally spaced $t \in [0.016, 0.06]$ in the basis $1, t^{1/3}, t^{2/3}, t$ gives constant terms -0.025 and -0.088 , and with the constant fixed at 0 a $t^{1/3}$ coefficient 1.34 against $\alpha = 1.24$ (maximal residual $3.6 \cdot 10^{-4}$). So K , K_F and α agree to the few-percent level these fits can see (an error of order $\log 2$ in a normalization would be visible); the fits are ill-conditioned and prove nothing. The approach is slow: $\log A(n) / \sqrt{n} = 2.1177$ at $n = 8000$ against $B = 2.2829$, which the term $-Dn^{1/6}$ explains.

The write (6 October 2026) re-read the five proofs and checked against the sources: both conjectures in the open-access text of [1], and its Theorem 2.1, whose display sums from $k \geq 1$ (Part V's first repair); the statements of Dousse–Mertens' Theorem 1.2 [2], Liu–Zhou's Theorem 1.3 [8], Zhou's Theorem 4.1 [9] (hypotheses at $k = 2$, $m = 0$, fixed $j = |r|$), Mallein's Lemmas 2.6 and 2.7 [11] and the q -Airy example of Štampach–Štovíček [12] (Part V's second repair), each in the arXiv version named in the notes, with the journal versions not compared; the arXiv record of Li's preprint (one version, v1 of 29 July 2026, no journal reference, on 6 October 2026). It recomputed the defect (9.1) numerically (with exact $p(n)$ and exact rank counts) and checked Proposition 1.3 by exhaustive search for $n \leq 24$. On copies of the arrival archives (Windows, Python 3.14.4) the five `counts` commands regenerated the shipped count tables (after removing the carriage returns of Windows text output), and Part V's 14 tests passed.

What was *not* done: the analytic estimates (the Rademacher and rank inputs and their specialization, the endpoint reduction and mixed bijection to every order, the spectral, Chernoff, local-limit and Fourier estimates, the killed-walk specialization, the first-pole transfer) were read and their displayed algebra checked, but they are the manuscripts' proofs, not an independent verification. Nobody verified the proofs of the external theorems: Hardy–Ramanujan–Rademacher (DLMF), Dousse–Mertens, Liu–Zhou, Zhou, Mallein, Štampach–Štovíček, and Li. Li's Theorem 1 and Corollary 2 were read at intake, where their hypotheses were found to be met; the write did not read them again. Zhou's coefficient recipe (11.8) was not re-derived from his Proposition 2.2.

Relation to neighbouring reports and formal projects

No other report of the repository treats A238873, A238875, the paper [1], superdiagonal or subdiagonal partitions in this sense, or the turning-point asymptotics of Ramanujan's entire function (repository search, 6 October 2026; the word “superdiagonal” elsewhere refers to matrices). The partition reports of this collection (for example `a022629-distinct-partition-norms`, `a033552-catalan-partitions`, `a097356-sqrt-restricted-partitions`, `a291698-moving-fugacity-partitions`) share only the generic Hardy–Ramanujan and Boltzmann methods. `a082161-airy-amplitudes` also meets the first Airy zero through a killed walk with an area or linear potential (its Dirichlet operator $-\partial_x^2 + 8x$); the mechanism is analogous, the theorems are not instances of each other. `a380274-mahonian-growing-powers/literature.md` cites R. Zhang's paper on Ramanujan's entire function $A_q(z)$ for a different problem. The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`

is related to the inverses as stated above. Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats these sequences, and no statement of this report is formalized.

Part I

Proofs of two conjectures on subdiagonal and superdiagonal partitions

Bundle Report 156, archive `Two_Partition_Conjectures_Proofs_and_Quantitative_Extensions_Source.zip`, delivered as `Report156.tex` (14 pages), the base of the merge. Delivered title block “Proofs of two conjectures on subdiagonal and superdiagonal partitions / Quantitative rank coupling and exponential separation / Report 156 / 3 October 2026”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery. Subsection 1.1 is added in the write.

Reading conventions of this Part	Elsewhere in the report
$b = \pi/\sqrt{3}$ (the rate of $q(n)$); B_* , B_0 , B_h lower rates	Parts III–V: $b = \log 2$, $B = 2\sqrt{C}$ the exact rate
$c = \pi/\sqrt{6}$; $\beta = c/\sqrt{n}$ (Section 4); $t = c/\sqrt{n}$ (Section 3)	Part II: $t = 6/\pi^2$, radial variable β
C a generic constant; C_M Catalan numbers; $D(n) = p(n) - p(n-1)$; K , m cutoffs; L the largest part	Parts III–V: $C = \pi^2/12 + \log^2 2$; Part IV: K the length

Abstract of this Part (as delivered). For weakly increasing partitions of n , let $s_r(n)$ count those satisfying $\lambda_i \leq i + r$ at every index, and let $A(n)$ count those satisfying $\lambda_i \geq i$. We prove both conjectures in Section 5 of Archibald, Blecher, Elizalde and Knopfmacher, *Subdiagonal and superdiagonal partitions*, Afrika Matematika **36**, 77 (2025).

An extension of their rank-zero injection, together with an elementary interior estimate and the classical Hardy–Ramanujan formula, gives

$$0 \leq \mathbb{P}_n(R \leq r) - \frac{s_r(n)}{p(n)} = O((\log n)^{-1}) \quad (r \in \mathbb{Z}_{\geq 0}),$$

uniformly over all nonnegative integer shifts, where R is Dyson’s rank. In particular $s_0(n)/p(n) = 1/2 + O((\log n)^{-1})$, proving their Conjecture 5.3. We derive its Lambert W inverse with the growing localization error $O(\log y / \log \log y)$. A separate application of the Dousse–Mertens rank theorem gives a uniform logistic crossover.

For superdiagonal partitions, Catalan prefixes followed by distinct large parts give explicit finite lower bounds. Restricting the prefix to fixed-height Dyck paths yields

$$\liminf_{n \rightarrow \infty} \frac{\log A(n)}{\sqrt{n}} \geq \sqrt{\pi^2/3 + 2(\log 2)^2} > \frac{\pi}{\sqrt{3}}.$$

Thus distinct partitions are exponentially negligible relative to $A(n)$ on the $\sqrt{n}$ scale, proving Conjecture 5.2. This is a lower bound, not an exact superdiagonal growth rate. No all-orders expansion is established. An exact, independently implemented companion checks sequence prefixes and the finite combinatorial mechanisms.

Attribution and scope. The definitions, generating functions, conjectures, and rank-zero injection are due to the cited sources. The argument below extends that injection to fixed ranks and supplies the quantitative coupling and prefix constructions needed for the two limits. A bounded search of the current article, OEIS entries, relevant literature, and the available report catalogue found no identical later solution as of 3 October 2026. This is not a claim of exhaustive scholarly priority. The constants and starting indices in asymptotic error terms are not numerically certified.

1 Definitions and the two conjectures

A partition will always be written in *weakly increasing* order:

$$\lambda = (\lambda_1, \dots, \lambda_k), \quad 1 \leq \lambda_1 \leq \dots \leq \lambda_k, \quad |\lambda| = n.$$

Let $p(n)$ be the ordinary partition number and $q(n)$ the number of partitions into distinct parts. For an integer $r \geq 0$, put

$$s_r(n) = \#\{\lambda \vdash n : \lambda_i \leq i + r \text{ for all } i\}, \quad (1.1)$$

$$A(n) = \#\{\lambda \vdash n : \lambda_i \geq i \text{ for all } i\}. \quad (1.2)$$

The empty partition gives $s_r(0) = A(0) = p(0) = q(0) = 1$. The unshifted sequences are $s_0(n) = \text{A238875}(n)$ and $A(n) = \text{A238873}(n)$ in OEIS [5, 6]. Changing the order convention changes the problem: weakly decreasing superdiagonal partitions belong to the Rogers–Ramanujan class A003114, not the class studied here.

Archibald et al. [1], Section 5, use exactly this increasing convention. Their Conjecture 5.2 asserts $q(n)/A(n) \rightarrow 0$; their Conjecture 5.3 asserts $s_0(n)/p(n) \rightarrow 1/2$. They prove $A(n)/p(n) \rightarrow 0$ in Proposition 5.1 and $\limsup s_0(n)/p(n) \leq 1/2$ in Proposition 5.5. The following quantitative results settle the two conjectures.

Theorem 1.1 (Uniform subdiagonal coupling). *For a uniformly random partition of n , let R be its largest part minus its number of parts. As $n \rightarrow \infty$,*

$$0 \leq \mathbb{P}_n(R \leq r) - \frac{s_r(n)}{p(n)} \leq \frac{C}{\log n} \quad \text{for every integer } r \geq 0, \quad (1.3)$$

with one constant C and one sufficiently large starting index, independent of r . In particular,

$$\frac{s_0(n)}{p(n)} = \frac{1}{2} + O((\log n)^{-1}), \quad (1.4)$$

$$s_0(n) = \frac{\exp(\pi\sqrt{2n/3})}{8\sqrt{3}n} \left(1 + O((\log n)^{-1})\right). \quad (1.5)$$

The one-sided upper error in (1.4) is $O(n^{-1/2})$.

Theorem 1.2 (Exponential separation). *Let*

$$b = \frac{\pi}{\sqrt{3}}, \quad B_* = \sqrt{\frac{\pi^2}{3} + 2(\log 2)^2}. \quad (1.6)$$

Then

$$\liminf_{n \rightarrow \infty} \frac{\log A(n)}{\sqrt{n}} \geq B_*. \quad (1.7)$$

For every $\eta > 0$, there is an index after which

$$\frac{q(n)}{A(n)} \leq \exp(-(B_* - b - \eta)\sqrt{n}). \quad (1.8)$$

Since $B_* > b$, this proves $q(n)/A(n) \rightarrow 0$.

Here and throughout, logarithms are natural. Both theorems are asymptotic statements with unspecified constants and onset. The finite inequalities used in the superdiagonal proof, however, hold at every integer satisfying their explicitly stated feasibility conditions. The remaining sections also distinguish the elementary proof of Theorem 1.1 from the external rank theorem used for the logistic corollary.

[write] 6 October 2026. The two conjectures are quoted verbatim from [1] in the front matter (“The two journal conjectures, and the OEIS entries”); Theorem 1.1 at $r = 0$ is Conjecture 5.3 and Theorem 1.2 implies Conjecture 5.2. Later Parts sharpen both theorems, as follows. *Part II* proves Theorem 1.1’s statement (1.4) again *at $r = 0$ only*, with the expansion $s_0(n)/p(n) = \frac{1}{2} - \frac{\beta}{8} + O(\beta^2)$, $\beta = \pi/\sqrt{6n}$, to every order (Theorem 10.1); so the error in (1.4) is $O(n^{-1/2})$ in both directions, (1.5) holds with relative error $O(n^{-1/2})$, and the ratio is below $\frac{1}{2}$ for all large n (by exact computation at the write, for every $10 \leq n \leq 400$; it equals $\frac{1}{2}$ at $n = 2$ and $n = 9$ and exceeds it at $n = 0, 1, 3, 5$; added after the independent check of 6 October 2026: it is below $\frac{1}{2}$ at $n = 4, 6, 7, 8$, so the list is complete for $n < 10$, and a floating-point dynamic program extends “below” to every $10 \leq n \leq 2000$). Part II says nothing about $r \geq 1$ or the logistic regime: for those, Theorem 1.1 and Corollary 4.1 are the only statements in this report. *Part III* replaces the lower rate B_* of Theorem 1.2 by the exact rate: $\log A(n)/\sqrt{n} \rightarrow B = \sqrt{\pi^2/3 + 4\log^2 2} = 2.2829\dots > B_* = 2.0617\dots$ (Theorem 20.1). Hence (1.8) holds with $B - b - \eta = 0.4691\dots - \eta$ in place of $B_* - b - \eta = 0.2479\dots - \eta$ (here $b = \pi/\sqrt{3}$). Part IV adds the term $-Dn^{1/6}$, and Part V, conditionally on an unrefereed preprint, the full equivalent. Theorem 1.2 and its proof stand as printed: they are the first proof of Conjecture 5.2, and the shortest.

1.1 [write] Choosable initial intervals

[write] 6 October 2026. This subsection is added in the write. It concerns the conjecture that OEIS A238873 displays (Gus Wiseman, 26 September 2025; quoted verbatim in the front matter), which no manuscript addresses: that $A(n)$ also counts the partitions y of n for which one can choose distinct positive integers $z_1, z_2, \dots$ with $z_i \leq y_i$ for every i . The proof below is the write’s; the same argument was posted earlier, on 31 August 2026, by Clément Garrot in OEIS A387112 [7], with the remark that it proves the corresponding conjectures in A238873, A387118 and A388711.

Proposition 1.3. *Let y be a partition of $n \geq 0$, with parts $y_1, \dots, y_k$ listed in any order, and let $\lambda_1 \leq \dots \leq \lambda_k$ be its weakly increasing arrangement. There are distinct positive integers $z_1, \dots, z_k$ with $z_i \leq y_i$ for every i if and only if $\lambda_i \geq i$ for every i . Consequently, for every $n \geq 0$, the number of partitions of n with choosable initial intervals is $A(n)$ of (1.2), and the number of those without is $p(n) - A(n)$.*

Proof. Whether such z_i exist depends only on the multiset of parts: if σ permutes the indices, a choice (z_i) for (y_i) gives the choice $(z_{\sigma(i)})$ for $(y_{\sigma(i)})$. So we may assume $y_i = \lambda_i$.

If $\lambda_i \geq i$ for every i , take $z_i = i$: these are distinct, and $z_i = i \leq \lambda_i$.

Conversely, let $z_1, \dots, z_k$ be distinct positive integers with $z_i \leq \lambda_i$, and fix $j \in \{1, \dots, k\}$. For $i \leq j$ we have $z_i \leq \lambda_i \leq \lambda_j$, so $z_1, \dots, z_j$ are j distinct elements of $\{1, \dots, \lambda_j\}$, a set with λ_j elements. Hence $j \leq \lambda_j$.

For $n = 0$ the empty partition satisfies both conditions vacuously (with the empty choice), in agreement with $A(0) = 1$. Since the two classes coincide partition by partition, their counts agree for every n , and the complement has $p(n) - A(n)$ members. $\square$

[write] 6 October 2026. Proposition 1.3 proves Wiseman’s conjecture in A238873, and also the one in A387118 (Wiseman, 4 October 2025: the partitions without choosable initial intervals are counted by the non-superdiagonal weakly increasing partitions). Since the equivalence holds partition by partition and preserves

the number of parts, it also proves the refinement by number of parts conjectured in A388711 (Wiseman's entry of 23 September 2025) and the prime-index form conjectured in A387112 [7] (the numbers whose weakly increasing prime indices y_i satisfy $y_i \geq i$), as Garrot's comment says (sentence added after the independent check of 6 October 2026). Exhaustive search at the write, with an independent backtracking search for the distinct choices, confirms the equivalence partition by partition and the counts $1, 1, 1, 2, 3, 3, 5, 7, \dots, 289$ of A238873 for $0 \leq n \leq 24$. Nothing was submitted to the OEIS.

2 Classical input and a fixed rank injection

Put $c = \pi/\sqrt{6}$, so $2c = \pi\sqrt{2/3}$. We use the classical refined Hardy–Ramanujan estimate [3], obtainable from the first term of the Rademacher expansion there,

$$p(n) = \frac{e^{2c\sqrt{n}}}{4\sqrt{3}n} \left(1 + O(n^{-1/2})\right). \quad (2.1)$$

For distinct partitions we need only the classical leading formula, also recalled in [1, Section 5.2],

$$q(n) \sim \frac{e^{b\sqrt{n}}}{4 \cdot 3^{1/4} n^{3/4}}, \quad \log q(n) = b\sqrt{n} + O(\log n). \quad (2.2)$$

These are inputs; no asymptotic enumeration of the constrained classes is being assumed.

For $n \geq 1$, write $L = \lambda_k$, $k = \ell(\lambda)$, $Z_1 = \#\{i : \lambda_i = 1\}$, and $R = L - k$. Let

$$N(r, n) = \#\{\lambda \vdash n : R(\lambda) = r\}, \quad D(n) = p(n) - p(n-1).$$

Deleting one part equal to 1 shows that $D(n)$ counts partitions of n with no ones.

Lemma 2.1 (Uniform rank atom bound). *For every $n \geq 2$ and every integer r ,*

$$N(r, n) \leq D(n), \quad \frac{D(n)}{p(n)} = O(n^{-1/2}). \quad (2.3)$$

Consequently the probability of any interval of m consecutive integer ranks is at most $mD(n)/p(n)$.

Proof. Fix the source rank r . For a partition other than the all-ones partition, delete its $j = Z_1$ ones and add j to one largest remaining part. For $j = 0$ this is the identity. The resulting partition ν has no ones, has the same weight, and satisfies

$$R(\nu) = (L + j) - (k - j) = r + 2j. \quad (2.4)$$

Thus ν and the specified rank r determine $j = (R(\nu) - r)/2$. If $j > 0$, the enlarged part is the unique largest part, even if several parts were largest before the map. Subtracting j from this part and restoring j ones recovers the source. If $j = 0$, the inverse is the identity. Outputs coming from these two cases cannot coincide at fixed r , by (2.4). This proves the injection into the $D(n)$ no-ones partitions.

The excluded partition has rank $1 - n$. This rank is attained only by the all-ones partition: $L \geq 1$ and $k \leq n$, with equality in $L - k \geq 1 - n$ only in that case. Since the singleton (n) has no ones for $n \geq 2$, $D(n) \geq 1$ and (2.3) holds at this rank too. Empty rank classes cause no difficulty.

Finally, directly taking the ratio in (2.1) gives

$$\frac{p(n-1)}{p(n)} = \frac{n}{n-1} e^{2c(\sqrt{n-1} - \sqrt{n})} \left(1 + O(n^{-1/2})\right) = 1 + O(n^{-1/2}).$$

Together with $D(n) \geq 0$, this proves the stated bound. No sharper one-step error is needed. Summing the atom bound gives the interval assertion. $\square$

Attribution. The rank-zero instance of this map is the injection in [1, Lemma 5.4]. Its extension to each specified source rank, using (2.4) to recover the number of deleted ones, is the step needed here.

[write] **6 October 2026.** At rank zero the bound reads $N(0, n) \leq D(n) = O(n^{-1/2}p(n))$; the true size is $N(0, n)/p(n) = \frac{\beta}{4} + O(\beta^2)$ (Liu-Zhou, used in Part II at (11.10)), so the order $n^{-1/2}$ of (2.3) is attained at $r = 0$.

Ferrers conjugation exchanges the largest part and the number of parts, so $N(r, n) = N(-r, n)$. Therefore

$$\mathbb{P}_n(R \leq 0) = \frac{1}{2} + \frac{N(0, n)}{2p(n)} = \frac{1}{2} + O(n^{-1/2}). \quad (2.5)$$

A subdiagonal partition necessarily has $L \leq k$, explaining the source paper's upper bound. The task is to show that the interior indices exclude only a negligible additional proportion.

3 Interior control and exceptional events

Lemma 3.1 (Three regions). *Let $r \geq 0$ and $m \geq 1$ be integers. Suppose that a partition $\lambda \vdash n$ has k parts and satisfies*

$$k \geq 2m + 1, \quad Z_1 \geq m, \quad L \leq k + r - m, \quad (m + 1)(k - m + 1) > n. \quad (3.1)$$

Then $\lambda_i \leq i + r$ for every index i .

Proof. For $i \leq m$, the part is 1. For $i > k - m$, the largest-part bound gives $\lambda_i \leq k + r - m \leq i + r$. If a remaining index $m < i \leq k - m$ violates the conclusion, integrality and $r \geq 0$ imply $\lambda_i \geq i + r + 1 \geq i + 1$. The final $k - i + 1$ parts then give

$$n \geq (i + 1)(k - i + 1).$$

This is a concave quadratic in i , so its minimum on $[m + 1, k - m]$ occurs at an endpoint. Those endpoint values are $(m + 2)(k - m)$ and $(m + 1)(k - m + 1)$; their difference is $k - 2m - 1 \geq 0$. Both are at least the last expression in (3.1), a contradiction. $\square$

[write] **6 October 2026.** Part II restates this interior step, with the same cutoffs (3.2) and the same short-length bound (3.7), in its Section 12 ((12.4), (12.5)). The restatement is kept there, because Part II's endpoint analysis continues from it.

For all sufficiently large n , choose

$$K = \left\lfloor \frac{\sqrt{n} \log n}{4c} \right\rfloor, \quad m = \left\lceil \frac{5c\sqrt{n}}{\log n} \right\rceil. \quad (3.2)$$

Then $K/m \rightarrow \infty$, $m = o(\sqrt{n})$, and

$$\frac{(m + 1)(K - m + 1)}{n} \rightarrow \frac{5}{4}. \quad (3.3)$$

Thus $K \geq 2m + 1$ and $(m + 1)(K - m + 1) > n$ eventually. Both conditions remain true with $k \geq K$ replacing K . The constants in (3.2) are convenient for the proof; no all- n feasibility assertion is made.

3.1 Too few ones

Deleting exactly m ones is a bijection between partitions of n with at least m ones and all partitions of $n - m$. Hence

$$\mathbb{P}_n(Z_1 < m) = 1 - \frac{p(n-m)}{p(n)} = O((\log n)^{-1}). \quad (3.4)$$

To justify the variable shift directly, $n - m \sim n$ and (2.1) give

$$\begin{aligned} \log \frac{p(n-m)}{p(n)} &= 2c(\sqrt{n-m} - \sqrt{n}) + \log \frac{n}{n-m} + O(n^{-1/2}) \\ &= O(m/\sqrt{n}) = O((\log n)^{-1}). \end{aligned}$$

Exponentiation and $p(n-m) \leq p(n)$ prove (3.4). This does not rely on subtracting two uncontrolled asymptotic remainders or on a sharpened one-step ratio.

3.2 Too few parts

By conjugation, $\mathbb{P}_n(k < K)$ is the probability that the largest part is less than K . Let $p_{\leq K}(n)$ count partitions whose parts are at most K . With $t = c/\sqrt{n}$ and $z = e^{-t}$, coefficient positivity in the ordinary partition product gives

$$p_{\leq K}(n) \leq z^{-n} P(z) \prod_{j>K} (1 - z^j), \quad P(z) = \prod_{j \geq 1} (1 - z^j)^{-1}. \quad (3.5)$$

The function $f(x) = -\log(1 - e^{-tx})$ is positive and decreasing; its logarithmic singularity at zero is integrable. Thus

$$\begin{aligned} \log P(e^{-t}) &= \sum_{j \geq 1} f(j) \leq \int_0^\infty f(x) dx \\ &= \sum_{\ell \geq 1} \frac{1}{\ell} \int_0^\infty e^{-t\ell x} dx = \frac{\pi^2}{6t}. \end{aligned}$$

The exchange uses nonnegative terms. In particular,

$$z^{-n} P(z) \leq e^{tn + \pi^2/(6t)} = e^{2c\sqrt{n}}, \quad \frac{z^{-n} P(z)}{p(n)} = O(n).$$

Furthermore, $\log(1 - u) \leq -u$ implies

$$\prod_{j>K} (1 - z^j) \leq \exp\left(-\frac{z^{K+1}}{1 - z}\right), \quad \frac{z^{K+1}}{1 - z} \sim c^{-1} n^{1/4}. \quad (3.6)$$

Indeed, the floor in (3.2) gives $z^{K+1} = n^{-1/4}(1 + O(n^{-1/2}))$, while $1 - z \sim c/\sqrt{n}$. For a fixed $c_0 > 0$ small enough,

$$\mathbb{P}_n(k < K) \leq \frac{p_{\leq K}(n)}{p(n)} = O(ne^{-c_0 n^{1/4}}). \quad (3.7)$$

3.3 The rank window

For every integer r , the interval $r - m < R \leq r$ contains exactly m integer ranks. Lemma 2.1 therefore gives

$$\mathbb{P}_n(r - m < R \leq r) \leq \frac{mD(n)}{p(n)} = O((\log n)^{-1}), \quad (3.8)$$

uniformly in r . The treatment of the all-ones partition in that lemma avoids any exceptional-rank convention in this bound.

3.4 Completion of the first conjecture

Every r -subdiagonal partition satisfies $L \leq k + r$, hence $R \leq r$. Conversely, among partitions satisfying $R \leq r$, remove the events $Z_1 < m$, $k < K$, and $r - m < R \leq r$. All remaining partitions satisfy Lemma 3.1. For every sufficiently large n and every integer $r \geq 0$, we obtain the exact union-bound inequality

$$\begin{aligned} 0 \leq \mathbb{P}_n(R \leq r) - \frac{s_r(n)}{p(n)} &\leq \mathbb{P}_n(Z_1 < m) + \mathbb{P}_n(k < K) \\ &\quad + \mathbb{P}_n(r - m < R \leq r). \end{aligned} \quad (3.9)$$

No independence among these events is assumed. Estimates (3.4), (3.7), and (3.8) prove the uniform coupling. Setting $r = 0$ and using (2.5) gives (1.4), including its sharper one-sided upper bound. Finally (2.1) gives (1.5). This completes the proof of Theorem 1.1, and thus of Conjecture 5.3.

4 A separate logistic crossover

The preceding proof uses no rank limit theorem. The following stronger description of growing shifts invokes the positive-rank part of Dousse–Mertens [2, Theorem 1.2].

Corollary 4.1. *Uniformly for all integers $r \geq 0$,*

$$\frac{s_r(n)}{p(n)} = \frac{1}{1 + \exp(-\pi r / \sqrt{6n})} + O((\log n)^{-1}). \quad (4.1)$$

Consequently, if $r_n / \sqrt{n} \rightarrow \alpha \in [0, \infty)$, the ratio tends to $(1 + e^{-c\alpha})^{-1}$. If $r_n / \sqrt{n} \rightarrow \infty$, the ratio tends to 1.

Proof. Set $\beta = c / \sqrt{n}$, $J = \lfloor \sqrt{n} \log n / (\pi \sqrt{6}) \rfloor$, $g(x) = \frac{1}{4} \operatorname{sech}^2(x/2)$, and $F(x) = (1 + e^{-x})^{-1}$. The cited theorem yields, uniformly for $1 \leq j \leq J$,

$$\frac{N(j, n)}{p(n)} = \beta g(\beta j) \left(1 + O(\beta^{1/2} j^{1/3})\right). \quad (4.2)$$

Only positive ranks are used here; the zero rank is controlled by Lemma 2.1. Since g decreases on $[0, \infty)$,

$$\left| \beta \sum_{j=1}^r g(\beta j) - \int_0^{\beta r} g(x) \, dx \right| \leq \beta g(0) \quad (0 \leq r \leq J). \quad (4.3)$$

In particular the main-term masses have bounded sum. Summing the relative error in (4.2) thus costs at most $O(\beta^{1/2}J^{1/3})$, not J times this quantity. By (2.5), (4.3), and $F' = g$,

$$\mathbb{P}_n(R \leq r) = F(\beta r) + O(n^{-1/12}(\log n)^{1/3}) \quad (0 \leq r \leq J).$$

At the endpoint $\beta J = (\log n)/6 + O(n^{-1/2})$, so $1 - F(\beta J) = O(n^{-1/6})$. For $r > J$, both distribution functions lie between their values at J and 1; monotonicity therefore extends the same error bound to all such r . This error is $o(1/\log n)$. Combining it with Theorem 1.1 proves (4.1). $\square$

[write] **6 October 2026.** The write read the statement of the cited theorem [2] in arXiv:1406.6848: for $|m| \leq \sqrt{n} \log n / (\pi\sqrt{6})$, $N(m, n) = \frac{\beta}{4} \operatorname{sech}^2(\frac{\beta m}{2}) p(n) (1 + O(\beta^{1/2}|m|^{1/3}))$ with $\beta = \pi/\sqrt{6n}$. With $g(x) = \frac{1}{4} \operatorname{sech}^2(x/2)$ this is (4.2) for $1 \leq j \leq J$. The theorem's range includes $m = 0$, where its displayed relative error vanishes; the proof above does not use it there, but takes the rank-zero term from Lemma 2.1. (Added after the independent check of 6 October 2026.) Read literally at $m = 0$, the statement would give $N(0, n) = \frac{\beta}{4} p(n)$ exactly for all large n , which is false: by Part II's (11.10), $N(0, n)/p(n) = \frac{\beta}{4} + a\beta^2 + O(\beta^3)$ with $a = \frac{3}{16} - \frac{3}{2\pi^2} = 0.0355\dots \neq 0$, and in exact arithmetic $N(0, 2000)/(\beta p(2000)) = 0.25103$. (Their Remark 1.3 obtains the factor $|m|^{1/3}$ from the choice $\alpha(m) = |m|^{-1/3}$, which is undefined at $m = 0$.) The rank-zero case of [2] is used nowhere in this report. Corollary 4.1 is the only statement of the report that depends on [2].

5 The inverse at the precision actually proved

For real $y > 1$, define the first threshold

$$T(y) = \min\{n \geq 0 : s_0(n) \geq y\}. \quad (5.1)$$

Prepending a part 1 is an injection from subdiagonal partitions of n to those of $n + 1$: each old index increases by one. Thus $s_0(n)$ is nondecreasing; (1.5) makes it unbounded. It is in fact strictly increasing for $n \geq 4$. At every odd target size at least 5, a partition $(1, 2, \dots, 2)$ has exactly one one; at every even target size at least 6, a partition $(1, 2, \dots, 2, 3)$ does too. Both are subdiagonal and outside the prepending map's image, whose members have at least two ones when the source is nonempty.

Theorem 5.1 (Limited-precision inverse). *Let W_{-1} be the real branch of Lambert's function with values at most -1 . For sufficiently large y , put*

$$x_0(y) = c^{-2} W_{-1} \left(-\frac{c}{\sqrt{8\sqrt{3}y}} \right)^2. \quad (5.2)$$

Then

$$\begin{aligned} T(y) &= x_0(y) + O \left(\frac{\sqrt{x_0(y)}}{\log x_0(y)} + 1 \right) \\ &= x_0(y) + O \left(\frac{\log y}{\log \log y} \right). \end{aligned} \quad (5.3)$$

Writing $H = \log y$, an equivalent explicit initializer is

$$T(y) = \frac{\left[H + 2 \log H + \log(12\sqrt{3}/\pi^2) \right]^2}{4c^2} + O \left(\frac{H}{\log H} \right). \quad (5.4)$$

Proof. For $x > 0$ define

$$M(x) = \frac{e^{2c\sqrt{x}}}{8\sqrt{3}x}, \quad h(x) = \log M(x) = 2c\sqrt{x} - \log x - \log(8\sqrt{3}).$$

The large solution of $M(x) = y$ is (5.2): put $u = \sqrt{x}$, take the positive square root, and rearrange as $(-cu)e^{-cu} = -c/\sqrt{8\sqrt{3}y}$. The branch W_{-1} selects $u \rightarrow \infty$. Also

$$\log s_0(n) = h(n) + O((\log n)^{-1}), \quad h'(x) = \frac{c}{\sqrt{x}} - \frac{1}{x} \sim \frac{c}{\sqrt{x}}. \quad (5.5)$$

Let $x_0 = x_0(y)$ and, for a fixed sufficiently large D , put $\Delta = D\sqrt{x_0}/\log x_0$. For $x \in [x_0 - \Delta - 1, x_0 + \Delta + 1]$, one has $x \sim x_0$, $\log x \sim \log x_0$, and $h'(x) \sim c/\sqrt{x_0}$ uniformly. Thus

$$h(x_0 \pm \Delta) - h(x_0) = \pm \frac{cD + o(1)}{\log x_0}.$$

Choose D large enough to dominate the fixed implicit constant in (5.5). Rounding the lower point down and the upper point up changes h by only $O(x_0^{-1/2}) = o(1/\log x_0)$. For all sufficiently large y , therefore,

$$s_0(\lfloor x_0 - \Delta \rfloor) < y, \quad s_0(\lceil x_0 + \Delta \rceil) \geq y.$$

Monotonicity brackets the first threshold and proves (5.3). Since $\sqrt{x_0} \sim H/(2c)$ and $\log x_0 \sim 2 \log H$, the second error form follows.

To obtain (5.4), set $v = 2c\sqrt{x_0}$. The model equation becomes

$$v - 2 \log v = H + \log(2\sqrt{3}/c^2).$$

It follows first that $v \sim H$, and substitution back into this equation gives

$$v = H + 2 \log H + \log(2\sqrt{3}/c^2) + O(\log H/H).$$

Squaring changes x_0 by $O(\log H)$, which is smaller than $O(H/\log H)$, and $2\sqrt{3}/c^2 = 12\sqrt{3}/\pi^2$. $\square$

The window in (5.3) grows. This theorem does not identify the exact integer threshold by rounding x_0 , does not produce two adjacent candidate integers, and does not supply a numerically certified interval for any specified y . Such statements would need explicit error constants and substantially sharper information. The elementary nested-log initializer is not an all-orders expansion for the constrained counting sequence.

[write] 6 October 2026. Part II's Theorem 10.2 sharpens Theorem 5.1, which concerns only $r = 0$: with the same centre x_0 , the threshold lies between $\lceil X_J - C_J x_0^{-J/2} \rceil$ and $\lceil X_J + C_J x_0^{-J/2} \rceil$ for every fixed J , with $X_1 = x_0 + \frac{7}{24} + \frac{3}{\pi^2}$. The centre x_0 is an instance of the transseries volume's Lambert core after the reduction $\xi = \sqrt{x}$ (front matter, "The inverses"). Strict monotonicity of s_0 from $n = 4$ on was confirmed by exact computation for $4 \leq n \leq 400$ (and $s_0(3) = s_0(4) = 2$, so it does not hold from $n = 3$). The injection that prepends a part 1 is used again in Part II, Section 17.

6 Catalan prefixes and distinct tails

We now prove Conjecture 5.2 independently of the subdiagonal argument. The first construction already yields a strict exponential separation. A refinement in the next section improves the constant.

For $M \geq 1$, let $\mathcal{C}_M$ consist of weakly increasing lists $\lambda = (\lambda_1, \dots, \lambda_M)$ satisfying

$$i \leq \lambda_i \leq M \quad (1 \leq i \leq M). \quad (6.1)$$

Reverse complementation, $\mu_j = M + 1 - \lambda_{M+1-j}$, is a bijection from $\mathcal{C}_M$ to weakly increasing lists with $1 \leq \mu_j \leq j$. Placing the j th north step at horizontal coordinate $\mu_j - 1$ and then completing with east steps gives the usual Dyck path from $(0, 0)$ to (M, M) . The inverse reads those coordinates. The reflection principle therefore gives

$$|\mathcal{C}_M| = C_M = \binom{2M}{M} - \binom{2M}{M+1} = \frac{1}{M+1} \binom{2M}{M}. \quad (6.2)$$

Every prefix has weight $w(\lambda) \leq M^2$.

Let $f_M(N)$ count distinct partitions of N with all parts greater than M , with $f_M(0) = 1$ and $f_M(N) = 0$ for negative N . For a prefix $\lambda \in \mathcal{C}_M$, append such a distinct tail $\tau_1 < \dots < \tau_\ell$. Its j th part is at least $M + j$, so the combined list is superdiagonal at index $M + j$ as well as at every prefix index. It is weakly increasing, since the prefix parts are at most M and the tail parts exceed M . The cutoff M recovers the prefix and tail uniquely. Consequently

$$A(n) \geq \sum_{\lambda \in \mathcal{C}_M} f_M(n - w(\lambda)). \quad (6.3)$$

Lemma 6.1 (Tail comparison). *For $N \geq M + 1$, the values $f_M(N)$ are positive and nondecreasing, and*

$$2^M f_M(N) \geq q(N). \quad (6.4)$$

Proof. The singleton (N) shows positivity. Increasing the unique largest part of a distinct partition by 1 preserves distinctness and the lower cutoff. It is injective: on its image, subtract 1 from the still unique largest part. Thus $f_M(N + 1) \geq f_M(N)$.

Every distinct partition is uniquely determined by its subset of small parts and its remaining large parts. Hence

$$q(N) = \sum_{S \subseteq \{1, \dots, M\}} f_M \left(N - \sum_{j \in S} j \right). \quad (6.5)$$

Each summand is at most $f_M(N)$. For arguments at least $M + 1$, use monotonicity; arguments between 1 and M contribute zero; argument zero contributes $1 \leq f_M(N)$; negative arguments contribute zero. There are 2^M subsets. $\square$

Proposition 6.2 (Finite Catalan lower bound). *For integers $M \geq 1$ and $n - M^2 \geq M + 1$,*

$$A(n) \geq C_M f_M(n - M^2) \geq C_M 2^{-M} q(n - M^2). \quad (6.6)$$

Proof. Every argument in (6.3) is at least $n - M^2$, inside the monotonicity range. Apply Lemma 6.1. $\square$

Fix $0 < \varepsilon < 1$ and take $M = \lfloor \varepsilon \sqrt{n} \rfloor$. The feasibility condition in Proposition 6.2 holds for all sufficiently large n . Stirling's formula in (6.2) and (2.2) give

$$\log(C_M 2^{-M}) = M \log 2 + O(\log M), \quad \log q(n - M^2) = b\sqrt{n - M^2} + O(\log n).$$

It follows that

$$\liminf_{n \rightarrow \infty} \frac{\log A(n)}{\sqrt{n}} \geq \varepsilon \log 2 + b\sqrt{1 - \varepsilon^2}. \quad (6.7)$$

The maximum of the right side is

$$B_0 = \sqrt{b^2 + (\log 2)^2} > b, \quad \varepsilon = \frac{\log 2}{B_0}. \quad (6.8)$$

This follows from Cauchy–Schwarz applied to the unit vector $(\varepsilon, \sqrt{1 - \varepsilon^2})$, with equality at the stated value. Thus the simple construction alone proves Conjecture 5.2. It makes no assertion that B_0 is the actual growth rate.

7 Fixed height prefixes improve the exponential rate

Fix an integer $h \geq 3$. Restrict the prefixes further to

$$i \leq \lambda_i \leq \min(M, i + h - 1). \quad (7.1)$$

Under the reverse complementation in Section 6, this becomes $1 \leq \mu_j \leq j$ and $\mu_j \geq j - h + 1$. In the associated Dyck path, the height $y - x$ just after its j th north step is $j - \mu_j + 1$. Maxima occur just after north steps, so (7.1) corresponds exactly to Dyck paths of semilength M with maximum height at most h .

Let $D_h(M)$ denote their number. If T_h is the adjacency matrix of the path graph on vertices $0, 1, \dots, h$, recording successive heights gives

$$D_h(M) = (T_h^{2M})_{00}. \quad (7.2)$$

The standard sine eigenvectors can also be checked directly: for $1 \leq j \leq h + 1$ and $0 \leq a \leq h$,

$$v_j(a) = \sqrt{\frac{2}{h+2}} \sin \frac{(a+1)j\pi}{h+2}, \quad T_h v_j = 2 \cos \frac{j\pi}{h+2} v_j.$$

The boundary values at $a = -1, h + 1$ vanish, and the sine addition formula verifies the eigenvalue equation. The vectors are an orthonormal basis by finite sine orthogonality. Therefore

$$D_h(M) = \frac{2}{h+2} \sum_{j=1}^{h+1} \sin^2 \frac{j\pi}{h+2} \left(2 \cos \frac{j\pi}{h+2} \right)^{2M}. \quad (7.3)$$

The eigenvalues of largest modulus are opposites, corresponding to $j = 1$ and $j = h + 1$. Since $2M$ is even, their contributions add. For each *fixed* h this proves

$$D_h(M) = c_h \rho_h^M (1 + O_h(\theta_h^M)), \quad \rho_h = 4 \cos^2 \frac{\pi}{h+2}, \quad c_h = \frac{4}{h+2} \sin^2 \frac{\pi}{h+2} > 0, \quad (7.4)$$

where $0 \leq \theta_h < 1$. In particular $\rho_h > 2$ for $h \geq 3$. No uniformity in h is asserted in this estimate.

Every restricted prefix has weight at most

$$W_h(M) = \frac{M(M+1)}{2} + (h-1)M. \quad (7.5)$$

This bound is deliberately simple; the upper cutoff M can make it nonsharp at the final indices. The same injection and tail comparison now give the finite inequality

$$A(n) \geq D_h(M)2^{-M}q(n - W_h(M)) \quad \text{if } n - W_h(M) \geq M + 1. \quad (7.6)$$

Here h, M, n are integers, and the statement itself is exact.

Keep h fixed and put $M = \lfloor \varepsilon\sqrt{n} \rfloor$, with $0 < \varepsilon < \sqrt{2}$. Then $W_h(M)/n \rightarrow \varepsilon^2/2$, so the feasibility condition eventually holds. Using (7.4), (2.2), and (7.6), we obtain

$$\liminf_{n \rightarrow \infty} \frac{\log A(n)}{\sqrt{n}} \geq \varepsilon \log(\rho_h/2) + b\sqrt{1 - \varepsilon^2/2}. \quad (7.7)$$

Write $a_h = \log(\rho_h/2) > 0$. Cauchy–Schwarz, now with the unit vector $(\varepsilon/\sqrt{2}, \sqrt{1 - \varepsilon^2/2})$, shows that the maximum is

$$B_h = \sqrt{b^2 + 2a_h^2}, \quad \varepsilon_h = \frac{2a_h}{\sqrt{b^2 + 2a_h^2}} \in (0, \sqrt{2}). \quad (7.8)$$

Thus the left side of (7.7) is at least B_h for *every fixed* integer $h \geq 3$. Taking the supremum of these already established lower bounds and using $\rho_h \rightarrow 4$ gives (1.7).

This order of limits is essential: first fix h , then take $n \rightarrow \infty$, then take the supremum over fixed heights. No estimate with a height growing with n is used or needed. By (2.2),

$$\limsup_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \log \frac{q(n)}{A(n)} \leq b - B_*.$$

This is equivalent to (1.8) and completes the proof of Theorem 1.2.

For orientation, the three constants are approximately

$$b = 1.8137993642, \quad B_0 = 1.9417314819, \quad B_* = 2.0617405660.$$

These decimals merely illustrate exact expressions. The trivial upper bound $A(n) \leq p(n)$ yields

$$B_* \leq \liminf \frac{\log A(n)}{\sqrt{n}} \leq \limsup \frac{\log A(n)}{\sqrt{n}} \leq \pi\sqrt{2/3} = 2.5650996603\dots \quad (7.9)$$

Neither existence of a limit nor equality with B_* follows from these inequalities.

[write] 6 October 2026. *Settled by Part III.* The limit exists and equals $B = \sqrt{\pi^2/3 + 4\log^2 2} = 2.2829104645\dots$, strictly between B_* and $\pi\sqrt{2/3}$ (Theorem 20.1). Part III removes both losses that Part I’s “further questions” name: an exponential tilt gives a matching upper bound, and a geometric tail with a positive survival probability replaces the distinct tail and its comparison $2^M f_M(N) \geq q(N)$. The bounded-height Dyck prefix of this section is Part III’s Lemma 23.1, used there with a height growing like $t^{-1/3}$ through the exact spectral bound (23.6), which is the sum (7.3) with all but two terms dropped.

8 Exact computation and reproducibility

The source package supplies a Python standard-library companion. Its exact integer computations test the finite mechanisms, not the truth of an asymptotic statement by numerical extrapolation. The empty-partition convention and increasing order are included in all sequence checks.

8.1 Independent enumeration

One method directly enumerates increasing partitions and tests the index inequalities. A second method processes part sizes and their multiplicities, recording the number of parts and total weight. For a block of a new parts equal to v following k smaller parts, the subdiagonal condition is $v \leq k + 1 + r$ when $a > 0$; the superdiagonal condition is $v \geq k + a$. These are respectively the strongest inequalities at the first and last index of the block. Thus the multiplicity method enforces the definitions without reusing the direct enumerator.

The original OEIS numeric prefixes have 57 entries for A238875 and 61 for A238873, both beginning at $n = 0$. The companion stores the extracted integers and source URLs, and compares them with its independently computed values. Exact data extend through $n = 200$. The following sample values are intended only as illustrations of the finite sequences.

n	$p(n)$	$s_0(n)$	$A(n)$	$q(n)$
25	1958	898	349	142
40	37338	17319	3997	1113
50	204226	95502	16582	3658
100	190569292	91171308	5323937	444793
200	3972999029388	1930845918142	25548856857	487067746

For example, at $n = 200$ the ratios are about 0.485992 and 0.0190642. These do not establish an error constant, a valid starting index, or the exact superdiagonal rate.

8.2 Finite mechanisms checked

The companion verifies the fixed-rank map and its inverse, detects collisions within each rank class, and treats the all-ones rank separately. It checks the deterministic interior lemma whenever its hypotheses hold; shifted subdiagonal counts are compared with the appropriate rank cumulative counts. The prefix tests compare Catalan enumeration with the exact binomial formula, and bounded height prefix enumeration with integer walk counts. Distinct-tail monotonicity, the small-subset decomposition, prefix–tail validity and disjointness, and both finite lower bounds are tested with integer arithmetic. The shipped verification exhausts 215,308 partitions, including the empty partition, through $n = 40$; checks 215,267 injection and inverse cases and 446,242 interior-lemma cases; and verifies 1,690 Catalan and 9,995 bounded-height finite lower bounds. The reproducible output records the remaining ranges and numbers of successful checks.

Finite verification does not replace any proof above. In particular, the asymptotic choices (3.2) need not satisfy the interior hypotheses at modest n , and finite data cannot certify the eventual spectral or partition asymptotics. The exact threshold command, where supported within a finite table, must search actual integer counts; the asymptotic center is not an integer-threshold certificate.

8.3 Package contract

The accompanying README gives bounded commands, exact result files, and regression tests that run both normally and with Python optimization enabled. Validation does not depend on Python `assert` statements. Output paths must be new regular files in trusted directories; existing paths, symlinks, and special-file targets are rejected. The software has explicit finite workload limits independent of the unrestricted mathematical theorems.

The PDF builder uses a clean exclusive directory, a fixed epoch, fixed metadata, an isolated LaTeX format, disabled shell escape, and stable references. A fixed allowlist and SHA-256 manifest control the source ZIP; working files and third-party journal PDFs are not included. Two clean builds and two archive replays must be byte-identical within the recorded toolchain. This is a same-toolchain reproducibility claim, not a promise of identical output across different TeX versions or fonts. The trusted-directory and modified-TeX qualifications are stated in the README.

[write] **6 October 2026.** The source package is shipped with the prefix `156-conjectures-`: `companion.py` and `test_companion.py`, `build_pdf.py`, `make_zip.py`, `release_tools.py` and `test_release.py` under `code/`, and the two result files, `data/oeis_prefixes.json` and `SOURCE_PROVENANCE.json` under `data/`. The PDF, the README (replaced by this report's) and `SHA256SUMS` are not shipped. The report's README gives the rerun route from the arrival archive. At intake, on a Windows copy, the exact data were regenerated byte for byte; two of the 37 companion tests fail there, because they exercise the POSIX-only safe-output code.

9 Further questions and present limits

The subdiagonal defect. The coupling reduces the half-partition conjecture to the fact that interior violations are rare. It does not identify the first term of

$$\mathbb{P}_n(R \leq 0) - s_0(n)/p(n).$$

The $O(1/\log n)$ estimate comes from a deliberately coarse union bound and uniform rank atom estimate. Determining the actual defect scale, its leading constant, or an all-orders expansion requires more precise joint information on the rank, small-part multiplicities, and interior shape. The known expansion of $p(n)$ alone does not provide an expansion of $s_0(n)$.

[write] **6 October 2026.** *Answered at $r = 0$ by Part II.* By (2.5), $\mathbb{P}_n(R \leq 0) = \frac{1}{2} + N(0, n)/(2p(n))$. Part II's (11.10) at $r = 0$ gives $N(0, n)/p(n) = \frac{\beta}{4} + a\beta^2 + O(\beta^3)$ with $a = \frac{3}{16} - \frac{3}{2\pi^2}$, and its (10.2) gives $s_0(n)/p(n) = \frac{1}{2} - \frac{\beta}{8} + c_2\beta^2 + O(\beta^3)$ with $c_2 = -\frac{11}{32} + \frac{3}{4\pi^2}$; here $\beta = \pi/\sqrt{6n}$. Since $\frac{a}{2} - c_2 = \frac{3}{32} - \frac{3}{4\pi^2} + \frac{11}{32} - \frac{3}{4\pi^2}$,

$$\mathbb{P}_n(R \leq 0) - \frac{s_0(n)}{p(n)} = \frac{\beta}{4} + \left(\frac{7}{16} - \frac{3}{2\pi^2}\right)\beta^2 + O(\beta^3). \quad (9.1)$$

So the defect has exact order $n^{-1/2}$, with leading term $\pi/(4\sqrt{6n})$; the $(\log n)^{-1}$ of the union bound (3.9) is not its true size. The leading term needs only Liu-Zhou's $N(0, n)/p(n) = \frac{\beta}{4} + O(\beta^2)$ and Part II's c_0, c_1 ; the β^2 term uses Part II's a and c_2 . A numerical check at the write, with exact $p(n)$, exact $N(0, n)$ (the Atkin-Swinnerton-Dyer rank formula, checked against enumeration for $n \leq 30$) and a floating-point dynamic program for $s(n)$: the defect divided by β is 0.2957, 0.2623, 0.2599 at $n = 100, 1000, 1500$, and $(\text{defect} - \beta/4)/\beta^2$ is 0.356, 0.303, 0.299 there, against $\frac{7}{16} - \frac{3}{2\pi^2} = 0.2855$. Part II also gives $s_0(n)$ itself to every order. The defect at fixed $r \geq 1$ and at growing r remains open (Section 47, item 1).

Sharper shifts. Theorem 1.1 is uniform over all nonnegative integer r , including n -dependent shifts. A natural refinement is an error sensitive to $r/\sqrt{n}$, with smaller error in regions where the logistic tail is small. Negative integer shifts do not give a symmetric extension of this definition: $r \leq -1$ already makes $\lambda_1 \leq 1 + r$ impossible for every nonempty partition.

[write] **6 October 2026.** Open; Section 47, item 1.

The superdiagonal exponential constant. The prefix construction sacrifices information twice: it replaces prefix weights by a common upper bound, and it bounds every small-part subset contribution by the largest tail count. Keeping the full weighted fixed-height transfer matrix, or using a sharper tail comparison, may improve B_* . A matching upper bound and a proof that $\log A(n)/\sqrt{n}$ has a limit remain open within this report. A leading amplitude, correction series, and precise inverse for $A(n)$ are therefore also beyond the present results.

[write] **6 October 2026.** *Answered by Part III, and in part by Parts IV and V.* The limit exists and equals B , with the matching upper bound (Theorem 20.1). The next term is $-Dn^{1/6}$, $D = \zeta bC^{-1/6}$, with an inverse to $o(x^{2/3})$ (Part IV, unconditional); the amplitude K , the power $n^{-11/12}$ and an inverse to $o(\sqrt{x})$ are Part V's, conditional on an unrefereed preprint. A correction series remains open (Section 47, items 5–7).

Explicit numerical onset. The superdiagonal finite inequalities are directly computable. The asymptotic results would become numerical certificates only after supplying explicit classical remainder bounds and tracking all constants through the coupling and inverse. In particular, the window in Theorem 5.1 cannot be turned into a specified finite search interval merely by choosing an arbitrary multiplier in its big- O term.

[write] **6 October 2026.** Open for every Part of this report; Section 47, item 2.

Part II

All orders asymptotics for subdiagonal partitions

Bundle Report 159, archive `Subdiagonal_Partitions_All_Orders_Expansion_and_Integer_Inverses_Source.zip`, delivered as `Report159.tex` (17 pages). Delivered title block “All orders asymptotics for subdiagonal partitions / Endpoint reductions and integer inverse localization / Report 159 / 3 October 2026”. Printed as delivered, with **[write]** notes. Delivered Section k is Section $k + 9$ here, and its statement or equation $k.j$ is $(k + 9).j$. It treats only $s = s_0$ (shift $r = 0$).

Reading conventions of this Part	Elsewhere in the report
$t = c^{-2} = 6/\pi^2$, a <i>number</i> ; $\beta = c/\sqrt{n}$ the radial variable	Parts I, III–V: t a variable
$b = \pi/\sqrt{6(n-1/24)}$ (Zhou); $a = \frac{3}{16} - \frac{t}{4}$, $B(r)$ rank coefficients; $h = \frac{7}{24} + \frac{t}{2}$	Part I: $b = \pi/\sqrt{3}$; Parts III–V: $b = \log 2$, B the rate
R the rank and a truncation order; C_i , G_b multiplicity and gap sums; E_n failure counts	Part I: R the rank; Parts III–V: C_j , G a buffer

Abstract of this Part (as delivered). Let $s(n)$ count weakly increasing partitions $1 \leq \lambda_1 \leq \dots \leq \lambda_k$ of n satisfying $\lambda_i \leq i$ for every index, and let $p(n)$ be the ordinary partition number. For $\beta = \pi/\sqrt{6n}$, we prove a unique, algorithmically computable formal expansion

$$\frac{s(n)}{p(n)} \sim \sum_{j \geq 0} c_j \beta^j, \quad c_j \in \mathbb{Q}[\pi^{-2}],$$

with a remainder $O_R(\beta^{R+1})$ after every fixed order R . The first four coefficients are

$$c_0 = \frac{1}{2}, \quad c_1 = -\frac{1}{8}, \quad c_2 = -\frac{11}{32} + \frac{3}{4\pi^2}, \quad c_3 = -\frac{317}{384} + \frac{329}{64\pi^2} - \frac{9}{8\pi^4}.$$

A uniform endpoint estimate reduces each requested order to finitely many rank counts with fixed shifts. A mixed row and column insertion bijection handles intersections of bottom and top obstructions, including explicitly enumerable, polynomially bounded exceptional classes. Zhou’s published complete fixed-rank expansion then supplies the analytic coefficients. Six small endpoint patterns explain the cubic term directly.

The ordinary threshold inverse has arbitrarily small algebraic windows about computable smooth centers, with ceilings retained on both endpoints. Its second explicit center is

$$x_0 + \frac{7}{24} + \frac{3}{\pi^2} + \left(\frac{23}{32} + \frac{45}{2\pi^4} \right) \frac{\pi}{\sqrt{6x_0}},$$

with window radius $O(x_0^{-1})$, where x_0 is the large Lambert W inverse of the leading model. Error constants and onset are not numerically specified. The companion uses exact integer and rational arithmetic and implements coefficient extraction only through order three.

Attribution and scope. The sequence is OEIS A238875. Archibald, Blecher, Elizalde and Knopfmacher formulated the half-proportion conjecture in their 2025 paper [1]. Report 156 [15] proved it with an $O(1/\log n)$ error by an elementary rank coupling. The present report is a distinct refinement: it proves an all-orders endpoint reduction and sharper integer inverse localization, using

additional published rank asymptotics. It makes no claim about a superdiagonal expansion. A bounded check of the source literature, OEIS data, and the available report catalogue found no duplicate of this refinement as of 3 October 2026; it is not exhaustive scholarly priority clearance.

10 Definitions and main results

A partition is written in weakly increasing order unless a decreasing row notation is explicitly used. Define

$$s(n) = \#\{(\lambda_1, \dots, \lambda_k) : 1 \leq \lambda_1 \leq \dots \leq \lambda_k, \quad \sum_i \lambda_i = n, \quad \lambda_i \leq i\},$$

and put $s(0) = p(0) = 1$ for the empty partition. The increasing order is essential. The source definition and numerical sequence are A238875 [5]; the first values for $n = 0, \dots, 10$ are

$$1, 1, 1, 2, 2, 4, 5, 7, 10, 15, 18.$$

For a nonempty partition let k be its length, L its largest part, and $R = L - k$ its Dyson rank. Write

$$N(r, n) = \#\{\lambda \vdash n : R = r\}, \quad P_r(n) = \sum_{u \leq r} N(u, n), \quad c = \frac{\pi}{\sqrt{6}}, \quad \beta = \frac{c}{\sqrt{n}}.$$

The empty partition has rank zero. Counts at negative weight are zero. Conjugation gives $N(r, n) = N(-r, n)$. All limits below have the integer truncation order fixed in advance.

Theorem 10.1 (Complete asymptotic expansion). *There is a unique sequence of computable coefficients $c_j \in \mathbb{Q}[\pi^{-2}]$ such that, for every integer $R \geq 0$,*

$$\frac{s(n)}{p(n)} = \sum_{j=0}^R c_j \beta^j + O_R(\beta^{R+1}). \quad (10.1)$$

In particular,

$$\begin{aligned} \frac{s(n)}{p(n)} &= \frac{1}{2} - \frac{\beta}{8} + \left(-\frac{11}{32} + \frac{3}{4\pi^2}\right) \beta^2 \\ &\quad + \left(-\frac{317}{384} + \frac{329}{64\pi^2} - \frac{9}{8\pi^4}\right) \beta^3 + O(\beta^4). \end{aligned} \quad (10.2)$$

At any requested order, a terminating finite combinatorial reduction followed by the published fixed-rank expansion computes the coefficients. No convergence or computational efficiency is asserted.

The first correction is negative; the half-proportion is approached from below eventually. In powers of n , the second-order statement is

$$\frac{s(n)}{p(n)} = \frac{1}{2} - \frac{\pi}{8\sqrt{6n}} + \left(\frac{1}{8} - \frac{11\pi^2}{192}\right) \frac{1}{n} + O(n^{-3/2}).$$

The full cubic formula has remainder $O(n^{-2})$.

[write] **6 October 2026.** “Report 156” [15] is Part I. Theorem 10.1 proves Conjecture 5.3 of [1] a second time, with a far sharper error, but only for the unshifted class $s = s_0$; Part I’s uniform statement for every shift r and its logistic Corollary 4.1 have no counterpart here. Combined with Part I’s (2.5) it gives the defect

formula (9.1). “Eventually” in the first sentence of this paragraph is needed: by exact computation (write $s(n)/p(n) < \frac{1}{2}$ for every $10 \leq n \leq 400$, while it equals $\frac{1}{2}$ at $n = 2$ and $n = 9$ and exceeds it at $n = 0, 1, 3, 5$. Numerically (intake, $n \leq 1500$), $(s/p - \sum_{j \leq 2} c_j \beta^j)/\beta^3 = -0.544, -0.629, -0.517, -0.441, -0.411, -0.388$ at $n = 100, 200, 400, 700, 1000, 1500$, drifting towards $c_3 = -0.3162$, and the remainder after the cubic term divided by β^4 stays near -2 to -3.5 ; this supports $c_0, \dots, c_3$ and proves nothing.

To state the inverse, set

$$F(x) = \frac{e^{2c\sqrt{x}}}{8\sqrt{3}x}, \quad x_0(y) = c^{-2}W_{-1}\left(-\frac{c}{\sqrt{8\sqrt{3}y}}\right)^2, \quad (10.3)$$

where W_{-1} is the real branch with values at most -1 . Thus $F(x_0(y)) = y$ on the large branch. Define the integer threshold

$$T(y) = \min\{n \in \mathbb{Z}_{\geq 0} : s(n) \geq y\}. \quad (10.4)$$

Theorem 10.2 (Integer inverse localization). *For each fixed $J \geq 1$ there are computable coefficients $b_0, \dots, b_{J-1} \in \mathbb{Q}[\pi^{-2}]$ such that, with $\beta_0 = c/\sqrt{x_0(y)}$ and*

$$X_J(y) = x_0(y) + \sum_{r=0}^{J-1} b_r \beta_0^r, \quad (10.5)$$

there exist constants $C_J, y_J > 0$ for which

$$\left\lceil X_J - C_J x_0^{-J/2} \right\rceil \leq T(y) \leq \left\lceil X_J + C_J x_0^{-J/2} \right\rceil \quad (y \geq y_J). \quad (10.6)$$

The first coefficients and second center are

$$b_0 = \frac{7}{24} + \frac{3}{\pi^2}, \quad b_1 = \frac{23}{32} + \frac{45}{2\pi^4}, \quad (10.7)$$

$$X_2 = x_0 + \frac{7}{24} + \frac{3}{\pi^2} + \left(\frac{23}{32} + \frac{45}{2\pi^4}\right) \frac{\pi}{\sqrt{6x_0}}. \quad (10.8)$$

The constants C_J, y_J are not numerically explicit here.

For $J \geq 1$, the two ceilings in (10.6) eventually agree or are adjacent. Agreement determines $T(y)$; disagreement leaves two candidates. This is not an assertion that $T(y) = X_J(y) + o(1)$, nor a certified finite-input rounding algorithm.

11 The analytic inputs and their fixed rank specialization

The combinatorial part of the proof is given in full below. Its analytic inputs are the classical partition expansion and Zhou’s complete expansion for each fixed Dyson rank. Their provenance and specialization matter, particularly at rank zero.

11.1 The smooth partition term

Put

$$H(x) = \frac{e^{2c\sqrt{x-1/24}}}{4\sqrt{3}(x-1/24)} \left(1 - \frac{1}{2c\sqrt{x-1/24}}\right). \quad (11.1)$$

The Rademacher expansion [3] gives, for every fixed $\varepsilon \in (0, c)$,

$$p(n) = H(n) + O\left(e^{(c+\varepsilon)\sqrt{n}}\right). \quad (11.2)$$

Expanding the explicit expression H gives the full algebraic expansion of $p(n)$ and each fixed shift. Finite differences may be computed from derivatives of H on fixed neighborhoods; the exponentially smaller error in (11.2) is harmless. In particular, for fixed positive $a_1, \dots, a_d$,

$$[q^n]P(q) \prod_{h=1}^d (1 - q^{a_h}) = O(\beta^d p(n)), \quad P(q) = \prod_{j \geq 1} (1 - q^j)^{-1}. \quad (11.3)$$

Indeed the finite difference of H is an integral of $H^{(d)}$ over a fixed box, and $H^{(d)}(x) = O(\beta^d H(n))$ there. This reasoning does not subtract uncontrolled big- O terms.

It is convenient to write $t = c^{-2} = 6/\pi^2$. Through degree three,

$$\begin{aligned} \frac{H'}{H} &= \beta - t\beta^2 + \left(\frac{t^2}{4} + \frac{t}{48}\right)\beta^3 + O(\beta^4), \\ \frac{H''}{H} &= \beta^2 - \frac{5t}{2}\beta^3 + O(\beta^4), \quad \frac{H'''}{H} = \beta^3 + O(\beta^4). \end{aligned} \quad (11.4)$$

Consequently, for every fixed $w \geq 0$,

$$\begin{aligned} \frac{p(n-w)}{p(n)} &= 1 - w\beta + \left(wt + \frac{w^2}{2}\right)\beta^2 \\ &\quad - \left[w\left(\frac{t^2}{4} + \frac{t}{48}\right) + \frac{5w^2t}{4} + \frac{w^3}{6}\right]\beta^3 + O_w(\beta^4). \end{aligned} \quad (11.5)$$

11.2 An explicit published recipe for the rank coefficients

Liu and Zhou [8, Theorem 1.3] give $N(r, n)/p(n) = \beta/4 + O_r(\beta^2)$ for every fixed integer r , including zero. For all orders we use Zhou [9, Theorem 4.1], specialized to his parameters $k = 2$, $m = 0$, $j = |r|$. These values satisfy his hypotheses. No rank increasing with n occurs in any finite reduction below.

For clarity, the specialized coefficient recipe can be written using only finite formal expansions. Set

$$b = \frac{\pi}{\sqrt{6(n-1/24)}}, \quad A = \frac{\sqrt{3}b^2e^{\pi^2/(3b)}}{2\pi^2}, \quad f(z) = \frac{1}{1+e^z},$$

and define

$$d_h(j) = (1/2 - j)^h - (-1)^h(1/2 + j)^h, \quad (11.6)$$

$$\gamma_\ell(\mu) = \frac{1}{8^\ell \ell!} \prod_{a=1}^{\ell} (4\mu^2 - (2a-1)^2), \quad \gamma_0(\mu) = 1. \quad (11.7)$$

The $m = 0$ specialization of the source operators gives the formal asymptotic recipe

$$\frac{N(r, n)}{A} \sim \sum_{h \geq 1} \frac{d_h(|r|)}{h!} (-b)^h \sum_{s, \ell \geq 0} \frac{(3/2)^s}{s!} \left(\frac{3}{\pi^2}\right)^\ell \gamma_\ell(-s - h - 3/2) f^{(h+2s)}(0) (-b)^{s+\ell}. \quad (11.8)$$

To any requested degree, the double sum is finite after retaining $h + s + \ell$ up to that degree. Formula (11.7) is the $v = 0$ coefficient from the source's Proposition 2.2. Terms containing positive powers of the evaluation variable vanish at $m = 0$. The derivatives $f^{(a)}(0)$ are rational and are computed by formal inversion of $1 + e^z$. Since $p(n)/A = 1 - 3b/\pi^2$ up to exponentially small error, division and the change from b to β give coefficients in $\mathbb{Q}[\pi^{-2}]$. Taking one additional finite order of the published expansion supplies every big- O remainder used below.

This is a use of an established coefficient-asymptotic theorem, not a general inference from a local generating-function expansion. In particular, the all-orders proof does not assume that an arbitrary local power series can be transferred termwise to coefficients.

[write] 6 October 2026. The write read the statements of both inputs in the cited arXiv versions. Liu–Zhou's Theorem 1.3 [8]: for $m = o(n^{3/4})$, $N_k(m, n)/p(n) = \frac{\pi}{4\sqrt{6n}} \operatorname{sech}^2\left(\frac{\pi m}{2\sqrt{6n}}\right) (1 + O(\frac{n+m^2}{n^{3/2}}))$, with N_2 the rank count; at fixed r this is $\beta/4 + O_r(\beta^2)$, as used, rank zero included. Zhou's Theorem 4.1 [9] expands $N_k(m + j, n)$ for fixed j , with $m + j \geq 0$ and $m^2\beta_n^3 = o(1)$; $k = 2$, $m = 0$, $j = |r|$ meets these hypotheses. The write did not re-derive the recipe (11.8) from Zhou's Proposition 2.2; the direct check of the cubic coefficient below is Part II's own. Zhou's variable b here is not Part I's $b = \pi/\sqrt{3}$ nor the $b = \log 2$ of Parts III–V, and $t = 6/\pi^2$ is a number in this Part (front matter, "Notation").

11.3 The rank input through cubic order

Let

$$a = \frac{3}{16} - \frac{t}{4}, \quad B(r) = \frac{53}{192} - \frac{r^2}{16} - \frac{89t}{192} + \frac{t^2}{16}. \quad (11.9)$$

The specialization just given yields

$$\frac{N(r, n)}{p(n)} = \frac{\beta}{4} + a\beta^2 + B(r)\beta^3 + O_r(\beta^4). \quad (11.10)$$

Here is a direct check of the third coefficient. Write $x = \pi^{-2}$. The b^2 coefficient inside the $h = 1$ source operator is

$$-\frac{9}{32} + \frac{27x}{8} - \frac{27x^2}{4},$$

using $f'(0) = -1/4$, $f'''(0) = 1/8$, $f^{(5)}(0) = -1/4$, $\gamma_1(-7/2) = 6$, and $\gamma_2(-5/2) = 3$. The $h = 2$ terms vanish at the needed orders because the positive even derivatives of f at zero vanish. Since $d_3(r) = 3r^2 + 1/4$, the $h = 3$ contribution to N/A is $-b^3(r^2/16 + 1/192)$. Thus its cubic coefficient is

$$\frac{53}{192} - \frac{r^2}{16} - \frac{27x}{8} + \frac{27x^2}{4}.$$

Dividing by $1 - 3bx$ and using $b = \beta + x\beta^3/8 + O(\beta^5)$ gives exactly (11.10). For fixed r, w it follows that

$$\begin{aligned} \frac{N(r, n - w)}{p(n)} &= \frac{\beta}{4} + \left(a - \frac{w}{4}\right)\beta^2 \\ &\quad + \left[B(r) - aw + \frac{w^2}{8} + \frac{3wt}{8}\right]\beta^3 + O_{r,w}(\beta^4). \end{aligned} \quad (11.11)$$

The rank-independent second coefficient and the polynomial degree of the cubic coefficient will force the endpoint cancellations.

12 Uniform localization of the obstructions

Write Z_a for the multiplicity of part a . Let E_n count partitions with $R \leq 0$, $Z_1 \geq 1$, and at least one failed subdiagonal inequality. Deleting one 1 gives, for $n \geq 2$, the exact identity

$$s(n) = P_1(n-1) - E_n. \quad (12.1)$$

For the all-ones partition of size at least two, deletion also preserves the rank-threshold equivalence; the singleton gives the same identity separately. Hence (12.1) also holds at $n = 1$. We only use it asymptotically.

12.1 Boltzmann conditioning

Under the partition Boltzmann probability Q_β , a partition λ has probability $q^{|\lambda|}/P(q)$ with $q = e^{-\beta}$. The random variables Z_a are independent and

$$Q_\beta(Z_a = z) = (1 - q^a)q^{az}, \quad z \geq 0.$$

For any set of partitions $\mathcal{E}$, positivity gives

$$\mathbb{P}_n(\mathcal{E}) \leq \frac{q^{-n}P(q)}{p(n)}Q_\beta(\mathcal{E}) = O(n)Q_\beta(\mathcal{E}). \quad (12.2)$$

To check the last estimate without a local limit theorem, the positive decreasing function $-\log(1 - e^{-\beta x})$ has integral $\pi^2/(6\beta)$, so

$$\log P(e^{-\beta}) \leq \frac{\pi^2}{6\beta}, \quad q^{-n}P(q) \leq e^{2c\sqrt{n}}, \quad p(n) \asymp n^{-1}e^{2c\sqrt{n}}.$$

Let $C_i = Z_1 + \cdots + Z_i$. There are $\binom{2i-1}{i} \leq 4^i$ nonnegative i -tuples with sum at most $i-1$. Therefore

$$Q_\beta(C_i \leq i-1) \leq 4^i \prod_{h=1}^i (1 - q^h) \leq 4^i \beta^i i! \leq (4\beta i)^i. \quad (12.3)$$

Conjugation preserves this probability and exchanges small-part multiplicities with the top gaps used below.

12.2 Excluding the middle and short partitions

Choose

$$K = \left\lfloor \frac{\sqrt{n} \log n}{4c} \right\rfloor, \quad m = \left\lceil \frac{5c\sqrt{n}}{\log n} \right\rceil. \quad (12.4)$$

Eventually $K \geq 2m + 1$ and $(m+1)(K-m+1) > n$, since the ratio of the latter product to n tends to $5/4$. A failure at an increasing index i forces

$$n \geq (i+1)(k-i+1).$$

If $k \geq K$ and $m < i \leq k-m$, concavity in i shows that the right side is at least $(m+1)(k-m+1) > n$. Thus every failure is at $i \leq m$ or at a decreasing index $j = k-i+1 \leq m$. This uses the same interior mechanism as Report 156, here retained with much finer endpoint information.

By conjugation and coefficient positivity, the short-length event satisfies

$$\begin{aligned}\mathbb{P}_n(k < K) &\leq \frac{q^{-n}P(q)}{p(n)} \prod_{a>K} (1 - q^a) \\ &\leq O(n) \exp\left(-\frac{q^{K+1}}{1-q}\right) = O(ne^{-c_0 n^{1/4}})\end{aligned}\tag{12.5}$$

for some $c_0 > 0$, because $q^{K+1}/(1-q) \sim c^{-1}n^{1/4}$. This is smaller than every fixed power of β .

[write] 6 October 2026. The cutoffs (12.4), the interior step of the preceding paragraph and the short-length bound (12.5) restate Part I's (3.2), Lemma 3.1 and (3.5)–(3.7). They are kept, since the endpoint analysis below starts from them; the conditioning bound (12.2) is the estimate Part I proves in Section 3 for $z^{-n}P(z)/p(n)$.

12.3 Fixed endpoint cutoffs with arbitrary accuracy

Use decreasing rows $\mu_1 \geq \mu_2 \geq \dots$, padded by zero, and gaps $G_b = \mu_b - \mu_{b+1}$. These gaps are the multiplicities of parts b in the conjugate partition. In the class counted by E_n , a bottom failure at i means $C_i \leq i - 1$; necessarily $i \geq 2$. At the top put $R = -d$, $d \geq 0$. A failure at decreasing index j is equivalent to

$$0 \leq d \leq j - 2, \quad G_1 + \dots + G_{j-1} \leq j - 2 - d, \tag{12.6}$$

since $L = k - d$ and $\mu_j \geq k - j + 2$.

Fix $M \geq 2$. Let $E_n^{(M)}$ be the union of the bottom events

$$R \leq 0, \quad Z_1 \geq 1, \quad C_i \leq i - 1 \quad (2 \leq i \leq M)$$

and the top events

$$R = -d, \quad Z_1 \geq 1, \quad \sum_{b=1}^{j-1} G_b \leq j - 2 - d, \quad 0 \leq d \leq j - 2, \quad 2 \leq j \leq M + 1.$$

These really are failure events for all $n > (M + 1)^2$: if a required index does not exist, $R \leq 0$ and length at most M force the weight to be at most M^2 .

For $a_i = (4\beta i)^i$ and $M + 1 \leq i < m$,

$$\frac{a_{i+1}}{a_i} = 4\beta(i+1)(1+1/i)^i \leq 4e\beta(m+1) \leq \frac{1}{2}$$

eventually. Hence

$$\sum_{i=M+1}^m (4\beta i)^i = O_M(\beta^{M+1}).$$

The omitted top events have $j \geq M + 2$, so their gap-prefix length $j - 1$ is also at least $M + 1$. Equations (12.2), (12.3), and (12.5) now prove

$$0 \leq E_n - E_n^{(M)} = O_M(\beta^{M-1}p(n)). \tag{12.7}$$

The loss of two powers comes from $O(n) = O(\beta^{-2})$ in conditioning. This is a uniform bound on a growing endpoint range; no fixed-rank expansion has been used outside a fixed range.

13 Exact mixed row and column reductions

The all-orders argument must handle simultaneous conditions at both ends. Separate bottom and top estimates do not suffice.

Lemma 13.1 (Small rectangle complements). *For fixed integers $B \geq 1$, $C \geq 0$, the number of partitions of n with $\mu_B \leq C$ is $O_{B,C}((n+1)^{B+C-1})$. These counts remain explicitly enumerable after imposing fixed rank or additional marked-coordinate conditions.*

Proof. Specify at most $B - 1$ initial row lengths, each at most n , and the multiplicities of the possible sizes $1, \dots, C$ below these rows. There are at most $(n+1)^{B+C-1}$ descriptions. Enforcing ordering and total weight only reduces this upper bound. All entries can be enumerated in finite boxes and tested against the extra conditions. $\square$

Such exceptional counts are polynomially bounded; they need not be polynomial functions of n . Dividing any fixed polynomial bound by $p(n)$ gives an error smaller than every algebraic power of β .

Lemma 13.2 (Mixed lower bounds). *Fix $A \geq 0$, $B \geq 1$, and nonnegative integer vectors $x = (x_1, \dots, x_A)$, $y = (y_1, \dots, y_B)$. Let $L_{x,y,r}(n)$ count partitions of rank r satisfying $Z_a \geq x_a$ for $a \leq A$ and $G_b \geq y_b$ for $b \leq B$. Set*

$$u = \sum_a x_a, \quad v = \sum_b y_b, \quad w = \sum_a a x_a + \sum_b b y_b.$$

Then the exact identity is

$$L_{x,y,r}(n) = N(r + u - v, n - w) - B_{\text{src}} + B_{\text{tgt}}, \quad (13.1)$$

where B_{src} counts cores of weight $n - w$, rank $r + u - v$, and $\nu_B \leq A$, while B_{tgt} counts the target partitions in $L_{x,y,r}(n)$ with $\mu_B \leq A + y_B$. Both complements are polynomially bounded and explicitly enumerable.

Proof. Start with a core ν having $\nu_B > A$. Insert x_a rows of length a for each $a \leq A$. They lie below the first B rows, which do not change. Now add y_b columns of height b , for $b \leq B$; equivalently, add $\sum_{j=b}^B y_j$ to row b . Each gap G_b increases by y_b in this second step. Inserting the small rows can change the boundary gap G_B , since row $B + 1$ can change, but the gap remains nonnegative. Thus the final lower bounds on every gap hold. No invariance of G_B under row insertion is required.

The affected top rows remain larger than A , so column insertion preserves all marked small-part multiplicities. Both operations act on the same top rows and inserted small rows in either order, so they commute on this region. Weight increases by w , length by u , and largest part by v . The output therefore has rank r and $\mu_B = \nu_B + y_B > A + y_B$.

Conversely, remove the indicated columns from any target with $\mu_B > A + y_B$. The gap lower bounds make all resulting gaps nonnegative. Its first B rows remain larger than A , hence no marked small-part multiplicity changes. Remove the specified small rows; the first B rows stay fixed and the original good core is recovered. No gap lower bound is imposed on that core. These maps are mutually inverse.

The good source and target sets are therefore equinumerous. Subtracting the bad-source complement and adding the independently defined bad-target complement gives (13.1). The latter are covered by Lemma 13.1. Empty cores and negative core weights cause no missing case under the stated conventions. In particular, targets with $n < w$ are necessarily bad targets. $\square$

When $B = 0$, the same argument inserts only small rows, using the good-core condition “largest part greater than A .” Both complements have largest part at most A and are polynomially bounded. The rank shift is $r + u$. This pure-row identity also holds for the cumulative condition $R \leq r$, with $P_{r+u}(n - w)$ in place of the rank atom.

Exact values are obtained by finite differences:

$$\mathbf{1}_{Z_a=x_a} = \mathbf{1}_{Z_a \geq x_a} - \mathbf{1}_{Z_a \geq x_a+1}, \quad \mathbf{1}_{G_b=y_b} = \mathbf{1}_{G_b \geq y_b} - \mathbf{1}_{G_b \geq y_b+1}. \quad (13.2)$$

Consequently every finite pattern of marked values at fixed rank is a finite integer combination of shifted $N(r, n - w)$ plus polynomially bounded complements. Pure-row cumulative patterns reduce similarly to $P_r(n - w)$. Rank symmetry gives the exact conversion, for every integer r ,

$$P_r(n) = \begin{cases} \frac{p(n)}{2} + \frac{N(0, n)}{2} + \sum_{j=1}^r N(j, n), & r \geq 0, \\ \frac{p(n)}{2} - \frac{N(0, n)}{2} - \sum_{j=1}^{-r-1} N(j, n), & r < 0. \end{cases} \quad (13.3)$$

Empty sums vanish. The factor $1/2$ is the only source of rational rather than integer coefficients in this combinatorial reduction.

14 The finite algorithm at every order

We now prove the existence assertion in Theorem 10.1. Apply inclusion–exclusion to the finitely many bottom and top events defining $E_n^{(M)}$.

In an intersection containing bottom indices I , let $A = \max I$. The largest-index condition bounds $Z_1 + \cdots + Z_A \leq A - 1$; together with $Z_1 \geq 1$ it leaves finitely many vectors x . The other bottom conditions select a subset of this finite list. If top indices J occur, let $B = \max J - 1$. Their ranks are $-d$, where $0 \leq d \leq \min J - 2$. The largest-index condition bounds $G_1 + \cdots + G_B \leq B - 1 - d$, leaving finitely many vectors y for each of finitely many ranks.

For a bottom-only intersection, use the cumulative pure-row reduction. For a top-only intersection, keep the exact gaps and rank and impose the single lower bound $Z_1 \geq 1$ in Lemma 13.2; no enumeration over arbitrarily many ones is needed. For a mixed intersection, use exact x and y at fixed rank. Thus (13.2) and (13.3) produce explicit finite lists with

$$E_n^{(M)} = \sum_t \alpha_t p(n - w_t) + \sum_t \eta_t N(r_t, n - w_t) + e_M(n), \quad (14.1)$$

where α_t, η_t are rational, all ranks and nonnegative shifts are fixed, and $e_M(n)$ is an explicitly enumerable signed combination of polynomially bounded complement counts. The function e_M itself is polynomially bounded, without being asserted polynomial.

For a requested $R \geq 0$, choose $M = R + 2$. The tail (12.7) is then $O_R(\beta^{R+1} p(n))$. The exceptional term in (14.1), divided by $p(n)$, is smaller than every fixed power of β . Expand its finitely many shifted partition counts by (11.1) and its finitely many rank atoms by Zhou’s theorem, or equivalently by its specialized recipe (11.8). Take enough source terms to leave an $O_R(\beta^{R+1})$ error. Finally subtract from $P_1(n - 1)/p(n)$ using (12.1) and (13.3). This proves (10.1), apart from the explicit coefficients.

For reference, the terminating procedure is:

1. Fix R , set $M = R + 2$, and list the endpoint events
2. Form their finite intersections by inclusion–exclusion
3. Enumerate each bounded multiplicity vector, gap vector, and rank
4. Replace exact coordinates by lower-bound differences
5. Apply the mixed or pure-row bijection, retaining its complements
6. Convert cumulative ranks to p and finitely many rank atoms
7. Expand the fixed shifts with the published analytic formulas and collect powers of β

Every operation is finite. Its naive running time may be large. All formal operations involve rational numbers and powers of π^{-2} , so the coefficient field is as claimed.

Finally, expansions derived with different endpoint cutoffs agree to their common order. Otherwise subtraction and division by the first differing power of β would give a nonzero constant converging to zero. The strict asymptotic scale $1, \beta, \beta^2, \dots$ therefore determines a single unique formal series. This proves the all-orders part of the theorem.

15 The first two corrections

The leading corrections can be isolated without enumerating the full finite reduction. This also shows why one must include both ends of the partition.

At fixed r , (11.10) begins $N(r, n)/p(n) = \beta/4 + a\beta^2 + O_r(\beta^3)$. For fixed positive A, B , deletion of small parts and inclusion–exclusion give, for all sufficiently large n ,

$$\begin{aligned} \#\{R = r, Z_1 < A, Z_2 < B\} &= N(r, n) - N(r + A, n - A) \\ &\quad - N(r + B, n - 2B) + N(r + A + B, n - A - 2B). \end{aligned} \quad (15.1)$$

The only largest-part exceptions are partitions supported on $\{1, 2\}$; at fixed rank these have bounded size. The leading terms and the shift-linear second coefficients cancel, so (15.1) is $O(\beta^3 p(n))$. Likewise, a fixed ordinary box in Z_1, Z_2, Z_3 has $O(\beta^3 p(n))$ members by (11.3).

Use $M = 4$ in the endpoint truncation. Bottom failures at $i = 3, 4$ lie in a bounded three-coordinate box. Top failures at $j = 3, 4, 5$, after conjugation, have rank in $\{0, 1, 2, 3\}$ and two bounded small-part multiplicities; (15.1) applies. Therefore only

$$\mathcal{B}_n = \{R \leq 0, Z_1 = 1, Z_2 = 0\}, \quad \mathcal{T}_n = \{R = 0, \mu_1 = \mu_2, Z_1 \geq 1\}$$

survive through order two. Their overlap is contained in a fixed-rank two-coordinate box and is $O(\beta^3 p(n))$. Hence

$$E_n = |\mathcal{B}_n| + |\mathcal{T}_n| + O(\beta^3 p(n)). \quad (15.2)$$

For all sufficiently large n , exact deletion identities are

$$|\mathcal{B}_n| = P_1(n - 1) - P_2(n - 2) - P_2(n - 3) + P_3(n - 4), \quad (15.3)$$

$$|\mathcal{T}_n| = N(1, n - 1) - N(0, n - 2). \quad (15.4)$$

For (15.3), first enforce exactly one 1, then exclude partitions having a 2. For (15.4), delete one 1, conjugate, and count rank -1 partitions without ones by deleting a 1 from the complementary class. The small largest-part exceptions are absent at sufficiently large weight.

Since

$$\frac{P_r(n)}{p(n)} = \frac{1}{2} + (r + 1/2)(\beta/4 + a\beta^2) + O_r(\beta^3) \quad (r \geq 0),$$

substitution of the fixed-shift formulas yields

$$\frac{|\mathcal{B}_n|}{p(n)} = \frac{\beta^2}{4} + O(\beta^3), \quad \frac{|\mathcal{T}_n|}{p(n)} = \frac{\beta^2}{4} + O(\beta^3).$$

Thus $E_n/p(n) = \beta^2/2 + O(\beta^3)$, while

$$\frac{P_1(n-1)}{p(n)} = \frac{1}{2} - \frac{\beta}{8} + \left(\frac{5}{32} + \frac{t}{8}\right)\beta^2 + O(\beta^3). \quad (15.5)$$

Subtracting proves c_0, c_1, c_2 as stated. In particular the first coefficient comes from the cancellation of the one-part shift $-\beta/2$ with the three half-rank contributions $3\beta/8$. The two surviving endpoint obstructions first affect the second coefficient.

16 Six patterns determine the cubic coefficient

Set $M = 5$, so the omitted endpoint tail is $O(\beta^4 p(n))$. Through cubic order, (11.11) has total degree at most two in the fixed rank and shift variables. The cumulative version for nonnegative threshold has total degree at most three: the sum of j^2 in (13.3) is cubic in its upper endpoint. Each exact coordinate in (13.2) is a finite difference translating the rank and shift by fixed constants, lowering total polynomial degree by at least one. Consequently:

- a fixed-rank pattern with at least three exact coordinates contributes $O(\beta^4 p(n))$
- a pure top pattern with at least three exact gaps has that bound, even with the lower bound of one 1
- a pure bottom pattern with at least four exact multiplicities has that bound

The polynomially bounded rectangle complements remain negligible. All cumulative thresholds in the pure-bottom reduction are nonnegative, so the polynomial statement applies throughout.

Classify failures by their first failing bottom or top index. Only the following first-failure patterns remain through degree three; within each orientation they are disjoint:

	Bottom exact multiplicities	Top rank and exact gaps
1	$(Z_1, Z_2) = (1, 0)$	$d = 0, (G_1) = (0)$
2	$(Z_1, Z_2, Z_3) = (1, 1, 0)$	$d = 0, (G_1, G_2) = (1, 0)$
3	$(Z_1, Z_2, Z_3) = (2, 0, 0)$	$d = 1, (G_1, G_2) = (0, 0)$

Here the bottom rank condition is $R \leq 0$, the top condition is $R = -d$, and every pattern has $Z_1 \geq 1$. For example, first bottom failure at $i = 3$ forces $Z_1 + Z_2 = 2$, $Z_3 = 0$, with $Z_1 \geq 1$; this gives exactly patterns 2 and 3. The top classification follows similarly from (12.6). Every bottom/top

intersection has at least two exact multiplicities and one exact gap at a fixed rank, hence contributes only $O(\beta^4 p(n))$.

Here are explicit finite formulas from which the coefficients can be checked without any numerical fit. For a vector v , write $|v| = \sum_a v_a$ and $\text{wt}(v) = \sum_a a v_a$. For a bottom vector x and top vector y , the relevant counts, up to polynomially bounded complements, are

$$\mathcal{B}(x; n) = \sum_{\epsilon \in \{0,1\}^A} (-1)^{|\epsilon|} P_{|x+\epsilon|}(n - \text{wt}(x + \epsilon)), \quad (16.1)$$

$$\mathcal{T}(d, y; n) = \sum_{\epsilon \in \{0,1\}^B} (-1)^{|\epsilon|} N(1 - d - |y + \epsilon|, n - 1 - \text{wt}(y + \epsilon)). \quad (16.2)$$

For the latter, the single mandatory 1 supplies the rank shift +1 and weight shift 1 in Lemma 13.2. Substitution of (11.5), (11.11), and (13.3) gives the same three normalized contributions on each side:

$$\frac{\beta^2}{4} - \left(\frac{1}{4} + \frac{5t}{8}\right) \beta^3, \quad \frac{3\beta^3}{8}, \quad \frac{3\beta^3}{8}, \quad \text{each up to } O(\beta^4). \quad (16.3)$$

For instance the first top expression is the difference $N(1, n-1) - N(0, n-2)$; its cubic coefficient from (11.11) is $-1/4 - 5t/8$. The other five follow from the same finite sums (16.1)–(16.2). They may equally be verified by exact rational polynomial arithmetic.

Adding both orientations and discarding their negligible intersections gives

$$\frac{E_n}{p(n)} = \frac{\beta^2}{2} + \left(1 - \frac{5t}{4}\right) \beta^3 + O(\beta^4). \quad (16.4)$$

The source term is

$$\begin{aligned} \frac{P_1(n-1)}{p(n)} &= \frac{1}{2} - \frac{\beta}{8} + \left(\frac{5}{32} + \frac{t}{8}\right) \beta^2 \\ &\quad + \left(\frac{67}{384} - \frac{151t}{384} - \frac{t^2}{32}\right) \beta^3 + O(\beta^4). \end{aligned} \quad (16.5)$$

Subtracting (16.4) yields

$$c_3 = -\frac{317}{384} + \frac{329t}{384} - \frac{t^2}{32} = -\frac{317}{384} + \frac{329}{64\pi^2} - \frac{9}{8\pi^4}.$$

This completes the explicit part of Theorem 10.1. The finite full-cutoff computation offers a second organization of the same algebra: at $M = 5$ it uses 22 bottom patterns, 64 top patterns, and 1408 mixed intersections, reducing to 201 nonzero shifted terms. The six-pattern argument explains why most of those terms cancel.

17 Smooth models and correctly rounded integer inverses

Prepending one 1 to a subdiagonal partition preserves all its inequalities and is injective. Thus $s(n)$ is nondecreasing. Theorem 10.1 and the partition asymptotic show that $s(n)$ is unbounded, so (10.4) exists for all large y . Putting $z = c\sqrt{x}$ in $F(x) = y$ gives $ze^{-z} = c/\sqrt{8\sqrt{3}y}$; the large solution is exactly (10.3) on the W_{-1} branch.

17.1 A general model window

Multiplying the complete ratio expansion by the classical partition expansion produces, for every fixed $J \geq 0$,

$$s(n) = F(n) \left(1 + \sum_{j=1}^J a_j \beta_n^j + O_J(\beta_n^{J+1}) \right), \quad \beta_x = \frac{c}{\sqrt{x}}. \quad (17.1)$$

Define the explicit smooth function

$$M_J(x) = F(x) \left(1 + \sum_{j=1}^J a_j \beta_x^j \right). \quad (17.2)$$

It is eventually positive and strictly increasing because

$$(\log M_J)'(x) = \frac{c}{\sqrt{x}} + O_J(x^{-1}) > 0. \quad (17.3)$$

Only an explicit finite model is differentiated. Let $\nu_J(y)$ be its unique sufficiently large solution of $M_J(\nu_J) = y$. Then $\nu_J \sim x_0$.

Lemma 17.1 (Ceiling window for the model root). *For some constants $D_J, y_J > 0$ and every $y \geq y_J$,*

$$\left\lceil \nu_J - D_J \nu_J^{-J/2} \right\rceil \leq T(y) \leq \left\lceil \nu_J + D_J \nu_J^{-J/2} \right\rceil. \quad (17.4)$$

Proof. Write $\nu = \nu_J$ and $\delta = D\nu^{-J/2}$. The logarithmic counting error in (17.1), relative to M_J , is $O_J(\nu^{-(J+1)/2})$ at integers in any fixed neighborhood of ν . Take the boundary integers

$$n_- = \lceil \nu - \delta \rceil - 1, \quad n_+ = \lceil \nu + \delta \rceil.$$

For fixed D they lie in such a neighborhood, including when $J = 0$. Their distances below and above ν are at least δ . By (17.3), the magnitude of the corresponding change in $\log M_J$ is at least a fixed positive multiple of $D\nu^{-(J+1)/2}$. Choose D sufficiently large to dominate the counting error. Then $s(n_-) < y$ and $s(n_+) \geq y$ for all sufficiently large y . Monotonicity gives $n_- + 1 \leq T(y) \leq n_+$, which is (17.4). The strict lower boundary is valid whether either real endpoint is integral or nonintegral. $\square$

17.2 The first two explicit centers

The smooth partition term gives

$$\frac{p(n)}{e^{2c\sqrt{n}}/(4\sqrt{3}n)} = 1 - \left(\frac{1}{24} + \frac{t}{2} \right) \beta + \left(\frac{1}{1152} + \frac{t}{16} \right) \beta^2 + O(\beta^3). \quad (17.5)$$

Combining it with the proved ratio expansion yields

$$a_1 = -h, \quad h = \frac{7}{24} + \frac{t}{2}, \quad a_2 = -\frac{779}{1152} + \frac{7t}{16}. \quad (17.6)$$

Both the ordinary partition correction and the subdiagonal correction contribute to h . The first root therefore satisfies $\nu_1 = x_0 + h + O(x_0^{-1/2})$. Lemma 17.1 gives

$$\left\lceil x_0 + h - Cx_0^{-1/2} \right\rceil \leq T(y) \leq \left\lceil x_0 + h + Cx_0^{-1/2} \right\rceil$$

for some fixed C and sufficiently large y .

For the next center put $x = x_0 + h + k\beta_0$. Direct expansion of the explicit functions gives

$$\begin{aligned}\log F(x) - \log F(x_0) &= h\beta_0 + (k - ht)\beta_0^2 + O(\beta_0^3), \\ \log(1 - h\beta_x + a_2\beta_x^2) &= -h\beta_0 + (a_2 - h^2/2)\beta_0^2 + O(\beta_0^3).\end{aligned}\tag{17.7}$$

Thus

$$k = ht - a_2 + h^2/2 = \frac{23}{32} + \frac{5t^2}{8} = \frac{23}{32} + \frac{45}{2\pi^4}.$$

The logarithmic residual of M_2 at this center is $O(\beta_0^3)$. Dividing by its derivative scale $\asymp \beta_0$ gives $\nu_2 - X_2 = O(\beta_0^2) = O(x_0^{-1})$. Absorbing this into the $J = 2$ model window proves

$$\lceil X_2 - C/x_0 \rceil \leq T(y) \leq \lceil X_2 + C/x_0 \rceil.\tag{17.8}$$

17.3 Every fixed order and the remaining rounding ambiguity

For general $J \geq 1$, substitute the ansatz (10.5) in $\log M_J(X_J) - \log y$ and equate coefficients through degree J in β_0 . The new coefficient b_r first enters with coefficient one at degree $r + 1$, because the leading logarithmic derivative is $\beta_0 + O(\beta_0^2)$. This is a triangular rational recursion. The resulting logarithmic residual is $O_J(\beta_0^{J+1})$, so the mean value theorem applied only to the explicit model gives

$$\nu_J - X_J = O_J(\beta_0^J) = O_J(x_0^{-J/2}).$$

Substitution in Lemma 17.1 proves Theorem 10.2, including the coefficient field. There is no uniformity claim with J growing with y .

[write] **6 October 2026.** The centre x_0 is Part I's (5.2), and the injection that prepends a 1 is Part I's (Section 5). Theorem 10.2 sharpens Part I's Theorem 5.1 from a window of width $O(\log y / \log \log y)$ to two adjacent integers. In the volume's terms (front matter, "The inverses"): x_0 is an instance of `p0:thm:lambert-core` after the reduction $\xi = \sqrt{x}$, and X_J is an instance of the reversion `p0:thm:lambert-centered` for the model (17.2) (the write checked $b_0 = h$ and $b_1 = ht - a_2 + h^2/2$ against the volume's recurrence); the ceiling windows of Lemma 17.1 are Part II's own, not an instance of `p0:thm:staircase`.

The ceilings cannot in general be removed. The integer threshold may stay an order-one distance from the smooth center even though the pre-rounding uncertainty tends to zero. Nor do unspecified C_J, y_J certify a numerical bracket at a chosen finite input. The exact threshold command in the companion instead searches actual integer counts in a bounded range and reports explicitly when the requested value has not been reached.

18 Exact computation and reproducibility

The source package contains a Python standard-library companion, exact deterministic JSON data, regression tests, a clean PDF builder, and a fixed-allowlist ZIP builder. The mathematical theorem holds at every fixed order; the software deliberately caps coefficient extraction at $R \leq 3$.

The counting algorithm records length k , weight n , and largest part a . If $D_k(n, a)$ counts increasing subdiagonal partitions with these parameters, then

$$D_k(n, a) = \sum_{b \leq a} D_{k-1}(n - a, b), \quad 1 \leq a \leq k, \quad D_0(0, 0) = 1.$$

Appending a enforces precisely the new inequality $a \leq k$. Prefix sums give an exact positive dynamic program. An independent recursive partition enumerator checks the resulting counts and directly tests endpoint events, mixed insertion maps, and exceptional-set identities over the recorded finite ranges. Numeric OEIS prefixes are attributed to their original entry; third-party journal PDFs and full OEIS entry text are not redistributed.

The coefficient implementation uses exact rational polynomials in $t = 6/\pi^2$ and the source-checked shifted formulas (11.5) and (11.11). It performs the finite endpoint construction, including mixed intersections, and checks agreement with the six-pattern cubic calculation. It neither fits coefficients to numerical data nor asserts validity of a derivative transfer rule beyond the checked order.

For orientation, exact counts are

n	$p(n)$	$s(n)$
25	1958	898
40	37338	17319
50	204226	95502
100	190569292	91171308
200	3972999029388	1930845918142

These finite values do not establish any asymptotic remainder constant, onset, or novelty claim. The exact verification outputs and the README state all workload limits and check counts.

Regression tests run normally and with Python optimization enabled; validation does not depend on Python `assert` statements. Output destinations must be fresh files in trusted directories. The release helpers reject existing output paths and static symlink components. Their guarantees assume no malicious same-user process concurrently replacing the parent or source tree; they are not an operating-system sandbox.

The PDF is built in an exclusive clean directory with isolated LaTeX state, disabled shell escape, fixed metadata and epoch, and references stabilized before the settled log is inspected. The ZIP uses a fixed allowlist and sorted SHA-256 manifest, excluding working files, review records, and other reports. Two clean PDF builds and two ZIP replays are required to agree byte for byte within the recorded toolchain. Reproducibility across arbitrary TeX versions, fonts, or operating systems is not promised. The README gives the exact replay commands and qualifications.

[write] **6 October 2026.** The package is shipped with the prefix `159-subdiag-` (code under `code/`; the three result files, the OEIS prefix and the provenance record under `data/`; the PDF, README and `SHA256SUMS` are not shipped). At intake, on a Windows copy, the exact data, including `endpoint_expansion.json`, were regenerated byte for byte; 31 companion tests ran without failure apart from three errors and eight skips of POSIX-only output tests. The report’s README gives the rerun route.

19 Source record and limits

The direct antecedent is Conjecture 5.3 in Archibald et al. [1], asserting $s(n)/p(n) \rightarrow 1/2$. Their definitions, generating functions, and rank-zero injection are prior results. Report 156 [15] supplies the earlier elementary quantitative proof and its coarse inverse; the present treatment repeats the needed localization estimates and does not require that report as an unproved lemma.

The additional analytic inputs are Liu–Zhou’s fixed-rank local estimate, expressly valid at zero, and Zhou’s complete expansion. The cited preprint versions are arXiv:1903.00835v3 and

arXiv:2110.11174v3. The latter’s Theorem 4.1 and Proposition 2.2 are the source of the general recipe and the direct cubic calculation. These results, and classical Hardy–Ramanujan and Rademacher theory, are not claimed as new.

A bounded catalogue and source check found no identical all-orders application to this exact subdiagonal sequence. This cannot establish first-ever priority or exclude uncatalogued work. The package’s provenance file records source identifiers, hashes where available, and inspection scopes without redistributing third-party full text.

The results proved here are an asymptotic expansion and integer localization at each fixed order. They do not establish convergence of the series, uniformity at growing truncation order, efficient high-order computation, explicit numerical error constants, or exact rounding for every input. The general algorithm is justified by the finite reduction and the published fixed-rank theorem. No coefficient beyond c_3 is stated or computationally certified in this package.

[write] **6 October 2026.** “Report 156” is Part I. The direct antecedent is quoted in the front matter. The open items of this section are collected in Section 47, item 3; the analytic inputs are external theorems whose statements, not proofs, the write read (note in Section 11). The abstract’s “Zhou’s published complete fixed-rank expansion” is such an input.

Part III

Exact exponential growth of increasing superdiagonal partitions

Bundle Report 158, archive `Superdiagonal_Partitions_Exact_Exponential_Growth_and_Inverse_Source.zip`, delivered as `Report158.tex` (13 pages). Delivered title block “Exact exponential growth of increasing superdiagonal partitions / An elementary proof and quantitative threshold inversion / Report 158 / 3 October 2026”. Printed as delivered, with `[write]` notes; the proof of Lemma 23.1 is replaced by a pointer to Part I. Delivered Section k is Section $k + 19$ here, and its statement or equation $k.j$ is $(k + 19).j$.

Reading conventions of this Part	Elsewhere in the report
$b = \log 2$; $C = \pi^2/12 + b^2$; $B = 2\sqrt{C}$; $L(x) = x \log 4 - x^2/2 + I(x)$	Part I: $b = \pi/\sqrt{3}$, $B_* < B$; Part V: $L = \log 4$
t the Boltzmann variable; M , h prefix length and height; $D_h(M)$, ρ_h , c_h Dyck data	Part I: the same $D_h(M)$, ρ_h , c_h
p a geometric parameter; $r = (1 - p)/p$; $K_0, \dots, K_3$ constants; $\mathbb{P}$ a Boltzmann law	Parts I, II: $p(n)$, $\mathbb{P}_n$ uniform

Abstract of this Part (as delivered). Let $A(n)$ count weakly increasing positive-integer partitions $\lambda_1 \leq \dots \leq \lambda_k$ of n satisfying $\lambda_i \geq i$, with $A(0) = 1$. This is OEIS A238873. We prove

$$\log A(n) = B\sqrt{n} + O(n^{1/3}), \quad B = \sqrt{\frac{\pi^2}{3} + 4\log^2 2} = 2.282910464597606\dots$$

The upper error is in fact $O(1)$. For $F(t) = \sum_{n \geq 0} A(n)e^{-tn}$ and $C = B^2/4 = \pi^2/12 + \log^2 2$, we establish

$$\frac{C}{t} - O(t^{-1/3}) \leq \log F(t) \leq \frac{C}{t} + O(1).$$

An exponential tilt gives the upper bound. A bounded-height Dyck prefix and a geometric tail with an explicit positive survival probability give the matching lower bound. An exact spectral inequality permits a growing strip height. Chernoff bounds and an elementary monotonicity injection then transfer the estimate to every coefficient without a Tauberian black box. The threshold inverse satisfies

$$\min\{n \geq 0 : A(n) \geq y\} = \left(\frac{\log y}{B}\right)^2 + O((\log y)^{5/3}).$$

The result determines the exponential rate. It does not determine a multiplicative equivalent, power prefactor, sharp second term, Airy correction, or all-orders expansion. Classical Dyck representations and the earlier lower bound in Report 156 are explicitly credited.

Scope and provenance. All logarithms are natural. The increasing order in the definition is essential: the weakly decreasing superdiagonal convention counts a different sequence. The proof is self-contained apart from elementary analysis, the Basel sum, and finite-dimensional linear algebra, all applications of which are made explicit. Finite exact computations are separate implementation checks, not evidence from which an asymptotic statement is inferred. A bounded primary-source search and report-catalogue check found no equivalent exact-rate result; this is not an exhaustive claim of scholarly priority. Section 29 records the scope.

20 The partition model and main results

A partition here is a finite weakly increasing list of positive integers. Its size is $|\lambda| = \sum_i \lambda_i$. Call it superdiagonal if

$$\lambda_i \geq i \quad (1 \leq i \leq k). \quad (20.1)$$

The empty partition is superdiagonal and has size zero. Write $A(n)$ for the number of such partitions of size n . The first values are

$$A(0), \dots, A(13) = 1, 1, 1, 2, 3, 3, 5, 7, 9, 11, 14, 19, 25, 31.$$

This is the class of Archibald, Blecher, Elizalde and Knopfmacher [1], Section 2, and OEIS A238873 [6]. Applying the same indexed condition to a decreasing list instead produces A003114; it is not the model of this report.

Set

$$b = \log 2, \quad C = \frac{\pi^2}{12} + b^2, \quad B = 2\sqrt{C} = \sqrt{\frac{\pi^2}{3} + 4b^2}. \quad (20.2)$$

For orientation, $C = 1.3029200473423146\dots$ and $B = 2.2829104645976064\dots$. The exact expressions, not their decimals, define these constants.

Theorem 20.1 (Generating function and coefficient estimates). *Let $F(t) = \sum_{n \geq 0} A(n)e^{-tn}$ for $t > 0$. There are positive constants K_0, K_1, K_2, K_3 and positive thresholds such that*

$$\frac{C}{t} - K_0 t^{-1/3} \leq \log F(t) \leq \frac{C}{t} + K_1 \quad \text{for all sufficiently small } t > 0, \quad (20.3)$$

$$B\sqrt{n} - K_2 n^{1/3} \leq \log A(n) \leq B\sqrt{n} + K_3 \quad \text{for all sufficiently large integers } n. \quad (20.4)$$

In particular,

$$\lim_{t \downarrow 0} t \log F(t) = C, \quad \lim_{n \rightarrow \infty} \frac{\log A(n)}{\sqrt{n}} = B, \quad A(n) = \exp\{B\sqrt{n} + O(n^{1/3})\}. \quad (20.5)$$

Theorem 20.2 (Threshold inverse). *For real $y > 1$, put $T(y) = \min\{n \geq 0 : A(n) \geq y\}$. This minimum exists, and as $y \rightarrow \infty$,*

$$\left(\frac{\log y}{B}\right)^2 - O(\log y) \leq T(y) \leq \left(\frac{\log y}{B}\right)^2 + O((\log y)^{5/3}). \quad (20.6)$$

Thus $T(y) = (\log y/B)^2 + O((\log y)^{5/3})$.

The constants and onset in these O -terms are unspecified. In particular, (20.6) is not a finite-input certified search bracket or an exact rounding rule. Likewise (20.4) does not imply $A(n) \sim cn^\alpha e^{B\sqrt{n}}$ for any specified c or α . The $O(n^{1/3})$ error is a rigorous bound, not an identified correction.

[write] **6 October 2026.** Theorem 20.1 replaces the lower rate B_* of Part I's Theorem 1.2. Part IV then shows that the lower error is exactly of order $n^{1/6}$: $B\sqrt{n} - \log A(n) = Dn^{1/6}(1 + o(1))$ with $D = \zeta b C^{-1/6} = 1.5507\dots$ (Theorem 30.1). So the $O(n^{1/3})$ in (20.4) is a valid but not optimal bound, and no bound $\log A(n) \geq B\sqrt{n} - o(n^{1/6})$ holds; the upper bound $B\sqrt{n} + O(1)$ stands. The approach is slow: $\log A(n)/\sqrt{n} = 1.5488, 1.9121, 2.1177$ at $n = 100, 800, 8000$ (intake), against $B = 2.2829$. Part V gives, conditionally, the full equivalent. Theorem 20.2 is refined by Part IV's Corollary 30.2 (unconditional) and Part V's Corollary 39.2 (conditional); it is not an instance of the transseries volume's inversion theorems.

Relation to prior results. The 2025 paper [1] develops generating functions and Dyck and ballot descriptions of this class, proves its density among all partitions is zero, and asks for its asymptotic enumeration. Report 156 [15] proves the distinct-partition density conjecture using the lower constant

$$B_* = \sqrt{\frac{\pi^2}{3} + 2\log^2 2} = 2.061740566010393\dots < B. \quad (20.7)$$

That report does not assert an exact exponential rate. The present report extends the prefix construction by a surviving geometric tail and adds a matching tilted upper bound. All steps needed for the new theorem are given below; no part of Report 156 is modified or silently strengthened.

[write] **6 October 2026.** “Report 156” [15] is Part I; (20.7) is Part I’s (1.6), and the bounded-height prefix construction is Part I’s Sections 6–7.

21 Cumulative multiplicities and an integral identity

Let Z_j be the multiplicity of the part j , and put $C_j = \sum_{i=1}^j Z_i$. Only finitely many Z_j are nonzero for a partition.

Lemma 21.1 (Equivalent barrier). *Condition (20.1) holds if and only if*

$$C_j \leq j \quad \text{for every integer } j \geq 1. \quad (21.1)$$

Proof. If $C_j = k > 0$, then $\lambda_k \leq j$, while superdiagonality gives $k \leq \lambda_k$, so $C_j \leq j$. The case $C_j = 0$ is immediate. Conversely, if $\lambda_i < i$, the value $j = \lambda_i$ has $C_j \geq i > j$, violating (21.1). The empty partition satisfies both conditions. $\square$

[write] **6 October 2026.** Printed once here. Part IV (Section 30) and Part V ((43.1)) use the same equivalence; Part IV’s proof of it is not reprinted.

The unrestricted partition product converges at e^{-t} for every $t > 0$:

$$\prod_{j \geq 1} (1 - e^{-tj})^{-1} < \infty.$$

Indeed, $-\log(1 - u) \leq 2u$ for $0 \leq u \leq 1/2$, so the logarithms have a summable tail, and the remaining finite set causes no difficulty. This product bounds $F(t)$ from above. Every generating sum below is thus finite or the monotone limit of finite nonnegative sums.

Define

$$f(x) = -\log(1 - e^{-x}), \quad I(a) = \int_a^\infty f(x) dx = \sum_{r \geq 1} \frac{e^{-ar}}{r^2} = \text{Li}_2(e^{-a}). \quad (21.2)$$

The nonnegative series for f justifies termwise integration. As f is decreasing, for every integer $M \geq 0$ and $x = tM > 0$,

$$\frac{I(x+t)}{t} \leq \sum_{j > M} f(tj) \leq \frac{I(x)}{t}. \quad (21.3)$$

If x stays in a fixed compact subinterval of $(0, \infty)$, either bound differs from $I(x)/t$ by $O(1)$, since $0 \leq I(x) - I(x+t) \leq tf(x)$.

For later use, put

$$L(x) = x \log 4 - \frac{x^2}{2} + I(x). \quad (21.4)$$

Then

$$L(b) = \frac{3}{2}b^2 + \text{Li}_2(1/2) = \frac{\pi^2}{12} + b^2 = C. \quad (21.5)$$

Here the standard dilogarithm value needs only elementary calculus: the derivative of $\text{Li}_2(u) + \text{Li}_2(1-u) + \log u \log(1-u)$ vanishes on $(0, 1)$. The limit at zero is $\sum_{r \geq 1} r^{-2} = \pi^2/6$; setting $u = 1/2$ gives $\text{Li}_2(1/2) = \pi^2/12 - b^2/2$.

[write] **6 October 2026.** This evaluation and the function L are used again in Part IV ((31.4), where $L(x) = 2bx - x^2/2 + I(x)$ is the same function) and in Part V ((43.8)); Part IV's proof of the evaluation is not reprinted.

22 A matching exponential tilt upper bound

For sufficiently small $t > 0$, take $M = \lfloor b/t \rfloor \geq 1$ and define $\eta_j = b - tj$ for $1 \leq j \leq M$. These multipliers are nonnegative and nonincreasing. Summation by parts and (21.1) give

$$\begin{aligned} \sum_{j=1}^M \eta_j Z_j &= \eta_M C_M + \sum_{j=1}^{M-1} (\eta_j - \eta_{j+1}) C_j \\ &\leq \eta_M M + \sum_{j=1}^{M-1} (\eta_j - \eta_{j+1}) j = \sum_{j=1}^M \eta_j. \end{aligned} \quad (22.1)$$

The signs of all coefficients matter: the cumulative upper bounds can be substituted in precisely this direction.

Since $e^{-tjZ_j} = 2^{-Z_j} e^{\eta_j Z_j}$ for $j \leq M$, bound the exponential factor with (22.1) and then discard every remaining constraint. Summing the independent geometric series yields

$$F(t) \leq \exp \left\{ \sum_{j=1}^M (\log 4 - tj) \right\} \prod_{j>M} (1 - e^{-tj})^{-1}. \quad (22.2)$$

For the first M variables, the extra $M \log 2$ comes from $\prod_{j=1}^M \sum_{z \geq 0} 2^{-z} = 2^M$; this accounts for the factor $\log 4$ rather than $\log 2$.

With $x = tM = b + O(t)$, equations (21.3) and (22.2) imply

$$\begin{aligned} \log F(t) &\leq M \log 4 - \frac{tM(M+1)}{2} + \sum_{j>M} f(tj) \\ &\leq \frac{L(x)}{t} - \frac{x}{2} = \frac{C}{t} + O(1). \end{aligned} \quad (22.3)$$

The last step uses smoothness of L near b and (21.5). For completeness,

$$L'(x) = \log(4e^{-x}(1 - e^{-x})), \quad L'(b) = L''(b) = 0, \quad L'''(b) = -2.$$

These cancellations improve the cutoff discrepancy, but they are not needed for the asserted upper bound.

23 Bounded height prefixes and an exact spectral bound

For positive integers M, h , let $\mathcal{P}_{M,h}$ consist of weakly increasing lists of exactly M parts satisfying

$$i \leq \lambda_i \leq \min(M, i + h - 1) \quad (1 \leq i \leq M). \quad (23.1)$$

Every such list has all parts at most M and weight at most

$$W_h(M) = \frac{M(M+1)}{2} + (h-1)M. \quad (23.2)$$

Write $D_h(M) = |\mathcal{P}_{M,h}|$.

Lemma 23.1 (Dyck representation). *$D_h(M)$ is the number of Dyck paths of semilength M whose maximum height is at most h .*

Proof. [write] **6 October 2026.** Report 158's proof is not reprinted: it is the argument of Part I, step by step. Part I's Section 6 maps a list to its reverse complement $\mu_j = M + 1 - \lambda_{M+1-j}$, a weakly increasing list with $1 \leq \mu_j \leq j$, and places the j th north step of a path from $(0,0)$ to (M,M) at horizontal coordinate $\mu_j - 1$, which gives the Dyck paths of semilength M bijectively (the inverse reads the coordinates). Part I's Section 7 adds the height: the upper bound in (23.1) becomes $\mu_j \geq j - h + 1$, the height just after the j th north step is $j - \mu_j + 1$, between 1 and h , and maxima occur just after north steps. The statement above is Report 158's.

□

This is a specialized form of the classical path encodings discussed in [1]; those encodings and Catalan enumeration are prior inputs. The bounded-height form was also used in Report 156 [15]. We include the spectral calculation to specify exactly what remains uniform when the height grows.

Let T_h be the adjacency matrix of the path on vertices $0, 1, \dots, h$. Reading Dyck heights gives

$$D_h(M) = (T_h^{2M})_{0,0}. \quad (23.3)$$

Its orthonormal eigenvectors have coordinates

$$v_r(a) = \sqrt{\frac{2}{h+2}} \sin \frac{(a+1)r\pi}{h+2} \quad (0 \leq a \leq h), \quad 1 \leq r \leq h+1,$$

with eigenvalues $2 \cos(r\pi/(h+2))$. The sine addition formula verifies the eigenvalue relation, including the zero boundary coordinates; finite sine orthogonality gives the normalization. Thus

$$D_h(M) = \frac{2}{h+2} \sum_{r=1}^{h+1} \sin^2 \frac{r\pi}{h+2} \left(2 \cos \frac{r\pi}{h+2} \right)^{2M}. \quad (23.4)$$

Set

$$\rho_h = 4 \cos^2 \frac{\pi}{h+2}, \quad c_h = \frac{4}{h+2} \sin^2 \frac{\pi}{h+2}. \quad (23.5)$$

All summands in (23.4) are nonnegative. Keeping the two extreme eigenvalues proves the exact, uniform inequality

$$D_h(M) \geq c_h \rho_h^M \quad (M, h \geq 1). \quad (23.6)$$

For fixed h , the remaining eigenvalues have smaller modulus, so

$$\log D_h(M) = M \log \rho_h + O_h(1) \quad (M \rightarrow \infty), \quad \rho_h \rightarrow 4 \quad (h \rightarrow \infty). \quad (23.7)$$

When $h = 1$, the two extreme eigenvalues exhaust the spectrum and the formula is already exact. We will use (23.7) only with fixed height, and (23.6) for growing height.

24 A geometric tail with an explicit survival bound

Lemma 24.1 (Uniform survival). *Let $Y_1, Y_2, \dots$ be independent with $\mathbb{P}(Y_i = z) = (1-p)p^z$ for $z = 0, 1, \dots$, where $0 < p < 1/2$. Then*

$$\mathbb{P} \left(\sum_{i=1}^j Y_i \leq j \text{ for every } j \geq 1 \right) \geq 1 - 2p > 0. \quad (24.1)$$

Proof. Put $S_n = \sum_{i=1}^n (Y_i - 1)$, $S_0 = 0$, and $r = (1-p)/p > 1$. Direct summation gives

$$\mathbb{E} r^{Y_i - 1} = \frac{1-p}{r(1-pr)} = 1.$$

Therefore r^{S_n} is a nonnegative mean-one martingale. If τ is the first time $S_n \geq 1$, finite-time stopping gives $1 = \mathbb{E} r^{S_{\tau \wedge N}} \geq r \mathbb{P}(\tau \leq N)$. This finite identity also follows by expanding the stopped increments: each increment multiplied by $\mathbf{1}_{\{\tau > j\}}$ has conditional mean zero. Hence no infinite-time optional-stopping assertion is needed. Letting $N \rightarrow \infty$ in the increasing events yields

$$\mathbb{P}(S_n < 1 \text{ for every } n \geq 1) \geq 1 - 1/r.$$

Now require $Y_1 = 0$ and impose this last event on the independent walk formed from $Y_2, Y_3, \dots$. Its partial sums are integers below 1, hence at most zero. The original partial sums are consequently at most -1 from time one onward. This is a subset of the event in (24.1), with probability at least $(1-p)(1-1/r) = 1 - 2p$. $\square$

[write] **6 October 2026.** Printed once here. Part IV restates it as (33.1)–(33.2) (its proof there is not reprinted) and uses the same martingale for the ruin bound (34.16); Part V uses it inside the proof of Lemma 43.1.

Fix $a > b$ and put $M = \lfloor a/t \rfloor$. Consider independent tail multiplicities Z_{M+j} with parameters $p_j = e^{-t(M+j)}$, $j \geq 1$. The strict endpoint inequality $t(M+1) > a$ implies

$$p_j < e^{-a} = p < 1/2.$$

A coordinatewise coupling gives $Z_{M+j} \leq Y_j$ for the iid geometrics in Lemma 24.1: their tail probabilities are $p_j^k \leq p^k$ for every k . Since survival is decreasing in each multiplicity,

$$\mathbb{P} \left(\sum_{i=1}^j Z_{M+i} \leq j \text{ for every } j \geq 1 \right) \geq 1 - 2e^{-a}. \quad (24.2)$$

The product measure is supported on finite partitions. In fact $\sum_{j>M} \mathbb{E} Z_j = \sum_{j>M} e^{-tj} / (1 - e^{-tj}) < \infty$, which implies a finite total number of parts almost surely.

Let $G_M(t)$ sum the weights $e^{-t|\lambda|}$ of tail partitions with all parts greater than M satisfying the barrier in (24.2). Normalizing the geometric product gives the precise inequality

$$G_M(t) \geq (1 - 2e^{-a}) \prod_{j>M} (1 - e^{-tj})^{-1}. \quad (24.3)$$

The normalizing product is finite and positive. This argument may also be obtained from finite products and monotone limits; it introduces no infinite formal measure with a missing normalization.

25 Concatenation and the exact generating function rate

Append a tail counted by $G_M(t)$ to a prefix in $\mathcal{P}_{M,h}$. The prefix has exactly M parts, all at most M . For a value $M + j$, the cumulative number of parts of the concatenated partition is at most $M + j$ by the tail barrier; below M , the prefix already satisfies Lemma 21.1. The result is superdiagonal. The cutoff at M uniquely recovers both pieces, so this construction is injective even when different prefixes have the same weight.

By (23.2) and (24.3), for $M = \lfloor a/t \rfloor$,

$$F(t) \geq D_h(M) e^{-tW_h(M)} (1 - 2e^{-a}) \prod_{j>M} (1 - e^{-tj})^{-1}. \quad (25.1)$$

Keep $a > b$ and h fixed while $t \downarrow 0$. Multiplying logarithms by t , and using $tM \rightarrow a$, (21.3) and (23.7), gives

$$\liminf_{t \downarrow 0} t \log F(t) \geq a \log \rho_h - \frac{a^2}{2} + I(a). \quad (25.2)$$

The prefix excess weight contributes $t^2(h-1)M \rightarrow 0$, and the survival factor contributes a fixed logarithm times t , also tending to zero. First let fixed $h \rightarrow \infty$, and then let $a \downarrow b$. The right side tends to $L(b) = C$. Together with (22.3), this proves

$$t \log F(t) \rightarrow C. \quad (25.3)$$

There is no interchange of a nonuniform pair of limits: each fixed (a, h) bounds the same $\liminf$ from below, and the supremum of these valid bounds is at least C .

26 Elementary transfer to every coefficient

Lemma 26.1 (Monotonicity). $A(n+1) \geq A(n)$ for every integer $n \geq 0$.

Proof. For a nonempty partition increase one largest part by 1. Its position at the end remains valid, the indexed lower bounds are preserved, and the resulting largest part is unique. Decreasing that unique largest part recovers the original list on the image. Thus this map is injective. For $n = 0$, send the empty list to (1). $\square$

[write] **6 October 2026.** Printed once here; Part IV (Section 35) and Part V (Section 44) use the same injection, and their proofs of it are not reprinted. The inequality is not strict in general: $A(0) = A(1) = A(2) = 1$ and $A(4) = A(5) = 3$.

Define a random nonnegative integer N_t by

$$\mathbb{P}(N_t = n) = \frac{A(n)e^{-tn}}{F(t)}. \quad (26.1)$$

For now assume only (25.3). Fix $0 < \epsilon < 1$ and choose a small fixed $s \in (0, 1)$. Exponential Markov bounds yield

$$\mathbb{P}\left(N_t \geq \frac{(1+\epsilon)C}{t^2}\right) \leq e^{-s(1+\epsilon)C/t} \frac{F((1-s)t)}{F(t)}, \quad (26.2)$$

$$\mathbb{P}\left(N_t \leq \frac{(1-\epsilon)C}{t^2}\right) \leq e^{s(1-\epsilon)C/t} \frac{F((1+s)t)}{F(t)}. \quad (26.3)$$

Using (25.3), their logarithms are at most, respectively,

$$\frac{Cs}{t} \left(\frac{s}{1-s} - \epsilon \right) + o(1/t), \quad \frac{Cs}{t} \left(\frac{s}{1+s} - \epsilon \right) + o(1/t).$$

The choice $s = \epsilon/4$ makes both leading coefficients negative. Thus the interval between $(1 - \epsilon)C/t^2$ and $(1 + \epsilon)C/t^2$ has probability tending to one exponentially fast. Real endpoints cause no difficulty: one sums the integer indices that lie between them.

For an arbitrary sufficiently large integer n , set $t = \sqrt{(1 + \epsilon)C/n}$. The upper endpoint equals n . Monotonicity and the lower endpoint bound on N_t give

$$(1 - o(1))F(t) \leq \sum_{(1-\epsilon)C/t^2 \leq m \leq n} A(m)e^{-tm} \leq (n+1)A(n)e^{-(1-\epsilon)C/t}. \quad (26.4)$$

After taking logarithms,

$$\log A(n) \geq \frac{(2 - \epsilon)C}{t} + o(1/t), \quad \liminf_{n \rightarrow \infty} \frac{\log A(n)}{\sqrt{n}} \geq (2 - \epsilon)\sqrt{\frac{C}{1 + \epsilon}}.$$

Let $\epsilon \downarrow 0$. Conversely $A(n)e^{-tn} \leq F(t)$, and $t = \sqrt{C/n}$ with (22.3) gives

$$\log A(n) \leq B\sqrt{n} + O(1). \quad (26.5)$$

This proves the coefficient limit in (20.5), pointwise for all sufficiently large indices. No derivative estimate, local limit theorem, complex-analytic saddle-point argument, or unstated Tauberian theorem is being used.

27 Quantitative errors and inverse localization

27.1 Growing height with a uniform lower bound

For sufficiently small $t > 0$, choose

$$a = b + t, \quad M = \lfloor a/t \rfloor = \lfloor b/t \rfloor + 1, \quad h = \lceil t^{-1/3} \rceil, \quad x = tM. \quad (27.1)$$

Then $b < x \leq b + t$. The survival factor in (25.1) satisfies

$$1 - 2e^{-a} = 1 - e^{-t} \geq t/2 \quad (0 < t \leq 1). \quad (27.2)$$

For $h \geq 2$, elementary trigonometric estimates imply

$$c_h \geq \frac{16}{(h+2)^3}, \quad 0 \leq \log 4 - \log \rho_h \leq \frac{2\pi^2}{(h+2)^2}. \quad (27.3)$$

Indeed, $\sin u \geq 2u/\pi$ on $[0, \pi/2]$ gives the first bound. For the second use $-\log \cos u \leq u^2$ on $[0, \pi/4]$, which follows by integrating $\tan u \leq 2u$ there. These inequalities apply uniformly to the actual height in (27.1).

Taking logarithms of (25.1), using the exact inequality (23.6), and applying the lower Riemann bound yields

$$\log F(t) \geq \frac{L(x)}{t} - O\left(\frac{M}{(h+2)^2}\right) - O(h) - O(\log(1/t)) - O(1). \quad (27.4)$$

Here $\log c_h$ costs $O(\log(h+2))$, the factor in (27.2) costs $O(\log(1/t))$, and the prefix weight beyond $x^2/(2t)$ costs $x/2 + (h-1)x = O(h)$. Finally, $I(x+t)/t \geq I(x)/t - f(x)$, with $f(x) = O(1)$ near b . This accounts separately for every loss in (27.4).

Since $x = b + O(t)$, $L(x)/t = C/t + O(1)$. Moreover $M/(h+2)^2 = O(t^{-1/3})$, $h = O(t^{-1/3})$, and $\log(1/t) = O(t^{-1/3})$. Thus

$$\log F(t) \geq C/t - O(t^{-1/3}). \quad (27.5)$$

Together with (22.3), this proves (20.3). The argument uses an exact all-height inequality, not the fixed-height asymptotic (23.7) with an uncontrolled height-dependent error.

27.2 Quantitative coefficient transfer

Write $\log F(t) = C/t + E(t)$, where

$$-Kt^{-1/3} \leq E(t) \leq K_1 \quad (27.6)$$

for all sufficiently small positive t . Take $\epsilon = L_0 t^{1/3}$ and $s = \epsilon/4$, with a fixed sufficiently large $L_0 > 0$. For $\epsilon \leq 1/2$, the leading terms of the logarithms in (26.2) and (26.3) are each at most

$$-\frac{C\epsilon^2}{8t}.$$

For example, $s/(1-s) \leq \epsilon/2$ in this range, and multiplication by $Cs/t = C\epsilon/(4t)$ gives the bound; the other sign is no larger. The error in either logarithm is at most

$$E((1 \pm s)t) - E(t) \leq Kt^{-1/3} + K_1.$$

Choose L_0 so that $CL_0^2/8 > K + 1$. Then both tail probabilities are at most $\exp(-ct^{-1/3})$ for some $c > 0$ and all sufficiently small t . The central probability is consequently bounded below by $1/2$ for all sufficiently small t .

For each sufficiently large integer n , choose the positive solution of

$$n = (1 + L_0 t^{1/3})C/t^2. \quad (27.7)$$

Existence and uniqueness follow because the right side is the strictly decreasing continuous function $Ct^{-2} + L_0 Ct^{-5/3}$ with range $(0, \infty)$. This solution tends to zero and satisfies $t \asymp n^{-1/2}$. Apply the central-sum argument (26.4), now with its probability at least $1/2$, to get

$$\log A(n) \geq \frac{(2-\epsilon)C}{t} - Kt^{-1/3} - \log(2(n+1)). \quad (27.8)$$

Since $B\sqrt{n} = 2C\sqrt{1+\epsilon}/t$ and $\sqrt{1+\epsilon} \leq 1 + \epsilon/2$, the difference between this target and the lower bound is

$$O(\epsilon/t + t^{-1/3} + \log n) = O(t^{-2/3}) = O(n^{1/3}).$$

This proves the lower half of (20.4); the upper half is (26.5). Theorem 20.1 is complete.

27.3 Inverting the threshold

By Lemma 26.1 and the lower bound in (20.4), $A(n)$ is nondecreasing and tends to infinity, so $T(y)$ exists. Put $u = \log y$ and $x = (u/B)^2$. For large y , any index with $A(n) \geq y$ lies past the finite onset of the asymptotic estimates. The upper bound $\log A(n) \leq B\sqrt{n} + K_3$ therefore gives

$$T(y) \geq \left(\frac{u - K_3}{B} \right)^2 = x - O(u). \quad (27.9)$$

For the other direction, choose a fixed $D > 0$ and set $n = \lceil x + Du^{5/3} \rceil$. Elementary Taylor expansion gives

$$B\sqrt{n} = u + \frac{DB^2}{2}u^{2/3} + O(u^{1/3}), \quad (27.10)$$

$$n^{1/3} = B^{-2/3}u^{2/3} + O(u^{1/3}). \quad (27.11)$$

The ceiling contributes a still smaller error. Choose $DB^2/2 > K_2B^{-2/3}$. The lower bound in (20.4) and these expansions then imply $\log A(n) \geq u$ for every sufficiently large y . Consequently $T(y) \leq x + O(u^{5/3})$. Together with (27.9) this proves Theorem 20.2. The permitted error grows, so no adjacent-integer localization is asserted.

28 Exact finite verification and reproducibility

The standard-library companion supplied with this report performs exact integer and rational checks. Its finite limits and output-safety contract are specified in the package README. It is useful to record why the algorithms correspond to the objects and inequalities in the proof.

28.1 Capped multiplicity recurrence

Fix a maximum weight N . After processing part values through v , let $d_v(w, k)$ count multiplicity vectors of weight w with k parts, satisfying every cumulative barrier so far. Starting from $d_0(0, 0) = 1$ and zero elsewhere, the fresh-layer recurrence is

$$d_v(w, k) = \mathbf{1}_{\{k \leq v\}} \sum_{z=0}^k d_{v-1}(w - vz, k - z), \quad (28.1)$$

where entries with negative weight are zero. Choosing z specifies the multiplicity of the new value v uniquely; the previous layer enforces all earlier barriers, and $k \leq v$ enforces the new one. Therefore $A(w) = \sum_k d_N(w, k)$ for $0 \leq w \leq N$. No part larger than N can occur in a partition of these weights. The bound $k(k+1)/2 \leq w \leq N$ permits the finite cap $k \leq \lfloor (\sqrt{8N+1} - 1)/2 \rfloor$, evaluated with integer arithmetic.

For a tail cutoff M , process values $v > M$ and replace the barrier by $k \leq v - M$. The same inductive proof counts exactly the tails in Section 24, with weight still measured using their actual part sizes. The stronger lower bound on a k -part tail is $Mk + k(k+1)/2$; using the preceding ordinary cap merely retains extra zero states and remains valid.

Direct enumeration of ascending partitions is independent of this recurrence. For each list it compares the indexed and cumulative tests, and checks that the largest-part injection is valid and collision-free. Prefix enumeration independently compares its count with an exact walk recurrence for $(T_h^{2M})_{0,0}$, checks the weight cap, and checks the coefficientwise prefix-tail inequalities.

28.2 Exact rational probability and product checks

At a finite horizon H , retain states $d = j - \sum_{i=1}^j Z_i \geq 0$. From a state d , a new geometric multiplicity z is admissible precisely when $0 \leq z \leq d + 1$. The transition has exact weight $(1 - p)p^z$, and the new state is $d + 1 - z$. Thus rational dynamic programming computes the finite survival probability exactly when p is rational. The companion compares it with $1 - 2p$ and verifies the martingale normalization algebraically. These finite-horizon tests do not themselves imply infinite survival; that claim is proved in Lemma 24.1.

For rational $q \in (0, 1)$ and a finite largest part J , another finite multiplicity recurrence computes the total rational generating weight. Choose M such that $q^j \geq 1/2$ for $j \leq M$ and $q^{M+1} < 1/2$. The finite version of (22.2) is

$$\sum_{\substack{\lambda \text{ superdiagonal} \\ \lambda_k \leq J}} q^{|\lambda|} \leq 4^M q^{M(M+1)/2} \prod_{j=M+1}^J (1 - q^j)^{-1}. \quad (28.2)$$

The sum includes the empty partition. For tails with cutoff M , the corresponding lower bound is

$$\sum_{\substack{\lambda \text{ tail admissible} \\ \lambda_k \leq J}} q^{|\lambda|} \geq (1 - 2q^{M+1}) \prod_{j=M+1}^J (1 - q^j)^{-1}. \quad (28.3)$$

It follows either by the same finite coupling or by applying the infinite iid survival event to its first $J - M$ coordinates. The companion compares these rational quantities exactly; decimal approximations are not used for pass/fail decisions.

28.3 Numerical values and scope

The exact count table contains $A(n)$ for $0 \leq n \leq 200$. Selected values are

n	20	40	60	100	200
$A(n)$	135	3997	61221	5323937	25548856857

The attributed OEIS numerical prefix is compared with the computed table. The package also supplies a bounded exact threshold command. If its search bound does not reach a requested threshold, it reports that fact; it does not substitute the asymptotic model for an exact answer.

Finite checks verify the selected integer and rational comparisons only. They establish neither the theorem's asymptotic onset nor a sharp error constant, and cannot prove a claim of priority. The proof of the theorem is Sections 21–27. The source package includes tests that remain effective under Python optimization, a clean PDF builder, and fixed-allowlist deterministic archive construction. Reproducibility is byte-for-byte within the recorded toolchain, not a promise across different TeX versions or platforms.

[write] 6 October 2026. The package is shipped with the prefix `158-growth-` (code under `code/`; the two result files, the OEIS prefix and the provenance record under `data/`; the PDF, README and `SHA256SUMS` are not shipped). At intake, on a Windows copy, the exact data were regenerated byte for byte; four of the 40 companion tests fail there, all of them CLI or POSIX safe-output tests. The threshold $T(10^6) = 84$ was reproduced at the write.

29 Source assessment and limits of the result

The full final 17-page 2025 article [1] was inspected, including its path preliminaries, Section 2 and the entire asymptotic Section 5. Section 2 identifies the present increasing class and supplies exact multivariate generating functions. Proposition 5.1 shows its density among unrestricted partitions tends to zero. Conjecture 5.2 asks whether distinct partitions have density zero within the superdiagonal class; the closing discussion asks for asymptotic enumeration. The exact constant (20.2) is not supplied there. The article’s Dyck and ballot encodings are credited rather than claimed as new.

The official OEIS A238873 page was checked on 3 October 2026. It states the indexed condition, distinguishes the decreasing class, gives the prefix through $n = 60$, and cites [1]; no exact exponential rate was displayed. The author’s primary publication list [10], including 2026 entries, and targeted searches for the sequence, the partition phrase, logarithmic asymptotics, $\log 2$, and the decimal rate found no equivalent later result. Superdiagonal compositions are a different ordered class and were not treated as evidence for this model.

The available report catalogues were refreshed on 3 October 2026 at 08:39 UTC; the relevant files were unchanged from the earlier inspection, and the targeted repository searches found no matching result. Report 156 is the explicitly distinguished antecedent (20.7). This bounded check supports proceeding with a separate exact-rate report, but it does not establish exhaustive worldwide priority. The source provenance file records hashes and inspection scopes without redistributing full journal PDFs or full OEIS entry text.

The present conclusions are precisely the logarithmic estimates (20.3)–(20.4) and the inverse localization (20.6). Determining a sharp second term, an amplitude, a power prefactor, or an all-orders expansion remains outside these proofs. No optimality claim is made for the error $O(n^{1/3})$. The growing-height argument is only a sufficient lower-bound construction; its loss is not identified with the true next-order asymptotics.

[write] 6 October 2026. *Answered by Part IV:* the next-order term is $-Dn^{1/6}$ (Theorem 30.1), so the error order $n^{1/3}$ is not optimal (Section 47, “Answered inside this report”). The repository search of the preceding paragraph is still accurate: at the write (6 October 2026) no other report treats this sequence. A sharp amplitude, a power prefactor and all orders: Section 47, items 5–7.

Part IV

Airy correction and limit shape for superdiagonal partitions

Bundle Report 160, archive `Superdiagonal_Partitions_Airy_Correction_and_Limit_Shape_Source.zip`, delivered as `Report160.tex` (17 pages). Delivered title block “Airy correction and limit shape for superdiagonal partitions / Pointwise enumeration and macroscopic structure / Report 160 / 3 October 2026”. Printed as delivered, with `[write]` notes; four restated proofs (barrier, dilogarithm, survival, monotonicity) are replaced by pointers to Part III. Delivered Section k is Section $k + 29$ here, and its statement or equation $k.j$ is $(k + 29).j$.

Reading conventions of this Part	Elsewhere in the report
$-\zeta$ the largest zero of Ai , $\zeta \approx 2.3381 > 0$	Part V: $a = a_1 = -\zeta$; Li: ζ an argument
K the number of parts; g, f the limit profiles;	Part V: K an amplitude; Part III:
$h(x) = -\log(1 - e^{-x})$; $m(x)$ mean profile	$f(x) = -\log(1 - e^{-x})$
$\beta = d - 2b - \log(1 - e^{-d})$; $D_M(t)$, $D = \zeta b C^{-1/6}$;	Parts I, II: $\beta = c/\sqrt{n}$; Part V: the same D
$\mathbb{P}_n$ uniform on superdiagonal partitions	

Abstract of this Part (as delivered). Let $A(n)$ count weakly increasing partitions $1 \leq \lambda_1 \leq \dots \leq \lambda_K$ of n with $\lambda_i \geq i$. Put $b = \log 2$, $C = \pi^2/12 + b^2$, and let $-\zeta$ be the largest negative zero of the Airy function Ai . We prove the pointwise expansion

$$\log A(n) = 2\sqrt{Cn} - \zeta b C^{-1/6} n^{1/6} + o(n^{1/6}).$$

Its radial analogue is $\log \sum_n A(n)e^{-tn} = C/t - \zeta b t^{-1/3} + o(t^{-1/3})$. Mallein’s published killed-walk area estimates supply the Airy input. A separate exact-weight argument proves the coefficient lower bound: it splits the geometric tail twice, creates a macroscopic buffer, proves a uniform lattice local limit on all Fourier arcs, retits at a fixed cutoff, and subtracts the remaining ruin probability. The ordinary threshold inverse has an explicit second term of order $(\log y)^{4/3}$.

For a uniform partition of n , at scale $t = \sqrt{C/n}$, the scaled length tK converges to $2b$. The cumulative multiplicity profile converges uniformly on the whole nonnegative half-line to $g(x) = x$ for $x \leq b$ and $g(x) = 2b + \log(1 - e^{-x})$ for $x \geq b$. Its inverse gives the row profile on compact subsets of $(0, 2b)$. All fixed-tolerance deviations have probability at most $e^{-c/t}$. The initial half of the limiting row profile follows the diagonal; this is not a statement about exact discrete contacts. No power prefactor, multiplicative equivalent, further asymptotic term, or fluctuation theorem is asserted.

Attribution and position. The class is OEIS A238873 [6], studied by Archibald, Blecher, Elizalde and Knopfmacher [1]. Their generating functions and ballot/Dyck encodings are prior results. Report 158 [16] established the exact exponential rate and a coarse pointwise lower bound. This report is a distinct refinement of its enumeration precision, with an additional macroscopic structural consequence. The shape proof needs only that earlier coarse bound; it does not need the Airy correction. Reports 158 and 159 are unchanged. A bounded literature and catalogue check found no duplicate of the results stated here in the sources inspected. It is not exhaustive scholarly priority clearance. The analytic Airy estimates are credited to Mallein [11]; the local smoothing needed here is proved explicitly rather than imported from a Tauberian assertion.

30 Definitions and main results

All partitions in this report are written in weakly increasing order. Define

$$A(n) = \# \left\{ (\lambda_1, \dots, \lambda_K) : 1 \leq \lambda_1 \leq \dots \leq \lambda_K, \quad \sum_i \lambda_i = n, \quad \lambda_i \geq i \right\}, \quad A(0) = 1.$$

The empty partition has length zero. This increasing-order convention is essential: the weakly decreasing superdiagonal class is different. Write

$$Z_j = \#\{i : \lambda_i = j\}, \quad C_j = \sum_{r=1}^j Z_r, \quad C_0 = 0.$$

Then the defining condition is equivalent to $C_j \leq j$ for every $j \geq 1$.

[write] **6 October 2026.** This is Part III’s Lemma 21.1. Report 160’s two-line proof of it, the same argument, is not reprinted.

We use the constants

$$b = \log 2, \quad C = \frac{\pi^2}{12} + b^2, \quad \zeta \approx 2.3381074105, \quad (30.1)$$

where $\text{Ai}(-\zeta) = 0$ and no Airy zero is closer to zero on the negative real axis; the displayed decimal is tabulated by NIST [4]. Generic positive constants will be denoted c_*, K_* or by indexed lowercase letters, never by the fixed C .

[write] **6 October 2026.** Part V writes $a = a_1 = -\zeta$ for the same zero, and Li’s preprint uses ζ for a scaled argument (front matter, “Notation”). “Report 158” is Part III and “Report 159” Part II.

Set $F(t) = \sum_{n \geq 0} A(n)e^{-tn}$ for $t > 0$. It is finite because it is bounded by the ordinary partition product.

Theorem 30.1 (Radial and pointwise Airy corrections). *As $t \downarrow 0$ and, respectively, as integer $n \rightarrow \infty$,*

$$\log F(t) = \frac{C}{t} - \zeta b t^{-1/3} + o(t^{-1/3}), \quad (30.2)$$

$$\log A(n) = 2\sqrt{Cn} - \zeta b C^{-1/6} n^{1/6} + o(n^{1/6}). \quad (30.3)$$

The second formula is pointwise in every sufficiently large integer n , not only on a subsequence or after summation. The error term does not claim an $O(\log n)$ bound. In particular it does not identify an amplitude or a power of n multiplying the exponential.

[write] **6 October 2026.** Theorem 30.1 is *unconditional*: its inputs are elementary or published (Mallein’s estimates, whose statements the write read; note in Section 32). Part V, conditionally on an unrefereed preprint, proves the stronger $A(n) \sim K n^{-11/12} e^{B\sqrt{n} - Dn^{1/6}}$ and $\log F(t) = C/t + ba_1 t^{-1/3} + \frac{1}{3} \log t + O(1)$; its constants agree with these, since $ba_1 = -\zeta b$ and Part V’s $D = -a_1 b C^{-1/6}$ is this $\zeta b C^{-1/6}$. As long as Part V is conditional, (30.2)–(30.3) are this report’s unconditional statements of the Airy term. Here $2\sqrt{C} = B$ of Part III.

Corollary 30.2 (Ordinary threshold inverse). *For $y > 1$, let $T(y) = \min\{n \geq 0 : A(n) \geq y\}$ and put*

$$x = \left(\frac{\log y}{2\sqrt{C}} \right)^2.$$

Then

$$T(y) = x + \zeta b C^{-2/3} x^{2/3} + o(x^{2/3}) \quad (y \rightarrow \infty). \quad (30.4)$$

This is an asymptotic localization of the integer threshold. It is not an exact rounding rule or a certified finite-input interval.

For the structural theorem, let $\mathbb{P}_n$ be the uniform law on the partitions counted by $A(n)$, and set $t = t_n = \sqrt{C/n}$. Define

$$m(x) = \begin{cases} 1, & 0 \leq x \leq b, \\ (e^x - 1)^{-1}, & x > b, \end{cases} \quad (30.5)$$

$$g(x) = \int_0^x m(u) du = \begin{cases} x, & 0 \leq x \leq b, \\ 2b + \log(1 - e^{-x}), & x \geq b, \end{cases} \quad (30.6)$$

$$f(y) = g^{-1}(y) = \begin{cases} y, & 0 \leq y \leq b, \\ -\log(1 - e^{y/4}), & b \leq y < 2b. \end{cases} \quad (30.7)$$

The function g is strictly increasing and 1-Lipschitz, with $g(\infty) = 2b$. Its inverse f diverges at $2b$.

Theorem 30.3 (Length and macroscopic shape). *For each fixed $\epsilon > 0$, there is $c_\epsilon > 0$ such that, for all sufficiently large n ,*

$$\mathbb{P}_n(|tK - 2b| > \epsilon) \leq e^{-c_\epsilon/t}, \quad (30.8)$$

$$\mathbb{P}_n\left(\sup_{x \geq 0} |tC_{\lfloor x/t \rfloor} - g(x)| > \epsilon\right) \leq e^{-c_\epsilon/t}. \quad (30.9)$$

Consequently

$$\frac{K}{\sqrt{n}} \longrightarrow \frac{2 \log 2}{\sqrt{C}} = 1.214497355562513 \dots \quad (30.10)$$

in probability and in every finite positive moment. For any closed interval $I = [a, d_0] \subset (0, 2b)$ and fixed $\epsilon > 0$, the probability that some index $k_t(y) = \lfloor y/t \rfloor$, $y \in I$, is outside $\{1, \dots, K\}$, or that

$$\sup_{y \in I} |t\lambda_{k_t(y)} - f(y)| > \epsilon,$$

is at most $e^{-c_{I,\epsilon}/t}$ for all sufficiently large n . The supremum is evaluated only if all the indicated rows exist; a missing row is already a bad event.

The same interior row assertion holds for ceilings or any indexing rule with uniform scaled discrepancy $O(t)$. No row-uniform claim is made at the divergent endpoint $y = 2b$.

31 Exact geometric change of measure

Fix an integer $M \geq 1$. For $j \leq M$, take independent geometric multiplicities with

$$\mathbb{P}_0(Z_j = z) = 2^{-z-1}, \quad z = 0, 1, \dots, \quad H_j = j - \sum_{i \leq j} Z_i, \quad H_0 = 0.$$

The increments $Y_j = H_j - H_{j-1} = 1 - Z_j$ have mean zero, variance 2, and a finite absolute exponential moment for every parameter below b . Introduce the left-endpoint area and the prefix weight

$$\mathcal{A}_M = \sum_{j=0}^{M-1} H_j, \quad W_P = \sum_{j=1}^M jZ_j, \quad \eta_j = b - tj.$$

Finite summation by parts gives the exact identities

$$\sum_{j=1}^M \eta_j Z_j = \sum_{j=1}^M \eta_j - t\mathcal{A}_M - \eta_M H_M, \quad (31.1)$$

$$W_P = \frac{M(M+1)}{2} + \mathcal{A}_M - MH_M. \quad (31.2)$$

They hold for every prefix multiplicity vector, with no barrier assumed.

For $j > M$, independently take

$$\mathbb{P}_t(Z_j = z) = (1 - e^{-tj})e^{-tjz}.$$

The sum of the tail means is finite, so only finitely many tail parts occur almost surely. Define

$$D_M(t) = \exp\left(2bM - \frac{tM(M+1)}{2}\right), \quad P_M(t) = \prod_{j>M} (1 - e^{-tj})^{-1}.$$

Let $Q_{t,M}(h)$ be the tail probability that

$$\sum_{i=1}^k Z_{M+i} \leq k + h \quad \text{for every } k \geq 1.$$

Combining the prefix geometric mass $2^{-M-\sum Z_j}$ with (31.1) yields

$$F(t) = D_M(t)P_M(t) \mathbb{E}_0 \left[e^{-t\mathcal{A}_M - \eta_M H_M} \mathbf{1}_{\{H_j \geq 0, 0 \leq j \leq M\}} Q_{t,M}(H_M) \right]. \quad (31.3)$$

The factor is 4^M before the staircase weight, not 2^M . All products and sums here converge absolutely, so this is an identity of ordinary positive quantities, not merely a formal series identity.

31.1 The deterministic exponential factor

Put

$$h(x) = -\log(1 - e^{-x}), \quad I(x) = \int_x^\infty h(u) du = \text{Li}_2(e^{-x}), \quad L(x) = 2bx - \frac{x^2}{2} + I(x).$$

For tM in a fixed positive compact interval, monotone Riemann bounds give $\sum_{j>M} h(tj) = I(tM)/t + O(1)$. Consequently

$$\log(D_M P_M) = \frac{L(tM)}{t} - \frac{tM}{2} + O(1) = \frac{C}{t} + O(1) \quad \text{if } tM = b + O(t). \quad (31.4)$$

Indeed,

$$L(b) = \frac{3}{2}b^2 + \text{Li}_2(1/2) = \frac{\pi^2}{12} + b^2 = C.$$

[write] **6 October 2026.** The evaluation is Part III's (21.5), and L is Part III's function (21.4) ($2b = \log 4$). Report 160's two-sentence proof of the evaluation, by the dilogarithm reflection formula as in Part III, is not reprinted.

Smoothness near b suffices for (31.4); no derivative of an unknown asymptotic remainder is involved.

32 The published Airy input and its specialization

We use Mallein's killed-walk area estimates [11]. The arXiv version 3 assumptions (2.12)–(2.13) and Lemma 2.6, equation (2.15), are on manuscript page 11; Lemma 2.7, equation (2.19), is on page 13. Both the arXiv and author manuscripts were checked, including the displayed source equations. The following is their homogeneous specialization.

For a centered independent-increment walk S with variance $\sigma^2 > 0$ and an absolute exponential moment, set

$$\mathcal{K}_{m,q}(x, J) = \mathbb{E}_x \left[\exp \left(-\frac{q}{m} \sum_{j=0}^{m-1} S_j \right); S_j \geq 0 \ (0 \leq j \leq m), \ S_m \in J \right].$$

For fixed $q > 0$, the source gives

$$\limsup_{m \rightarrow \infty} m^{-1/3} \log \sup_{x \geq 0} \mathcal{K}_{m,q}(x, [0, \infty)) \leq -\zeta(\sigma^2/2)^{1/3} q^{2/3}. \quad (32.1)$$

For fixed $0 < a < c$ and $0 < a' < c'$, it also gives

$$\liminf_{m \rightarrow \infty} m^{-1/3} \log \inf_{x \in [am^{1/3}, cm^{1/3}]} \mathcal{K}_{m,q}(x, [a'm^{1/3}, c'm^{1/3}]) \geq -\zeta(\sigma^2/2)^{1/3} q^{2/3}. \quad (32.2)$$

We require only integer starting states. The original assumptions allow a centered independent triangular array with a positive continuous variance profile and a uniform absolute exponential moment; there is no symmetry or nonlattice requirement. Our homogeneous walk $1 - \text{Geom}(1/2)$ satisfies every hypothesis, with $\sigma^2 = 2$. The multiplier $(\sigma^2/2)^{1/3}$ is therefore exactly one. The left-endpoint area and killing through time m match (31.3).

The potential used below is $t = q_m/m$, with $q_m \rightarrow b$. This does not require an unquoted uniform version of the source: on nonnegative paths, sandwich the exponential between those with fixed $q = b - \delta$ and $q = b + \delta$, apply (32.1)–(32.2), then let $\delta \downarrow 0$. The starting and endpoint intervals remain fixed when the source is applied.

[write] **6 October 2026.** The write read the two cited statements in arXiv:1307.4496v3 [11] (journal version not compared). For a triangular array of independent centred variables with $\mathbf{E}[X_{n,k}^2] = \sigma_{k/n}^2$, σ continuous and positive (2.12), and a uniform exponential moment (2.13), Lemma 2.6 (p. 11) bounds $\limsup n^{-1/3} \log \sup_{x \in \mathbb{R}} \mathbf{E}_x[e^{-\frac{h}{n} \sum_{j < n} S_j}; S_j \geq 0, j \leq n]$ by $\frac{\alpha_1}{2^{1/3}}(h\bar{\sigma})^{2/3}$, and Lemma 2.7 (p. 13) bounds the corresponding $\liminf$ of the infimum over starting points in $[an^{1/3}, bn^{1/3}]$, with endpoint in $[a'n^{1/3}, b'n^{1/3}]$, from below by $\frac{\alpha_1}{2^{1/3}}(h\bar{\sigma})^{2/3}$; $\alpha_1 \approx -2.3381$ is the largest zero of Ai , $\underline{\sigma}$ and $\bar{\sigma}$ the minimum and maximum of σ . For a constant σ both constants equal $-\zeta(\sigma^2/2)^{1/3} h^{2/3}$, as in (32.1)–(32.2) (whose upper supremum over $x \geq 0$ is weaker than Mallein's over $x \in \mathbb{R}$). With $\sigma^2 = 2$ the factor is 1. The write did not verify Mallein's proofs. Part V obtains the same Airy constant by an independent route (ba_1 in (39.4)), under its extra input.

[write] **7 October 2026, batch 108 (reciprocal note).** The same two inequalities of Mallein, (2.15) of Lemma 2.6 and (2.19) of Lemma 2.7, are the Airy input of the collection's report `a397711-bounded-i`

ndegree-dags (labelled acyclic digraphs with every indegree at most two, OEIS A397711; its $n^{1/3}$ term $-3\zeta 2^{-1/3} n^{1/3}$ in $\log a_n$). There they are applied to the walk $\text{Poisson}(1) - 1$, with $\sigma^2 = 1$ and hence the factor $(\sigma^2/2)^{1/3} = 2^{-1/3}$, at fixed homogeneous tilts on the blocks of a geometric grid, to reach a potential of order $1/i$; here the walk is $1 - \text{Geom}(1/2)$, $\sigma^2 = 2$, with one tilt sequence $q_m \rightarrow b$. As here, only fixed tilts and fixed scaled intervals are used, and no strip estimate or time-inhomogeneous case is taken from Mallein. Its statement of the two lemmas agrees with the note above (its independent check re-read them on pp. 11 and 13 of arXiv:1307.4496v3); Mallein's proofs were verified in neither report. No object or theorem is shared.

33 The radial theorem and a localized prefix

For the upper bound, take $M = \lfloor b/t \rfloor$. Then $\eta_M \geq 0$ and $Q_{t,M} \leq 1$. Dropping both factors in (31.3) and applying (32.1) gives

$$\limsup_{t \downarrow 0} t^{1/3} \left(\log F(t) - \frac{C}{t} \right) \leq -\zeta b.$$

Here $t^{1/3} M^{1/3} \rightarrow b^{1/3}$, while the walk exponent on the $M^{1/3}$ scale is $-\zeta b^{2/3}$.

33.1 A tail survival bound

For a geometric variable Z with parameter $p < 1/2$, put $r = (1 - p)/p > 1$. Directly,

$$\mathbb{E} r^{Z-1} = \frac{1 - p}{r(1 - pr)} = 1. \quad (33.1)$$

[write] **6 October 2026.** Report 160 next proves, for iid such variables, the bound (33.2) below. It is Part III's Lemma 24.1, and (33.1) is the normalization in its proof. Report 160's proof (the nonnegative martingale $r^{\sum (Z_i - 1)}$, the finite-horizon maximal inequality with monotone convergence, and a forced first zero) is Part III's and is not reprinted. Report 160 remarks that no infinite-time optional-stopping equality is used and that upward jumps need not hit a prescribed level exactly; both hold for Part III's proof as well.

$$\mathbb{P} \left(\sum_{i=1}^k Z_i \leq k \text{ for all } k \geq 1 \right) \geq (1 - p)(1 - 1/r) = 1 - 2p. \quad (33.2)$$

For the lower bound choose

$$M = \lceil b/t \rceil + 1. \quad (33.3)$$

Then $-2t \leq \eta_M \leq -t$, and all tail geometric parameters are at most $p = e^{-t}/2$. Coordinatewise coupling and (33.2) therefore give, uniformly for $h \geq 0$,

$$Q_{t,M}(h) \geq 1 - e^{-t} \geq t/2 \quad (33.4)$$

for sufficiently small t . A nonnegative initial buffer only helps.

33.2 Repairing the start at zero

The lower source estimate starts at positive height on the $M^{1/3}$ scale. Fix $0 < \rho \leq 1$ and force $\ell = \lceil \rho M^{1/3} \rceil$ initial increments to be 1. The probability is $2^{-\ell}$, the path is a nonnegative ramp, its area is $\ell(\ell - 1)/2$, and it reaches height ℓ . Let $m = M - \ell$. For large M , the height lies in

$[(\rho/2)m^{1/3}, 2\rho m^{1/3}]$. Apply (32.2) to the remaining m steps with endpoint interval $[m^{1/3}, 2m^{1/3}]$. The potential satisfies $tm \rightarrow b$. The ramp area cost is $O(M^{-1/3})$, whereas $\ell b/M^{1/3} \rightarrow \rho b$. Thus

$$\liminf_{t \downarrow 0} M^{-1/3} \log \mathbb{E}_0[e^{-t\mathcal{A}_M}; H_j \geq 0 \ (j \leq M)] \geq -\zeta b^{2/3} - \rho b. \quad (33.5)$$

Taking the limit with ρ fixed and then letting $\rho \downarrow 0$ removes the boundary cost. In (31.3), $e^{-\eta_M H_M} \geq 1$. Equations (31.4) and (33.4), whose logarithmic loss is $o(t^{-1/3})$, now prove the radial lower bound and hence (30.2).

33.3 A full fixed event for the pointwise argument

Choose a fixed sufficiently large K_0 with $K_0 b > \zeta b^{2/3} + b + 2$. At cutoff (33.3), define

$$E_t = \{H_j \geq 0 \ (0 \leq j \leq M), \quad 0 \leq H_M \leq 2M^{1/3}, \quad \mathcal{A}_M \leq K_0 M^{4/3}\}. \quad (33.6)$$

This is the entire three-constraint event, independent of the ramp parameter ρ . The ramp subevents above end below $2M^{1/3}$. On a nonnegative path the weighted mass discarded by imposing the area constraint is at most $\exp(-tK_0 M^{4/3})$. For every fixed $\rho \in (0, 1]$, our choice of K_0 makes this exponentially smaller than the ramp lower bound on the $M^{1/3}$ scale. Subtract it first, then let $\rho \downarrow 0$. The matching upper bound is (32.1). We obtain

$$\log \mathbb{E}_0[e^{-t\mathcal{A}_M}; E_t] = -\zeta b t^{-1/3} + o(t^{-1/3}). \quad (33.7)$$

Moreover, (31.2) gives the deterministic localization

$$\left| W_P - \frac{M(M+1)}{2} \right| \leq (K_0 + 2)M^{4/3} = O(t^{-4/3}) \quad \text{on } E_t. \quad (33.8)$$

The area restriction is an extra ingredient for pointwise extraction; radial asymptotics alone do not supply it.

34 Exact weight localization in the geometric tail

Retain the cutoff (33.3) and independent tail law $\mathbb{P}_t$. Put

$$W_T = \sum_{j>M} jZ_j, \quad \mu_T = \mathbb{E}_t W_T, \quad B_k = k - \sum_{i=1}^k Z_{M+i}.$$

Lemma 34.1 (Uniform exact weight and survival). *For every fixed $K_1 > 0$ there are positive constants c_1, K_2 such that, uniformly over integers w with $|w - \mu_T| \leq K_1 t^{-4/3}$ and every $h \geq 0$,*

$$\mathbb{P}_t(W_T = w, B_k + h \geq 0 \text{ for all } k \geq 1) \geq c_1 t^{5/2} \exp(-K_2 t^{-1/6}) \quad (34.1)$$

for sufficiently small t .

Its logarithmic cost is $o(t^{-1/3})$, strictly smaller than the Airy scale. We prove every component, including lattice smoothing.

34.1 A second cutoff and a surviving head with a buffer

Fix $d > b$, for example $d = b + 1$, and set $J = \lfloor d/t \rfloor$ and $N = J - M$. Split the tail into

$$U = \sum_{j=M+1}^J jZ_j, \quad G = N - \sum_{j=M+1}^J Z_j, \quad V = \sum_{j>J} jZ_j, \quad W_T = U + V.$$

Write $\mu_U = \mathbb{E}_t U$ and $\mu_V = \mathbb{E}_t V$. The stricter zero-buffer head barrier $\sum_{i=1}^k Z_{M+i} \leq k$ for $1 \leq k \leq N$ has probability at least $t/2$ by (33.4). Riemann sums on $[b, d]$ give

$$\mathbb{E}_t G = \frac{\beta}{t} + O(1), \quad \beta = (d - b) - \int_b^d \frac{du}{e^u - 1} = d - 2b - \log(1 - e^{-d}) > 0. \quad (34.2)$$

Strict positivity follows from $(e^u - 1)^{-1} < 1$ for $u > b$. Set

$$a(t) = t^{-19/12}. \quad (34.3)$$

We claim

$$\mathbb{P}_t(G < \beta/(2t)) \leq e^{-c_2/t}, \quad (34.4)$$

$$\mathbb{P}_t(|U - \mu_U| > a(t)) \leq 2e^{-c_3 t^{-1/6}}. \quad (34.5)$$

There are $O(1/t)$ head variables, all with geometric parameter at most $1/2$. Their unweighted centered logarithmic moment generating functions sum to at most $K_3 s^2/t$ on a fixed neighborhood of zero. Since $\mathbb{E}G \geq 3\beta/(4t)$ eventually, a fixed sufficiently small tilt proves (34.4). For the weighted head sum,

$$\frac{d^2}{ds^2} \log \mathbb{E}_t e^{sU} = \sum_{j=M+1}^J \frac{j^2 e^{-(t-s)j}}{(1 - e^{-(t-s)j})^2} \leq K_4 t^{-3}, \quad |s| \leq t/4.$$

The scaled arguments stay away from zero and $j = O(1/t)$. Taylor's theorem bounds the centered log moment generating function by $K_4 s^2 t^{-3}/2$. Tilts of either sign, of magnitude a suitable constant times $a(t)t^3 = t^{17/12} = o(t)$, prove (34.5) because $a(t)^2 t^3 = t^{-1/6}$.

Let E_H require the zero-buffer head barrier, $G \geq \beta/(2t)$, and $|U - \mu_U| \leq a(t)$. A union bound gives

$$\mathbb{P}_t(E_H) \geq t/2 - e^{-c_2/t} - 2e^{-c_3 t^{-1/6}} \geq t/3. \quad (34.6)$$

This does not assume concentration after conditioning on survival. The surviving head produces a buffer of order $1/t$ for the independent far tail.

34.2 The far tail local limit on every Fourier arc

Keep $J = \lfloor d/t \rfloor$ fixed while varying a parameter $s > 0$. For $j > J$, take independent geometrically distributed variables with parameter e^{-sj} . Define

$$\begin{aligned} Z_J(s) &= \prod_{j>J} (1 - e^{-sj})^{-1}, & K_J(s) &= \log Z_J(s), \\ \mu_J(s) &= \sum_{j>J} \frac{j}{e^{sj} - 1} = -K'_J(s), & v_J(s) &= \sum_{j>J} \frac{j^2 e^{-sj}}{(1 - e^{-sj})^2} = K''_J(s). \end{aligned}$$

All differentiated sums converge locally uniformly for $s > 0$. Uniformly whenever $s/t \rightarrow 1$, Riemann sums and exponential tails give

$$v_J(s) \sim v_d t^{-3}, \quad v_d = \int_d^\infty \frac{x^2 e^{-x}}{(1 - e^{-x})^2} dx > 0. \quad (34.7)$$

The sum of the third-cumulant majorants is $O(t^{-4})$, uniformly in a small relative neighborhood of t .

We prove that whenever $k = \mu_J(s)$ is an integer and $s/t \rightarrow 1$,

$$\mathbb{P}_s(V = k) = \frac{1 + o(1)}{\sqrt{2\pi v_J(s)}}. \quad (34.8)$$

The characteristic function is the absolutely convergent product

$$\phi_s(\theta) = \prod_{j>J} \frac{1 - e^{-sj}}{1 - e^{-sj} e^{ij\theta}}.$$

With $p_j = e^{-sj}$, its centered logarithm has the absolutely convergent double series

$$\log \phi_s(\theta) - i\mu_J(s)\theta = \sum_{j>J} \sum_{r \geq 1} \frac{p_j^r}{r} (e^{irj\theta} - 1 - irj\theta). \quad (34.9)$$

The logarithm here is defined by this convergent expansion, so no branch ambiguity is introduced. Taylor's elementary remainder bound gives

$$\begin{aligned} \log \phi_s(\theta) - i\mu_J(s)\theta &= -\frac{1}{2}v_J(s)\theta^2 + O(t^{-4}|\theta|^3), \\ \sum_{j>J} \frac{j^3 p_j(1 + p_j)}{(1 - p_j)^3} &= O(t^{-4}). \end{aligned} \quad (34.10)$$

The last bound follows by summing $j^3 e^{-sj}$, since sj is bounded away from zero. It is uniform for the tilts under consideration.

For the consecutive block $j = J + 1, \dots, J + \lfloor 1/t \rfloor$, the parameters p_j lie in one fixed compact subinterval of $(0, 1)$. The exact identity

$$\left| \frac{1 - p}{1 - pe^{iu}} \right|^2 = \left(1 + \frac{4p \sin^2(u/2)}{(1 - p)^2} \right)^{-1}$$

therefore implies

$$|\phi_s(\theta)| \leq \exp \left(-c_4 \sum_{\text{block}} \sin^2(j\theta/2) \right). \quad (34.11)$$

We now cover the entire inversion interval $[-\pi, \pi]$.

1. Choose a fixed small $c_0 > 0$. If $|\theta| \leq c_0 t$, all block values $|j\theta|$ are below π , whence

$$\sum_{\text{block}} \sin^2(j\theta/2) \geq c_5 \theta^2 \sum_{\text{block}} j^2 \geq c_6 \theta^2 t^{-3}.$$

2. If $A_0 t \leq |\theta| \leq \pi$, the finite geometric series gives

$$\sum_{\text{block}} \sin^2(j\theta/2) \geq \frac{\lfloor 1/t \rfloor}{2} - \frac{1}{2|\sin(\theta/2)|} \geq c_7/t$$

for sufficiently large fixed A_0 , using $|\sin(\theta/2)| \geq |\theta|/\pi$.

3. In the remaining region $c_0 t \leq |\theta| \leq A_0 t$, write $u = \theta/t$. Uniform Riemann sums yield

$$t \sum_{\text{block}} \sin^2(j\theta/2) \longrightarrow \int_d^{d+1} \sin^2(ux/2) dx$$

uniformly on $c_0 \leq |u| \leq A_0$. This continuous integral has a strictly positive minimum there: a nonzero-frequency sine cannot vanish throughout an interval. Thus the sum is again at least c_8/t .

There are no unexamined secondary or rational arcs. At $\theta = t^{3/2}x$, the remainder in (34.10) is $O(t^{1/2}|x|^3)$ on bounded x -sets. Equations (34.7) and (34.11) give integrable Gaussian domination on the inner arc after this rescaling; the total outer-arc integral is $O(e^{-c_9/t}) = o(t^{3/2})$. Fourier inversion for the integer-valued V and dominated convergence prove (34.8). All estimates are uniform on shrinking relative tilt bands $s/t \rightarrow 1$. In particular they cover every tilt below. Consecutive allowed weights also give span one; all integer targets at least $J+1$ are represented by a single allowed part.

[write] **6 October 2026.** Part V, Section 43, uses the same cutoff $J = \lfloor d/t \rfloor$, the same block of $\lfloor 1/t \rfloor$ consecutive indices and the same three-arc cover of $[-\pi, \pi]$ ((43.15)–(43.18)), but assembles them differently: a moment-generating-function limit and a total-variation cutoff instead of the tilted local limit theorem with exact weights used here. Both are printed, as two routes.

34.3 Moderate retilting with a fixed cutoff

Suppose k is an integer with

$$|k - \mu_J(t)| \leq 2a(t). \quad (34.12)$$

It is positive and of order t^{-2} , because $a(t) = o(t^{-2})$. The mean $\mu_J(s)$ is continuous and strictly decreasing from infinity to zero. In a small relative neighborhood of t , its negative derivative $v_J(s)$ is bounded above and below by positive multiples of t^{-3} . Integration of the lower bound brackets the unique root of $\mu_J(s) = k$ and gives

$$|s - t| = O(a(t)t^3) = O(t^{17/12}) = o(t). \quad (34.13)$$

Here J remains fixed; it is not redefined as $\lfloor d/s \rfloor$. The exact likelihood ratio is

$$\mathbb{P}_t(V = k) = \mathbb{P}_s(V = k) \exp\{(s - t)k + K_J(s) - K_J(t)\}. \quad (34.14)$$

Expanding $K_J(t)$ about s and using $K'_J(s) = -k$ shows that the exponent equals

$$-\frac{1}{2}K''_J(\xi)(s - t)^2 \geq -K_5 t^{-3} a(t)^2 t^6 = -K_5 t^{-1/6}.$$

Its sign is nonpositive, as convexity also confirms. Combining this with (34.8) proves the uniform bound

$$\mathbb{P}_t(V = k) \geq c_{10} t^{3/2} e^{-K_5 t^{-1/6}} \quad \text{throughout (34.12)}. \quad (34.15)$$

No asymptotic remainder has been differentiated.

34.4 Restoring the far tail barrier by subtraction

Choose fixed $p_* \in (e^{-d}, 1/2)$ and let $r_* = (1 - p_*)/p_* > 1$. The original t -law far-tail variables are dominated coordinatewise by iid geometrics with parameter p_* . From (33.1) and the same maximal inequality, ruin from a buffer $g \geq 0$ obeys

$$\mathbb{P}_t \left(\exists k \geq 1 : \sum_{i=1}^k (Z_{J+i} - 1) > g \right) \leq r_*^{-\lfloor g \rfloor - 1}. \quad (34.16)$$

On E_H , the buffer G is at least $\beta/(2t)$, so ruin has probability at most $e^{-c_{11}/t}$. An additional prefix buffer $h \geq 0$ cannot worsen the bound. For a retained head weight u and any target of Lemma 34.1,

$$|(w - u) - \mu_V| \leq K_1 t^{-4/3} + a(t) \leq 2a(t)$$

for sufficiently small t . Apply (34.15) to this integer weight and subtract (34.16). The latter is negligible relative to the exact local probability, not just relative to one. Uniformly in the good head realization, the joint exact-weight and barrier probability is at least

$$\frac{c_{10}}{2} t^{3/2} e^{-K_5 t^{-1/6}}.$$

Both probabilities being subtracted are under the original t law. Integrate over the independent head event of probability at least $t/3$ in (34.6); this proves (34.1). Neither weight and survival nor survival and head concentration have been assumed independent.

35 Pointwise extraction and integer inversion

The radial expansion and coefficient positivity immediately give $A(n)e^{-tn} \leq F(t)$. Taking $t = \sqrt{C/n}$ proves the upper half of (30.3). For the lower half use the localized prefix and Lemma 34.1.

An integration by parts gives

$$\int_b^\infty \frac{u}{e^u - 1} du = b h(b) + I(b) = b^2 + \text{Li}_2(1/2).$$

Smooth Riemann sums away from zero and $tM = b + O(t)$ yield

$$\begin{aligned} \mu_T &= t^{-2}(b^2 + \text{Li}_2(1/2)) + O(t^{-1}), \\ \frac{M(M+1)}{2} + \mu_T &= \frac{C}{t^2} + O(t^{-1}) = n + O(t^{-1}). \end{aligned} \quad (35.1)$$

By (33.8), each retained prefix therefore leaves an integer residual target $w = n - W_P$ within a fixed $O(t^{-4/3})$ window of μ_T . This target is positive for small t .

The exact coefficient version of (31.3) is

$$\begin{aligned} A(n)e^{-tn} &= D_M P_M \mathbb{E}_0 [e^{-t\mathcal{A}_M - \eta_M H_M} \mathbf{1}_{\{H_j \geq 0, j \leq M\}} \\ &\quad \times \mathbb{P}_t(W_T = n - W_P, B_k + H_M \geq 0 \ \forall k \geq 1)]. \end{aligned} \quad (35.2)$$

Restrict to E_t . Since $\eta_M \leq 0$, its terminal factor is at least one. Apply (31.4), (33.7), and (34.1), uniformly over all retained prefixes. The tail logarithmic loss satisfies

$$\log(c_1 t^{5/2} e^{-K_2 t^{-1/6}}) = o(t^{-1/3}).$$

Thus

$$\log A(n) - tn \geq C/t - \zeta b t^{-1/3} + o(t^{-1/3}).$$

At $t = \sqrt{C/n}$, $tn = C/t = \sqrt{Cn}$ and $t^{-1/3} = C^{-1/6} n^{1/6}$. This proves the matching lower bound and completes Theorem 30.1. The compatible localization scales are worth recording:

$$t^{-4/3} \ll t^{-3/2} \ll t^{-19/12} \ll t^{-5/3}, \quad a(t)^2 t^3 = t^{-1/6} \ll t^{-1/3}.$$

The proof has not inferred a pointwise coefficient lower bound from a radial expansion by itself.

35.1 Threshold inversion without differentiating an error

[write] **6 October 2026.** Report 160 first shows by the largest-part injection that $A(n)$ is nondecreasing. This is Part III's Lemma 26.1, and Report 160's proof, the same injection, is not reprinted.

Thus $A(n)$ is nondecreasing; (30.3) also shows it tends to infinity. Put $B = 2\sqrt{C}$, $D = \zeta b C^{-1/6}$, $u = \log y$, and $x = (u/B)^2$. For fixed real k and any integer $n = x + kx^{2/3} + O(1)$, expansion of the explicit terms gives

$$B\sqrt{n} - Dn^{1/6} = u + (Bk/2 - D)x^{1/6} + o(x^{1/6}).$$

The unspecified remainder in (30.3) remains $o(x^{1/6})$ on either such sequence because $n/x \rightarrow 1$. For fixed $\epsilon > 0$, take

$$n_- = \left\lfloor x + (2D/B - \epsilon)x^{2/3} \right\rfloor, \quad n_+ = \left\lceil x + (2D/B + \epsilon)x^{2/3} \right\rceil.$$

Eventually $A(n_-) < y \leq A(n_+)$, whence $n_- < T(y) \leq n_+$. Letting $\epsilon \downarrow 0$ after the limit proves Corollary 30.2, since $2D/B = \zeta b C^{-2/3}$. Integer rounding contributes only $O(1)$.

[write] **6 October 2026.** Corollary 30.2 refines Part III's Theorem 20.2; Part V's Corollary 39.2 extends it, conditionally, with the same coefficient $2D/B = 1.3585577\dots$. Because of the growing term $n^{1/6}$ it is not an instance of the transseries volume's theorems (front matter, "The inverses").

36 Macroscopic concentration from a product reference law

We now prove Theorem 30.3. The only enumerative input needed is the weaker statement

$$\log A(n) \geq 2\sqrt{Cn} - K_6 n^{1/3} \quad \text{for all sufficiently large integers } n. \quad (36.1)$$

It was already established in Report 158 and also follows from Theorem 30.1. Thus the structural argument does not depend on the Airy correction or its local-limit proof. Even the exact logarithmic rate with an $o(\sqrt{n})$ error would suffice.

[write] **6 October 2026.** "Report 158" is Part III; (36.1) is the lower half of its (20.4). Theorem 30.3 is therefore unconditional and independent of Section 32.

36.1 Exact conditioning and its cost

Take $M = \lfloor b/t \rfloor$ and let $\mathbb{Q}_t$ be the independent product law with geometric parameters

$$p_j = \begin{cases} 1/2, & j \leq M, \\ e^{-tj}, & j > M. \end{cases}$$

Every parameter is at most $1/2$, and the means are exactly $\mathbb{E}_{\mathbb{Q}_t} Z_j = m(tj)$, including a possible equality $tM = b$. Since $\sum_j m(tj) < \infty$, this law is supported on finite partitions. Set

$$\mathcal{U}_M = D_M P_M, \quad \mathcal{W}_t = e^{-tA_M - \eta_M H_M}, \quad \mathcal{B} = \{H_j \geq 0 \text{ for all } j \geq 1\}, \quad \mathcal{N} = \sum_{j \geq 1} j Z_j.$$

The pointwise change of measure is

$$e^{-t|\lambda|} = \mathcal{U}_M \mathbb{Q}_t(\{\lambda\}) \mathcal{W}_t(\lambda). \quad (36.2)$$

On the full barrier $\mathcal{B}$, $0 < \mathcal{W}_t \leq 1$ because $\eta_M \geq 0$. For every event E in the multiplicities, including a t -dependent event,

$$\begin{aligned} \mathbb{P}_n(E) &= \frac{\mathbb{E}_{\mathbb{Q}_t}[\mathbf{1}_E \mathbf{1}_{\mathcal{B}} \mathbf{1}_{\{\mathcal{N}=n\}} \mathcal{W}_t]}{R_n(t)}, \\ R_n(t) &= \frac{A(n)e^{-tn}}{\mathcal{U}_M}, \quad \mathbb{P}_n(E) \leq \frac{\mathbb{Q}_t(E)}{R_n(t)}. \end{aligned} \quad (36.3)$$

The barrier includes every tail index. At $t = \sqrt{C/n}$, (31.4) and (36.1) give

$$\log R_n(t) \geq -K_7 t^{-2/3} - K_8, \quad \mathbb{P}_n(E) \leq e^{K_7 t^{-2/3} + K_8} \mathbb{Q}_t(E). \quad (36.4)$$

Therefore any product-law bound $A_* e^{-c_*/t}$ transfers to $e^{-c_*/(2t)}$ for sufficiently small t . No separate estimate of $\mathbb{Q}_t(\mathcal{N} = n)$ or independence after conditioning is required.

36.2 Exponential moments uniformly over infinite subsets

Put $s_0 = b/2$. For a geometric variable of parameter $p \leq 1/2$ and mean $\mu_p = p/(1-p)$, its centered log moment generating function is

$$\ell_p(s) = \log(1-p) - \log(1-pe^s) - s\mu_p, \quad |s| \leq s_0.$$

It has value and derivative zero at zero and

$$\ell_p''(s) = \frac{pe^s}{(1-pe^s)^2} \leq \frac{e^{s_0}\mu_p}{(1-e^{s_0}/2)^2}.$$

Taylor's theorem thus gives $\ell_p(s) \leq a_0 s^2 \mu_p$, where $a_0 = e^{s_0}/[2(1-e^{s_0}/2)^2]$. Independence implies the same summed inequality over any finite index set. It extends to an infinite set by increasing finite subsets: use monotone convergence for uncentered exponentials when $s \geq 0$, and dominated convergence (the exponentials are at most one) when $s < 0$. The corresponding deterministic mean sums converge. Since m is nonincreasing,

$$t \sum_{j \geq 1} m(tj) \leq \int_0^\infty m(x) dx = 2b.$$

With $V_0 = 2ba_0$, for every finite or infinite subset $I_0 \subset \mathbb{N}$,

$$\log \mathbb{E}_{\mathbb{Q}_t} \exp \left(s \sum_{j \in I_0} (Z_j - m(tj)) \right) \leq V_0 s^2 / t, \quad |s| \leq s_0. \quad (36.5)$$

For fixed $r > 0$, choose $s_r = \min(s_0, r/(2V_0))$. Chernoff's inequality with both signs yields

$$\mathbb{Q}_t \left(t \left| \sum_{j \in I_0} (Z_j - m(tj)) \right| \geq r \right) \leq 2e^{-\kappa_r/t}, \quad \kappa_r = rs_r/2 > 0. \quad (36.6)$$

The constants work simultaneously for every prefix and for the infinite total length.

36.3 Global Riemann bounds and length moments

For $J_0 = \lfloor x/t \rfloor$, monotonicity of m gives

$$0 \leq \int_0^{tJ_0} m(u) du - t \sum_{j=1}^{J_0} m(tj) \leq t(m(0) - m(tJ_0)) \leq t.$$

The remaining interval has length below t and $m \leq 1$, so

$$0 \leq g(x) - t\mathbb{E}_{\mathbb{Q}_t} C_{\lfloor x/t \rfloor} \leq 2t \quad \text{for every } x \geq 0. \quad (36.7)$$

This estimate is global, even for x growing as t decreases. Letting the integer cutoff tend to infinity also gives

$$0 \leq 2b - t\mathbb{E}_{\mathbb{Q}_t} K \leq t. \quad (36.8)$$

Equations (36.6), (36.8) and (36.4) prove (30.8). Under $\mathbb{P}_n$, the deterministic constraint implies $K(K+1)/2 \leq n$, hence $K/\sqrt{n} \leq \sqrt{2}$. Convergence in probability therefore implies convergence of every finite positive moment, proving (30.10) in the stated sense.

36.4 Uniform cumulative concentration

Fix $L > 0$ and $\epsilon > 0$. Choose a finite grid of $[0, L]$ with mesh at most $\epsilon/4$. If the deviation in (30.9) is at most $\epsilon/2$ at all grid points, monotonicity of the cumulative count and the 1-Lipschitz property of g bound it by $3\epsilon/4$ between adjacent points. By (36.7), a grid deviation exceeding $\epsilon/2$ implies a centered deviation exceeding $\epsilon/4$ for sufficiently small t . A finite union bound in (36.6) consequently proves

$$\mathbb{Q}_t \left(\sup_{0 \leq x \leq L} |tC_{\lfloor x/t \rfloor} - g(x)| > \epsilon \right) \leq A_{L,\epsilon} e^{-c_{L,\epsilon}/t}.$$

The grid is fixed after L, ϵ are fixed. To reach the whole half-line, choose fixed L with $2b - g(L) < \epsilon/4$. On the event that the compact supremum is at most $\epsilon/4$ and $|tK - 2b| \leq \epsilon/4$, for $x \geq L$,

$$\begin{aligned} tC_{\lfloor x/t \rfloor} - g(x) &\leq tK - g(L) \leq \epsilon/2, \\ g(x) - tC_{\lfloor x/t \rfloor} &\leq 2b - tC_{\lfloor L/t \rfloor} \leq \epsilon/2. \end{aligned}$$

The complement has product-law probability at most $A_\epsilon e^{-c_\epsilon/t}$. Transfer by (36.4) proves (30.9).

37 Row inversion and structural consequences

Solving (30.6) for x gives (30.7); both pieces and their first derivatives agree at b . We give the integer-index argument for the claimed compact-interior row convergence. Fix $I = [a, d_0] \subset (0, 2b)$ and $\epsilon > 0$. Put $\delta = \min(\epsilon/2, a/2) > 0$. Since $f(y) \geq y \geq a$, $f(y) - \delta$ is nonnegative. The two continuous positive functions

$$y - g(f(y) - \delta), \quad g(f(y) + \delta) - y$$

have a common positive lower bound Δ on I . Set $r = \Delta/3$ and suppose

$$\sup_{x \geq 0} |tC_{\lfloor x/t \rfloor} - g(x)| \leq r, \quad t < \min(a/2, r).$$

For $y \in I$ put $k = \lfloor y/t \rfloor \geq 1$ and $x_{\pm} = f(y) \pm \delta$. As $y - t < tk \leq y$,

$$tC_{\lfloor x_-/t \rfloor} \leq y - 2r < tk, \quad tC_{\lfloor x_+/t \rfloor} \geq y + 2r > tk.$$

Thus $C_{\lfloor x_-/t \rfloor} < k \leq C_{\lfloor x_+/t \rfloor}$. The right inequality proves the row exists. Since the list is sorted,

$$\lfloor x_-/t \rfloor < \lambda_k \leq \lfloor x_+/t \rfloor, \quad x_- < t\lambda_k \leq x_+.$$

This holds simultaneously for all $y \in I$ and bounds the row supremum by $\delta < \epsilon$. Its bad event is contained in the global cumulative bad event at tolerance r , proving the row part of Theorem 30.3. An $O(t)$ change of scaled indices is absorbed by the same strict gaps.

37.1 Half of the limiting row mass follows the diagonal

At $M = \lfloor b/t \rfloor$, (30.9) gives $tC_M \rightarrow b$ while (30.8) gives $tK \rightarrow 2b$. Therefore

$$\frac{C_{\lfloor b/t \rfloor}}{K} \rightarrow \frac{1}{2} \tag{37.1}$$

with exponentially small probability of every fixed-tolerance deviation. This counts rows whose part sizes are at most the critical scale b/t . Equivalently, the initial half of the limiting row curve is $f(y) = y$. For every $0 < a < d_0 < b$,

$$\sup_{y \in [a, d_0]} t(\lambda_{\lfloor y/t \rfloor} - \lfloor y/t \rfloor) \rightarrow 0$$

with the same kind of fixed-tolerance exponential bound. These are macroscopic statements. They do not determine the number or density of exact equalities $\lambda_i = i$.

37.2 Area normalization and scope

Changing variables $y = g(x)$ gives

$$\begin{aligned} \int_0^{2b} f(y) dy &= \int_0^\infty xm(x) dx \\ &= \frac{b^2}{2} + \int_b^\infty \frac{x}{e^x - 1} dx = \frac{3}{2}b^2 + \text{Li}_2(1/2) = C. \end{aligned}$$

This matches the deterministic rescaled area $t^2n = C$. It is a consistency check, not a substitute for concentration or an assertion of unproved extreme-row control. The constants in the exponential probability bounds depend on the fixed tolerance and, for rows, the fixed interior interval. No optimal constant, central limit theorem, covariance process, diagonal fluctuation scale, largest-part centering, or extreme-part law is proved here.

[write] **6 October 2026.** These open items are collected in Section 47, item 4.

38 Reproducibility and evidentiary limits

The accompanying source archive contains the editable L^AT_EX, a bounded standard-library Python companion, deterministic JSON outputs, source provenance, tests, and exclusive clean PDF/ZIP

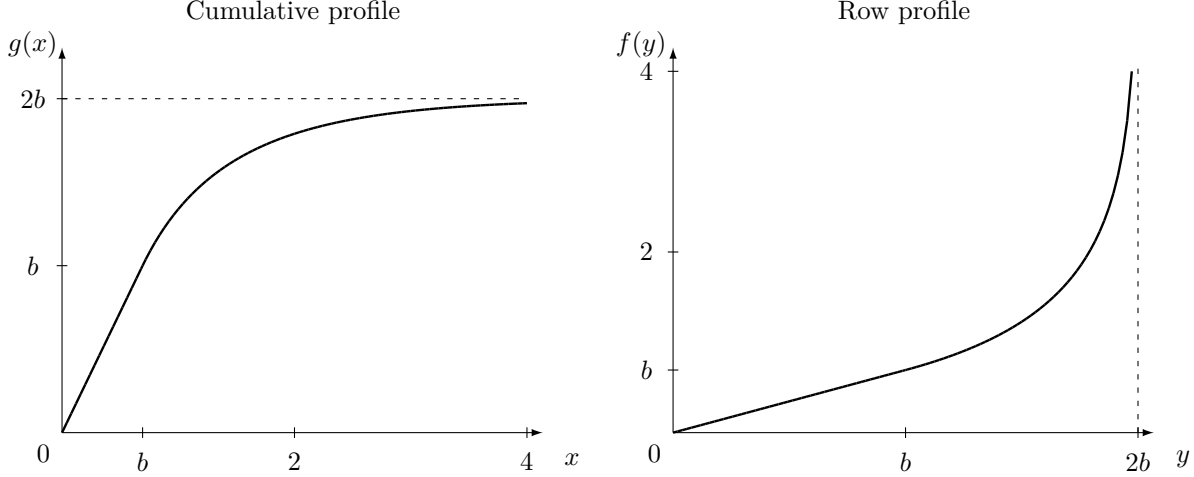


Figure 1: The analytic limiting cumulative curve g and its inverse row curve f . The solid curves follow the formulas in (30.6)–(30.7); the junction is $b = \log 2$. Dashed lines mark $g(\infty) = 2b$ and the divergent endpoint of f . Numerical drawing coordinates are illustrative, not sampled data or probability certificates.

builders. Its exact finite checks test enumeration, the equivalence of indexed and cumulative constraints, the largest-part injection, and rational change-of-measure identities. Separate numerical illustrations evaluate the explicit constants and shape curves. They do not certify the Airy zero, an asymptotic remainder, a probability bound, or an onset threshold.

The analytic proof depends on the explicitly stated Mallein estimates and elementary arguments supplied here. The numerical computation is independent supporting evidence, never a replacement for those estimates. The same distinction applies to finite Fourier or likelihood-ratio probes. No finite list of counts proves a pointwise asymptotic theorem.

The package fixes a finite allowlist and SHA-256 manifest. It excludes third-party papers, full OEIS text, working reviews, unrelated reports, and unlisted files. The archive is reproducible byte for byte under the recorded Python and $\text{\TeX}$ toolchain; cross-toolchain identity is not promised. Build commands disable shell escape and reject existing output targets, but modified $\text{\TeX}$ is not thereby made a security sandbox. The README gives exact bounds, replay commands, and the trusted-filesystem assumptions.

38.1 What the refinement establishes

The pointwise term $-\zeta b C^{-1/6} n^{1/6}$ strengthens the earlier exact exponential rate; the radial term alone would not have justified it. The explicit second-order inverse follows from monotonicity and bracketing. The length and shape conclusions are structural consequences already available from a coarse coefficient lower bound. The diagonal row segment has half the limiting mass, while its exact discrete contacts remain an open microscopic issue in this report. The published partition class, path encodings, probability estimates, and general tilting methods are not claimed as inventions here.

[write] 6 October 2026. The package is shipped with the prefix `160-airy-`: its companion README at the report root, the companion and its tests, the PDF builder, archive packer and release tools under `code/`,

and the four result files, the OEIS prefix and the provenance record under **data/** (the PDF, README and **SHA256SUMS** are not shipped). At intake, on a Windows copy, the four result files were regenerated byte for byte; one of the 23 companion tests fails there (it relies on POSIX open flags). “Reports 158 and 159” are Parts III and II.

Part V

Full prefactor and refined inverse for superdiagonal partitions (conditional)

Bundle Report 162, archive `Superdiagonal_Partitions_Full_Prefactor_and_Refined_Inverse_Source.zip`, delivered as `Report162.tex` (15 pages). Delivered title block “Full prefactor and refined inverse for superdiagonal partitions / Exact coefficient asymptotics for A238873 / Report 162 / 3 October 2026”. **This Part’s main theorem and corollary are conditional on the unrefereed preprint arXiv:2607.27504v1 [13]**; the unconditional superdiagonal statements of the report are those of Parts III and IV. Printed as delivered, with `[write]` notes and a “conditional” tag in the two statement headings; the proof of the monotonicity injection is replaced by a pointer to Part III. Delivered Section k is Section $k + 38$ here, and its statement or equation $k.j$ is $(k + 38).j$. Section 47, which closes the report, is added in the write.

Reading conventions of this Part	Elsewhere in the report
$a = a_1 < 0$ the largest zero of A_i ; s the scaled argument (Li’s ζ); $L = \log 4$	Part IV: $\zeta = -a$; Parts III–IV: $L(x)$ a function
$H_q(y)$ Ramanujan’s entire function (Li’s $A_q(z)$); ρ_j its zeros; K, K_F amplitudes; α	Part IV: K the length; Part I: ρ_h
$D = -abC^{-1/6}$; D_M ; d a cutoff and an inverse coefficient; p a geometric parameter and $p = \frac{11}{12}$; N the size	Part IV: the same $D, D_M(t)$; Parts I, II: $N(r, n)$

Abstract of this Part (as delivered). Let $A(n)$ count weakly increasing positive-integer partitions $\lambda_1 \leq \dots \leq \lambda_k$ with $\lambda_i \geq i$, including the empty partition at $n = 0$. Put $b = \log 2$, $C = \pi^2/12 + b^2$, and let $a < 0$ be the largest negative zero of A_i . We prove

$$A(n) \sim Kn^{-11/12} \exp(B\sqrt{n} - Dn^{1/6}), \quad K = \frac{\sqrt{2}C^{5/12}}{\sqrt{\pi} \operatorname{Ai}'(a)},$$

where $B = 2\sqrt{C}$ and $D = -abC^{-1/6}$. The proof uses the 2025 exact partition generating function, the 2017 positive simple-zero product for Ramanujan’s entire function, and Yu-Tian Li’s quantitative turning-point theorem in *arXiv:2607.27504v1, submitted 29 July 2026*. The latter is an explicit preprint dependency; no journal publication of it has been verified.

A positive first-pole estimate is uniform over the relevant moving length saddle. It gives the complete radial amplitude and its first relative correction. A separate, self-contained coefficient transfer uses a real moment-generating-function limit, exponentially accurate removal of far-tail constraints, and geometric Fourier damping on all arcs. It does not assume complex- q turning-point asymptotics. The threshold inverse is obtained through the $\sqrt{x} \log x$ and $\sqrt{x}$ terms, with $x = (\log y/B)^2$, by monotonicity and integer bracketing. No pointwise first relative correction, effective finite crossover, all-orders expansion, or exhaustive priority claim is made.

Scope and attribution. This report is self-contained after its explicitly stated source inputs. The exact combinatorial identity is due to Archibald, Blecher, Elizalde and Knopfmacher [1]; the zero product is due to Štampach and Štovíček [12]; the quantitative Airy expansion is due to Li [13]. The coefficient local limit and their assembly for this partition class are proved below. The two source convention defects are repaired explicitly, so neither can hide a normalization factor. Reports 156, 158 and 160 are unchanged and are not used as proof dependencies. The companion performs bounded finite checks; no numerical fit is used to determine an asymptotic constant.

39 Definitions and main conclusions

Throughout, partitions are written in increasing order. Let $\mathcal{S}_n$ be the finite set of lists

$$1 \leq \lambda_1 \leq \cdots \leq \lambda_k, \quad \sum_{i=1}^k \lambda_i = n, \quad \lambda_i \geq i.$$

The empty list belongs to $\mathcal{S}_0$, and $A(n) = |\mathcal{S}_n|$. This is the class associated with OEIS A238873 [6]. Set

$$b = \log 2, \quad L = 2b = \log 4, \quad C = \frac{\pi^2}{12} + b^2, \quad (39.1)$$

$$a = a_1 < 0, \quad B = 2\sqrt{C}, \quad D = -abC^{-1/6}, \quad (39.2)$$

$$K_F = \frac{2\sqrt{2}}{\text{Ai}'(a)}, \quad K = \frac{\sqrt{2}C^{5/12}}{\sqrt{\pi} \text{Ai}'(a)}, \quad \alpha = a^2 \left(\frac{1}{4} - \frac{b}{30} \right). \quad (39.3)$$

Here $a_1 > a_2 > \cdots$ are the negative Airy zeros, and $\text{Ai}'(a_1) > 0$. All constants in (39.2)–(39.3) except a are positive. The radial generating function is

$$F(t) = \sum_{n \geq 0} A(n) e^{-tn}, \quad t > 0.$$

It is finite because $A(n)$ is bounded by the ordinary partition number and $\prod_{j \geq 1} (1 - e^{-tj})^{-1} < \infty$.

Theorem 39.1 (Full equivalent and radial correction; [write] conditional on arXiv:2607.27504v1). *Using the source inputs stated in Section 40, as $t \downarrow 0$,*

$$F(t) = K_F t^{1/3} \exp \left(\frac{C}{t} + bat^{-1/3} \right) \left(1 + \alpha t^{1/3} + O(t^{2/3}) \right). \quad (39.4)$$

As integers $n \rightarrow \infty$,

$$A(n) \sim K n^{-11/12} \exp \left(B\sqrt{n} - Dn^{1/6} \right). \quad (39.5)$$

Equivalently,

$$\log A(n) = B\sqrt{n} - Dn^{1/6} - \frac{11}{12} \log n + \log K + o(1). \quad (39.6)$$

Corollary 39.2 (Refined threshold inverse; [write] conditional on arXiv:2607.27504v1). *For real $y > 1$, let $T(y) = \min\{n \geq 0 : A(n) \geq y\}$ and put $x = (\log y/B)^2$. Then, as $y \rightarrow \infty$,*

$$T(y) = x + \frac{2D}{B} x^{2/3} + \frac{11}{6B} \sqrt{x} \log x - \frac{2 \log K}{B} \sqrt{x} + o(\sqrt{x}). \quad (39.7)$$

For orientation only, the constants have numerical values

$$\begin{aligned} C &\approx 1.30292004734231, & B &\approx 2.28291046459761, & D &\approx 1.55073274812252, \\ K_F &\approx 4.03363301463591, & K &\approx 1.27049761356030, & \alpha &\approx 1.24037790708076. \end{aligned}$$

These decimal evaluations are not certified intervals and play no role in the proof. In particular (39.4) does not by itself prove (39.5); Section 43 supplies the local transfer. Nor does its displayed correction establish a corresponding pointwise relative correction. The local-limit error below is only $o(1)$.

[write] **6 October 2026.** *Conditional.* Theorem 39.1 and Corollary 39.2 use Theorem 1 and Corollary 2 of Li's preprint arXiv:2607.27504v1 [13], an unrefereed version 1 of 29 July 2026 (on 6 October 2026 arXiv lists no later version and no journal reference); they hold *if* those two statements hold. The heading tags were added by the write. The input enters only through (40.7)–(40.9): the normalized expansion on compact s -sets with an $O(t)$ remainder, and the global indexing of the first two positive zeros. How the other steps depend on it: Lemma 41.1 uses (40.7)–(40.9) for (41.5); the proof of Lemma 43.1 divides by (39.4), but only needs a lower bound $\log F(t) \geq C/t - O(t^{-1/3})$, which Part III's (20.3) gives unconditionally; Proposition 43.2 uses (39.4) at two nearby arguments to precision $o(1)$ (proof of (43.2)), which the unconditional Parts do not supply. The exact identity (40.2), the product (40.3), the Fourier bounds and the inverse argument of Section 44 do not use the preprint. Until the preprint is confirmed, this report's unconditional statements about $A(n)$ are Part III's (the rate) and Part IV's (the Airy term, the limit shape). The statements agree: (39.6) implies Part IV's (30.3), since $D = -abC^{-1/6} = \zeta bC^{-1/6}$, and (39.4) implies (30.2). Numerically the intake found K , K_F and α consistent with exact counts to a few percent (front matter, "What was checked").

40 Exact identity and carefully normalized source inputs

40.1 The exact superdiagonal generating function

For $0 < q < 1$, define

$$H_q(y) = \sum_{j \geq 0} \frac{(-1)^j q^{j^2} y^j}{(q; q)_j}, \quad (q; q)_j = \prod_{r=1}^j (1 - q^r), \quad c_k(q) = [y^k] \frac{1}{H_q(y)}. \quad (40.1)$$

The empty product is one. Theorem 2.1 and equation (6) of [1], with the empty term restored, give

$$F(t) = \sum_{k \geq 0} e^{-tk(k-1)/2} c_k(e^{-t}). \quad (40.2)$$

Here is the underlying combinatorial normalization. Subtract $i - 1$ from the i th increasing part and reverse the resulting list. The result is a positive composition $(\mu_1, \dots, \mu_k)$ satisfying $\mu_{i+1} \leq \mu_i + 1$. This map is invertible and reduces the weight by exactly $k(k-1)/2$. The paper's elementary functional equation for these 1-compositions gives the series $1/H_q(y)$ when the empty composition is included. Thus $c_k(q) \geq 0$ and (40.2) has exactly the normalization $A(0) = 1$.

The published display in [1] begins at $k = 1$, although its preceding generating series begins with 1. Starting at $k = 0$ repairs this empty-object convention. It does not multiply F by any factor. All ensuing summations are over nonnegative combinatorial weights; finiteness was checked above.

[write] **6 October 2026.** Confirmed in the paper: the display of Theorem 2.1 of [1] is $A(q, t) = \sum_{k \geq 1} t^k q^{\binom{k}{2}} [y^k] \sum_{i \geq 0} (\sum_{j \geq 1} (-1)^{j-1} q^{j^2} y^j / (q; q)_j)^i$, starting at $k = 1$; the inner sum is $1/H_q(y)$. Restoring the $k = 0$ term adds the empty partition and gives $A(0) = 1$, the convention of every Part of this report. This repairs a convention, not a mathematical error.

40.2 The simple positive-zero product and a source typo

For each $0 < q < 1$, the theorem and example in [12, Theorem 5.2, pp. 164–167, Example 5.2] give

$$H_q(y) = \prod_{j \geq 1} \left(1 - \frac{y}{\rho_j} \right), \quad 0 < \rho_1 < \rho_2 < \dots, \quad \sum_{j \geq 1} \rho_j^{-1} < \infty. \quad (40.3)$$

The zeros are simple, and the product converges locally uniformly. For clarity, the relevant hypotheses and the exact normalization can be checked directly. Take $x_k = q^{(2k-1)/4}$; then $x_k x_{k+1} = q^k > 0$ and $\sum_k x_k x_{k+1} < \infty$. The paper's associated positive-off-diagonal Jacobi matrix is self-adjoint and Hilbert–Schmidt. Its nonzero eigenvalues are real and simple. The paired characteristic product in w is a genus-zero product in $y = w^2$, with no free exponential factor.

The intermediate identification in the published Example 5.2 prints an extraneous factor q in $A_q(w^2) = q \mathfrak{F}(\{wq^{(2k-1)/4}\})$. At $w = 0$ that display would say $1 = q$. The correct identity has no factor q , as the defining independent-set sum shows: the coefficient of $(-w^2)^m$ in $\mathfrak{F}$ is

$$\sum_{\substack{1 \leq k_1 < \dots < k_m \\ k_{i+1} \geq k_i + 2}} q^{k_1 + \dots + k_m} = q^{m^2} \sum_{0 \leq r_1 \leq \dots \leq r_m} q^{r_1 + \dots + r_m} = \frac{q^{m^2}}{(q; q)_m}. \quad (40.4)$$

We used $k_i = 2i - 1 + r_i$ and the ordinary partition product with at most m parts. This is precisely (40.1). The final unit-normalized product in the source is already correct; (40.4) repairs the faulty intermediate display independently before we use it.

[write] 6 October 2026. *Checked in the arXiv version.* The write read arXiv:1510.01454v1 [12] (the journal version was not available to it): there the factorization theorem is Theorem 15, the positive case is discussed after it, and the q -Airy example is Example 19 (p. 16), which defines $A_q(z) = \sum_{n \geq 0} q^{n^2} (-z)^n / (q; q)_n$, i.e. H_q , and prints $A_q(w^2) = q \mathfrak{F}(\{wq^{(2k-1)/4}\}_{k=1}^\infty)$, with the extraneous factor q described above, before the unit-normalized product $A_q(z) = \prod_k (1 - z/\iota_k(q))$. The definition of $\mathfrak{F}$ there, $\mathfrak{F}(x) = 1 + \sum_{m \geq 1} (-1)^m \sum x_{k_1} x_{k_1+1} \dots x_{k_m} x_{k_m+1}$ over $k_{i+1} \geq k_i + 2$, gives (40.4), which the write re-derived. Whether the journal's Example 5.2 has the same display was not checked; Part V's identification of it with the journal numbering is the manuscript's.

40.3 The quantitative turning-point theorem

The newer input is Yu-Tian Li, *Airy Turning-Point Asymptotics for Ramanujan's Entire Function* $A_q(z)$, arXiv:2607.27504v1, submitted 29 July 2026, Theorem 1 and Corollary 2 [13]. The exact defining series is (40.1); it is not a different q -Airy function. The referenced version is a preprint. Its variable $q = e^{-t}$ has real $t > 0$. The complex uniformity below is in the scaled argument s , not in t or q .

Put $E_0 = \pi^2/12 - b^2$ and

$$y_t(s) = \frac{1}{4} \exp(-t/2 - t^{2/3}s), \quad (40.5)$$

$$N_t(s) = \frac{t^{1/6}}{2\sqrt{\pi}} \exp\left(\frac{E_0}{t} - bst^{-1/3} - \frac{s^2 t^{1/3}}{4}\right), \quad U_t(s) = N_t(s) H_{e^{-t}}(y_t(s)). \quad (40.6)$$

For every fixed compact set $\mathcal{K} \subset \mathbb{C}$, that theorem states, uniformly for $s \in \mathcal{K}$,

$$U_t(s) = \text{Ai}(s) + \frac{t^{2/3}}{30} (4s \text{Ai}(s) + s^2 \text{Ai}'(s)) + O_{\mathcal{K}}(t). \quad (40.7)$$

Its corollary identifies the *globally* first positive zero as $\rho_1 = y_t(s_t)$, where

$$s_t = a - \frac{a^2}{30} t^{2/3} + O(t). \quad (40.8)$$

The second positive zero has coordinate $a_2 + O(t^{2/3})$. Only these two fixed indices are needed. No growing-index uniformity is inferred.

The remainder in (40.7) is holomorphic in s . Cauchy's integral formula on a slightly larger fixed disk therefore differentiates it with the same $O(t)$ order on the smaller disk. Since $\text{Ai}''(a) = a \text{Ai}(a) = 0$,

$$U'_t(s_t) = \text{Ai}'(a) \left(1 + \frac{a}{5} t^{2/3} + O(t) \right). \quad (40.9)$$

Indeed the derivative of the correction at a is $[6a \text{Ai}'(a) + a^2 \text{Ai}''(a)]/30 = a \text{Ai}'(a)/5$; the displacement of $\text{Ai}'(s_t)$ begins only at order $t^{4/3}$.

Remark 40.1 (What the source input supplies). The normalization (40.6), the $O_K(t)$ error, and global zero indexing are essential. An unnormalized Airy approximation or a fixed-argument oscillatory formula would not suffice. The preprint obtains these through a Gaussian integral representation, pole-free contour deformation, uniform product expansion, cubic descent and zero counting. Its order- $t^{1/3}$ local density integrates to zero; its next density gives the correction in (40.7). The proof here uses the stated theorem, not a new proof of that turning-point result. The older fixed- $x > 1/4$ result of Li and Wong [14] is not a substitute for its moving-turning-point error estimate.

[write] **6 October 2026.** At intake Theorem 1 and Corollary 2 of [13] were read in the arXiv HTML text (5 October 2026): with $q = e^{-\varepsilon}$, $z = \frac{\sqrt{q}}{4} e^{-\varepsilon^{2/3}} \zeta$, ζ in a compact set and $\varepsilon \downarrow 0$ real, $N_\varepsilon(\zeta) A_q(z) = \text{Ai}(\zeta) + \frac{\varepsilon^{2/3}}{30} (4\zeta \text{Ai}(\zeta) + \zeta^2 \text{Ai}'(\zeta)) + O(\varepsilon)$ with $N_\varepsilon(\zeta) = \frac{\varepsilon^{1/6}}{2\sqrt{\pi}} \exp\{\frac{\pi^2 - 3L^2}{12\varepsilon} - \frac{L\zeta}{2\varepsilon^{1/3}} - \frac{\varepsilon^{1/3}\zeta^2}{4}\}$, $L = \log 4$; and the n th positive zero satisfies $\rho_n(e^{-\varepsilon}) = \frac{1}{4} \exp(-a_n \varepsilon^{2/3} - \frac{\varepsilon}{2} + \frac{a_n^2 \varepsilon^{4/3}}{30} + O_n(\varepsilon^{5/3}))$, globally indexed. With $\varepsilon = t$, $\zeta = s$ these are (40.5)–(40.8): $(\pi^2 - 3L^2)/12 = \pi^2/12 - b^2 = E_0$, $L\zeta/2 = bs$. The hypotheses are met (real $t > 0$, compact s -sets, the fixed indices 1 and 2). Li's proof was not verified, at intake or at the write.

41 Uniform positive first-pole transfer

Write $\rho = \rho_1$ and remove the first zero from (40.3):

$$G_q(y) = \prod_{j \geq 2} (1 - y/\rho_j)^{-1} = \sum_{j \geq 0} g_j y^j, \quad S_q = G_q(\rho) = -\frac{1}{\rho H'_q(\rho)} > 0. \quad (41.1)$$

The product convergence proves that G_q is analytic for $|y| < \rho_2$, that S_q is finite, and that all $g_j \geq 0$. Convolution with $(1 - y/\rho)^{-1}$ gives the exact identity and bound

$$c_k(q) = \rho^{-k} \sum_{j=0}^k g_j \rho^j, \quad 0 \leq c_k(q) \leq S_q \rho^{-k}. \quad (41.2)$$

For any $\rho < R < \rho_2$, the positive omitted tail satisfies

$$0 \leq 1 - \frac{c_k(q)}{S_q \rho^{-k}} \leq \left(\frac{\rho}{R} \right)^{k+1} \frac{G_q(R)}{S_q}. \quad (41.3)$$

To see the last inequality, write each omitted term as $g_j R^j (\rho/R)^j$ and use $j \geq k+1$. This step avoids using a fixed- q residue estimate nonuniformly.

Choose one fixed $s \in (a_2, a)$ and set $R = y_t(s)$. Since y_t decreases with its argument and the first two zero coordinates converge to a, a_2 , we have $\rho < R < \rho_2$ for small t . At a zero, differentiation of (40.6) has no contribution from N'_t , and $y'_t(s) = -t^{2/3} y_t(s)$. Thus the following identities are exact:

$$G_q(R) = \frac{1 - R/\rho}{H_q(R)}, \quad S_q = \frac{t^{2/3} N_t(s_t)}{U'_t(s_t)}. \quad (41.4)$$

Let $\Delta_t = s_t - s \rightarrow a - s > 0$. Equations (40.7)–(40.9) show

$$\log \frac{G_q(R)}{S_q} = b\Delta_t t^{-1/3} + O(1), \quad (41.5)$$

$$\log \frac{\rho}{R} = -\Delta_t t^{2/3}. \quad (41.6)$$

For (41.5), all factors not displayed in its exponent have nonzero finite limits: $(1 - R/\rho)/t^{2/3} \rightarrow s - a < 0$, $\text{Ai}(s) < 0$ on (a_2, a) , and $U'_t(s_t) \rightarrow \text{Ai}'(a) > 0$. The $s^2 t^{1/3}$ contribution to the normalization ratio is bounded. The logarithms are taken of positive real quantities.

Lemma 41.1 (Uniform dominance above the barrier scale). *For every fixed $\delta > 0$, there is $c_\delta > 0$ such that, uniformly in integers $k \geq 0$ with $kt \geq b + \delta$,*

$$c_k(e^{-t}) = S_{e^{-t}} \rho^{-k} \left(1 + O(e^{-c_\delta t^{-1/3}})\right). \quad (41.7)$$

The upper bound in (41.2) holds for every k .

Proof. The logarithm of the right side of (41.3) is at most

$$-((k+1)t - b)\Delta_t t^{-1/3} + O(1).$$

For the stated k , the first parenthesis is at least δ and Δ_t is bounded away from zero. Decreasing a positive constant absorbs the bounded error uniformly. The bound is one-sided before being written in relative-error notation. $\square$

The threshold b is important. The length saddle will have $kt \rightarrow 2b$, so a fixed choice $0 < \delta < b$ places its central region strictly inside the uniform domain. We never need an infinite sum of residues or asymptotics uniform over an increasing number of zeros.

42 The Gaussian length sum and the radial amplitude

Put

$$\lambda = -\log \rho = L + t/2 + t^{2/3}s_t, \quad m_t = \frac{\lambda + t/2}{t} = \frac{L + t + t^{2/3}s_t}{t}.$$

Completing the square gives

$$T_t := \sum_{k \geq 0} e^{-tk(k-1)/2} \rho^{-k} = e^{(\lambda+t/2)^2/(2t)} \sum_{k \geq 0} e^{-t(k-m_t)^2/2}. \quad (42.1)$$

The elementary Gaussian Fourier transform and Poisson summation give

$$\sum_{k \in \mathbb{Z}} e^{-t(k-m_t)^2/2} = \sqrt{\frac{2\pi}{t}} \sum_{\ell \in \mathbb{Z}} e^{-2\pi^2 \ell^2/t} e^{2\pi i \ell m_t} = \sqrt{\frac{2\pi}{t}} \left(1 + O(e^{-2\pi^2/t})\right),$$

uniformly in the fractional part of m_t . Since $tm_t \rightarrow L > 0$, the $k < 0$ tail is exponentially small relative to the full sum. Hence, for some $c > 0$,

$$T_t = \sqrt{\frac{2\pi}{t}} e^{(\lambda+t/2)^2/(2t)} (1 + O(e^{-c/t})). \quad (42.2)$$

Fix $0 < \delta < b$. The region $kt < b + \delta$ is a positive macroscopic distance from the center $tm_t \rightarrow 2b$; its Gaussian mass is also $O(e^{-c/t})$ relative to (42.2). On the complement use Lemma 41.1. On the discarded part use the global upper bound (41.2). These two-sided positive comparisons establish

$$F(t) = S_{e^{-t}} T_t \left(1 + O(e^{-ct^{-1/3}}) \right). \quad (42.3)$$

In particular there is no uncontrolled small- k contribution.

The exact chain-rule expression (41.4) reads

$$S_{e^{-t}} = \frac{t^{5/6}}{2\sqrt{\pi} U'_t(s_t)} \exp \left(\frac{E_0}{t} - bs_t t^{-1/3} - \frac{s_t^2 t^{1/3}}{4} \right).$$

Substitute this in (42.3) and expand only the Gaussian square. Since $E_0 + L^2/2 = C$, the result is

$$F(t) = \frac{t^{1/3}}{\sqrt{2} U'_t(s_t)} \exp \left(\frac{C}{t} + bs_t t^{-1/3} + L + \frac{s_t^2 t^{1/3}}{4} + s_t t^{2/3} + \frac{t}{2} \right) \cdot \left(1 + O(e^{-ct^{-1/3}}) \right). \quad (42.4)$$

This identity at the asymptotic precision shown keeps every constant factor visible. The algebraic powers are $t^{5/6} t^{-1/2} = t^{1/3}$. The constant exponential factor is $e^L = 4$. Thus the leading amplitude is $4/(\sqrt{2} \text{Ai}'(a)) = K_F$.

For the next radial term, (40.8) gives

$$bs_t t^{-1/3} = bat^{-1/3} - \frac{ba^2}{30} t^{1/3} + O(t^{2/3}), \quad \frac{s_t^2 t^{1/3}}{4} = \frac{a^2}{4} t^{1/3} + O(t).$$

The derivative correction in (40.9) and the term $s_t t^{2/3}$ begin at order $t^{2/3}$. Exponentiating the remaining order- $t^{1/3}$ term proves (39.4), including $\alpha = a^2(1/4 - b/30)$. Dropping the zero displacement would produce an incorrect radial correction.

43 A self contained exact coefficient local limit

This section proves the required transfer from the real radial formula to an exact coefficient. Its ingredients are elementary probability, an exact change of measure, and Fourier inversion. No complex- q version of (40.7) is assumed.

For a partition, let Z_j be the multiplicity of the part j and $C_j = \sum_{i \leq j} Z_i$. The indexed condition $\lambda_i \geq i$ is equivalent to

$$C_j \leq j \quad \text{for every } j \geq 1. \quad (43.1)$$

Indeed, if $C_j = r > j$, then the r th part is at most $j < r$. Conversely, if $\lambda_i < i$, take $j = \lambda_i$; then $C_j \geq i > j$.

[write] **6 October 2026.** This is Part III's Lemma 21.1; the short proof is kept here, inside the setting of this section.

Define heights $H_j = j - C_j$, with $H_0 = 0$. Under the Boltzmann law $\mathbb{P}_t$, the probability of an admissible partition is $e^{-tN}/F(t)$, where $N = \sum_{j \geq 1} jZ_j$.

43.1 The real moment generating function limit

For fixed real u , $t - ut^{3/2} > 0$ when t is sufficiently small, and

$$\mathbb{E}_t \exp \left(ut^{3/2}(N - C/t^2) \right) = e^{-uCt^{-1/2}} \frac{F(t - ut^{3/2})}{F(t)} \longrightarrow e^{Cu^2}. \quad (43.2)$$

To verify the limit without differentiating a remainder, apply (39.4) at the two real arguments. The leading exponent gives

$$\frac{C}{t - ut^{3/2}} - \frac{C}{t} - uCt^{-1/2} = Cu^2 + O(t^{1/2}).$$

The difference of the Airy exponents is $O(t^{1/6})$, and that of the logarithmic powers is $O(t^{1/2})$. The radial relative errors tend to zero. The mgfs are finite on every fixed real neighborhood of zero for small t , so the mgf continuity theorem yields

$$t^{3/2}(N - C/t^2) \Rightarrow \mathcal{N}(0, 2C). \quad (43.3)$$

Consequently the centered characteristic functions converge pointwise to e^{-Cu^2} . Centering at C/t^2 , rather than asserting an unproved differentiated mean formula, is sufficient at this scale.

43.2 Removing the constraints after a fixed cutoff

Fix $d > b$ and put $M = \lfloor b/t \rfloor$ and $J = \lfloor d/t \rfloor$. Let $F_J(t)$ sum the weights of all multiplicity vectors satisfying (43.1) only for $j \leq J$. The size law normalized by F_J is a convolution of a prefix law and the independent unrestricted tail

$$V = \sum_{j>J} jZ_j, \quad \mathbb{P}(Z_j = z) = (1 - e^{-tj})e^{-tjz} \quad (z = 0, 1, \dots). \quad (43.4)$$

Both its normalizer and its random tail are well-defined: for fixed $t > 0$, the ordinary partition product converges and $\sum_j j\mathbb{E}Z_j < \infty$.

Lemma 43.1 (Exponentially accurate cutoff). *For fixed $d > b$, there exists $c > 0$ such that, for all sufficiently small $t > 0$,*

$$0 \leq \frac{F_J(t)}{F(t)} - 1 \leq e^{-c/t}. \quad (43.5)$$

The total variation distance between the two size laws is at most $e^{-c/t}$, after possibly decreasing c .

Proof. Use the independent reference law with $Z_j \sim \text{Geom}(1/2)$ for $j \leq M$, and $Z_j \sim \text{Geom}(e^{-tj})$ for $j > M$; all geometrics have support $\{0, 1, \dots\}$ and parameter p means probability $(1 - p)p^z$ at z . Set

$$\mathcal{A}_M = \sum_{j=0}^{M-1} H_j, \quad D_M = 4^M e^{-tM(M+1)/2}, \quad P_M = \prod_{j>M} (1 - e^{-tj})^{-1}, \quad \delta_M = b - tM \geq 0.$$

Summation by parts gives

$$\sum_{j=1}^M jZ_j = \frac{M(M+1)}{2} - MH_M + \mathcal{A}_M. \quad (43.6)$$

The ratio of the unnormalized Boltzmann weight to the reference probability is therefore exactly

$$D_M P_M \exp(-t \mathcal{A}_M - \delta_M H_M). \quad (43.7)$$

For example, the prefix ratio before simplification is $2^{M+C_M} \exp(-t \sum_{j \leq M} j Z_j)$, and (43.6) gives (43.7). On admissibility through M , both $\mathcal{A}_M$ and H_M are nonnegative, so the final exponential factor is at most one.

A monotone integral comparison for $f(x) = -\log(1 - e^{-x})$ gives

$$\log P_M = \frac{1}{t} \int_b^\infty f(x) dx + O(1) = \frac{\text{Li}_2(1/2)}{t} + O(1).$$

The lower limit differs from tM by at most t and f is bounded near b . Also $\log D_M = 3b^2/(2t) + O(1)$. Using $\text{Li}_2(1/2) = \pi^2/12 - b^2/2$, we obtain

$$D_M P_M = \exp(C/t + O(1)). \quad (43.8)$$

Let $G = \sum_{j=M+1}^J (1 - Z_j)$. Under the reference product law,

$$\mathbb{E}G = \frac{\beta}{t} + O(1), \quad (43.9)$$

$$\beta = \int_b^d \left(1 - \frac{1}{e^x - 1}\right) dx = d - 2b - \log(1 - e^{-d}) > 0. \quad (43.10)$$

The integrand is strictly positive for $x > b$. All geometric parameters in this block are at most $1/2$; their centered exponential moments therefore have uniformly bounded second logarithmic derivatives on a fixed neighborhood of zero. There are $O(1/t)$ variables, giving

$$\log \mathbb{E}e^{s(G - \mathbb{E}G)} \leq K_0 s^2/t \quad (|s| \leq s_0).$$

Choose a fixed sufficiently small negative s . Markov's inequality, (43.9), and $\beta > 0$ give

$$\mathbb{P}\{G < \beta/(2t)\} \leq e^{-c_1/t} \quad (43.11)$$

for some $c_1 > 0$ and all small t .

If the prefix through J is admissible but $H_J < \beta/(2t)$, then $H_M \geq 0$ and $H_J = H_M + G$ force the event in (43.11). Its tilted mass is at most $D_M P_M e^{-c_1/t}$ by (43.7). Otherwise $H_J \geq \beta/(2t)$. Choose $e^{-d} < p_* < 1/2$. For small t , every parameter after J is at most p_* . Monotone coupling bounds subsequent ruin by that for iid $Z \sim \text{Geom}(p_*)$. For $r = (1 - p_*)/p_* > 1$,

$$\mathbb{E}r^{Z-1} = \frac{1 - p_*}{r(1 - p_*r)} = 1.$$

Thus $r^{\sum_{i=1}^\ell (Z_i - 1)}$ is a nonnegative martingale of mean one. The maximal inequality, first to a finite horizon and then by monotone convergence, bounds ruin from height h by r^{-h} (and $r^{-\lfloor h \rfloor}$ is a harmless weaker bound). From the stated buffer this is at most $e^{-c_2/t}$ after reducing $c_2 > 0$. No optional-stopping equality is required. Conditional on the prefix, the tail is independent; integrate the ruin bound and use again that the factor in (43.7) is at most one on the admissible prefix. The lost mass is at most $D_M P_M e^{-c_2/t}$.

Combining the two cases with (43.8) gives

$$0 \leq F_J - F \leq \exp(C/t + O(1))e^{-c_3/t}$$

for a positive c_3 , with a constant factor absorbed. Divide by (39.4); the loss is only a power of t and $\exp(O(t^{-1/3}))$. A smaller positive c proves (43.5). The true partition law is exactly the cutoff law conditioned on all remaining constraints, an event of probability F/F_J . Total variation decreases under the size map, and is at most $1 - F/F_J \leq F_J/F - 1$, proving the final assertion. $\square$

[write] **6 October 2026.** Restated material in this proof: (43.6) is Part IV's (31.2), and (43.7) is the density in Part IV's change of measure (31.3) (Part IV's η_M is δ_M here, and its $D_M(t) = e^{2bM - tM(M+1)/2}$ is this D_M); (43.8) is Part IV's (31.4) with Part III's (21.5); the buffer (43.10) is Part IV's (34.2); the martingale ruin bound is Part III's Lemma 24.1 and Part IV's (34.16). They are kept, because the proof is assembled from them in its own order.

43.3 Fourier bounds on the full circle

Let $\phi_t(\theta) = \mathbb{E}_t e^{i\theta N}$. The exact convolution in (43.4) and Lemma 43.1 imply

$$|\phi_t(\theta)| \leq |\mathbb{E} e^{i\theta V}| + 2e^{-c/t}. \quad (43.12)$$

The factor 2 is valid for the convention $d_{\text{TV}}(P, Q) = \sup_E |P(E) - Q(E)|$. For a geometric variable of parameter $p = e^{-tj}$,

$$\left| \frac{1-p}{1-pe^{ij\theta}} \right| = \left(1 + \frac{4p}{(1-p)^2} \sin^2(j\theta/2) \right)^{-1/2}. \quad (43.13)$$

On the consecutive block $\mathcal{B}_t = \{J+1, \dots, J + \lfloor 1/t \rfloor\}$, the parameters stay in a fixed compact subinterval of $(0, 1)$. Use $\log(1+v) \geq cv$ for bounded $v \geq 0$ in (43.13); all other characteristic factors have modulus at most one. This yields

$$|\mathbb{E} e^{i\theta V}| \leq \exp \left(-c_4 \sum_{j \in \mathcal{B}_t} \sin^2(j\theta/2) \right). \quad (43.14)$$

We now cover every angle in $[-\pi, \pi]$ explicitly.

Small arc. Choose $c_0 > 0$ sufficiently small depending on d . For $|\theta| \leq c_0 t$ and all $j \in \mathcal{B}_t$, the quantities $|j\theta/2|$ lie in a fixed interval where $|\sin v| \geq c|v|$. Since $\sum_{j \in \mathcal{B}_t} j^2 \asymp t^{-3}$,

$$|\mathbb{E} e^{i\theta V}| \leq e^{-c_5 \theta^2 t^{-3}} \quad (|\theta| \leq c_0 t). \quad (43.15)$$

Angles separated from the small arc. Put $m = \lfloor 1/t \rfloor$. The finite geometric-sum formula gives

$$\sum_{j \in \mathcal{B}_t} \sin^2(j\theta/2) \geq \frac{m}{2} - \frac{1}{2|\sin(\theta/2)|}. \quad (43.16)$$

For $At \leq |\theta| \leq \pi$, use $|\sin(\theta/2)| \geq |\theta|/\pi$. For a fixed sufficiently large A and sufficiently small t , (43.16) is at least c/t .

The intervening compact range. For $\theta = ut$ with $c_0 \leq |u| \leq A$, the Riemann sums converge uniformly:

$$t \sum_{j \in \mathcal{B}_t} \sin^2(utj/2) \longrightarrow \int_d^{d+1} \sin^2(ux/2) dx. \quad (43.17)$$

The integrand is nonnegative; for $u \neq 0$ its integral over a nondegenerate interval is strictly positive. Continuity and compactness give a positive minimum on this u -range. Combining the three ranges, for some $c_6 > 0$,

$$|\mathbb{E}e^{i\theta V}| \leq e^{-c_6/t} \quad (c_0 t \leq |\theta| \leq \pi). \quad (43.18)$$

This includes rational and secondary arcs. The consecutive tail indices also rule out a residual nontrivial lattice span.

[write] **6 October 2026.** The same three-arc cover appears in Part IV, Section 34.2, in a different assembly (note there).

43.4 Fourier inversion and the exact coefficient

Proposition 43.2 (Local limit). *For any integer targets $m = m(t)$ such that $(m - C/t^2)t^{3/2} \rightarrow z \in \mathbb{R}$,*

$$\mathbb{P}_t\{N = m\} \sim \frac{t^{3/2}}{\sqrt{4\pi C}} \exp\left(-\frac{z^2}{4C}\right). \quad (43.19)$$

Proof. Fourier inversion on the integer lattice gives

$$\mathbb{P}_t\{N = m\} = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-im\theta} \phi_t(\theta) d\theta.$$

By (43.12) and (43.18), the contribution from $|\theta| \geq c_0 t$ is $O(e^{-c_7/t}) = o(t^{3/2})$. On the central arc set $\theta = t^{3/2}u$. The centered characteristic function converges to e^{-Cu^2} by (43.3); multiplying by the target phase gives pointwise limit $e^{-izu - Cu^2}$.

For a literal integrable majorant, retain the additive error in (43.12). In this central range $|u| \leq c_0 t^{-1/2}$,

$$e^{-c/t} \leq e^{-(c/c_0^2)u^2}.$$

Together with (43.15), this bounds the absolute value of the rescaled integrand, extended by zero outside the central range, by $3e^{-c_8 u^2}$ for a positive c_8 independent of small t . Alternatively the additive error can be integrated separately over the circle; its total is $o(t^{3/2})$. Dominated convergence is now valid with no unseparated constant on the whole real line. It gives

$$t^{-3/2} \mathbb{P}_t\{N = m\} \rightarrow \frac{1}{2\pi} \int_{\mathbb{R}} e^{-izu - Cu^2} du = \frac{1}{\sqrt{4\pi C}} e^{-z^2/(4C)}.$$

The limiting density is positive for finite z , yielding the stated equivalent. $\square$

Take $t = \sqrt{C/n}$ and $m = n$, so $m = C/t^2$ exactly and $z = 0$. The defining Boltzmann law gives

$$A(n) = e^{tn} F(t) \mathbb{P}_t\{N = n\}. \quad (43.20)$$

Using (39.4) and (43.19), the power of t in (43.20) is $1/3 + 3/2 = 11/6$. Hence

$$A(n) \sim \frac{K_F C^{11/12}}{\sqrt{4\pi C}} n^{-11/12} \exp\left(2\sqrt{Cn} + baC^{-1/6} n^{1/6}\right).$$

The coefficient in front is precisely K in (39.3). This finishes Theorem 39.1. Neither this argument nor the value of K uses fitted count data.

44 The refined threshold inverse

First note that $A(n)$ is nondecreasing.

[write] **6 October 2026.** This is Part III's Lemma 26.1. Report 162's proof, the same largest-part injection from $\mathcal{S}_n$ to $\mathcal{S}_{n+1}$, is not reprinted.

Theorem 39.1 also shows $A(n) \rightarrow \infty$, so $T(y)$ is finite.

Let $p = 11/12$, $u = \log y$, and $x = (u/B)^2$. For fixed real c, d, e , consider an integer target with

$$n = x + cx^{2/3} + d\sqrt{x} \log x + e\sqrt{x} + O(1). \quad (44.1)$$

Since $(n - x)/x = O(x^{-1/3})$, elementary Taylor expansions in (39.6) give

$$\log A(n) - u = \left(\frac{Bc}{2} - D\right)x^{1/6} + \left(\frac{Bd}{2} - p\right)\log x + \frac{Be}{2} + \log K + o(1). \quad (44.2)$$

For example, the square-root quadratic error is $O((n - x)^2/x^{3/2}) = O(x^{-1/6})$ and tends to zero. The variation of $n^{1/6}$ is $O(x^{-1/6})$, and $\log n - \log x = O(x^{-1/3})$. Thus no unreported constant is concealed in the cross terms.

Choose

$$c = \frac{2D}{B}, \quad d = \frac{2p}{B} = \frac{11}{6B}, \quad e = -\frac{2 \log K}{B}. \quad (44.3)$$

For any fixed $\varepsilon > 0$, let $n_-(y)$ be the floor of the right side of (44.1) without $O(1)$, replacing e by $e - \varepsilon$; let $n_+(y)$ be its ceiling with $e + \varepsilon$. Equation (44.2) gives

$$\log A(n_-(y)) - u \rightarrow -B\varepsilon/2 < 0, \quad \log A(n_+(y)) - u \rightarrow B\varepsilon/2 > 0.$$

For all sufficiently large y , monotonicity therefore brackets the threshold as $n_-(y) < T(y) \leq n_+(y)$. Divide the difference from the central expression by $\sqrt{x}$, let $y \rightarrow \infty$, and then let $\varepsilon \downarrow 0$. This proves Corollary 39.2. Rounding contributes only $O(1)$; there is no differentiation of the unknown $o(1)$ remainder.

It is useful to identify the first omitted deterministic cross term, without claiming a further inverse term. With (44.3), the square-root quadratic contribution and the first Airy variation sum to

$$-\left(\frac{Bc^2}{8} + \frac{Dc}{6}\right)x^{-1/6} = -\frac{5D^2}{6B}x^{-1/6} = o(1). \quad (44.4)$$

The logarithmic-shift cross terms are smaller still. A formal inverse term of scale $x^{1/3}$ might compensate (44.4), but the available $o(1)$ coefficient-log error can induce an error as large as $o(\sqrt{x})$. It does not certify such an $x^{1/3}$ term, an exact rounding rule, or a numerical finite-input threshold window.

[write] **6 October 2026.** Corollary 39.2 is conditional, like Theorem 39.1; its first two terms are Part IV's unconditional Corollary 30.2, with the same $2D/B = \zeta bC^{-2/3} = 1.3585577\dots$. It is not an instance of the transseries volume's theorems (front matter, "The inverses"). The $x^{1/3}$ term: Section 47, item 8.

45 Finite checks and reproducible companion

The companion uses bounded standard-library Python and exact integers or rational arithmetic for its combinatorial and algebraic checks. Its results are implementation diagnostics, not proofs of the asymptotic estimates or of a finite eventual-error bound.

45.1 Two exact counting routes and direct enumeration

The main recurrence stores $d(k, w)$ after processing the part sizes $1, \dots, v$. The entry counts weight- w , length- k multiplicity vectors with every processed cumulative barrier satisfied. Initially $d(0, 0) = 1$. At part size v , update in increasing weight order

$$d(k, w) += d(k - 1, w - v) \quad (w \geq v, 1 \leq k \leq \min(v, K_N)),$$

where $K_N = \max\{k : k(k + 1)/2 \leq N\}$. The increasing order allows an arbitrary multiplicity of v . The restriction $k \leq v$ enforces the new cumulative constraint; earlier ones are unchanged. Summing the final $d(k, n)$ over k gives $A(n)$. Exact Python integers avoid a fixed-limb overflow assumption. The workspace is $O(N\sqrt{N})$ cells and the arithmetic operation count is $O(N^2\sqrt{N})$; the public bound is $N \leq 400$.

An independent route truncates (40.1) as a formal q -series, forms $c_0 = 1$ and $c_k = -\sum_{j=1}^k h_j c_{k-j}$, and applies the weight shift $k(k - 1)/2$ in (40.2). All coefficients are integers. The supplied verification compares this route with the positive recurrence through $n = 80$. Direct recursive enumeration of ordinary increasing partitions through weight 28 checks the indexed and cumulative conditions, the resulting counts, and the largest-part injection independently.

The data file contains the exact recurrence output through 400. The threshold command searches these actual counts for the first crossing. Value one has threshold zero. If the requested bound has not reached a target, the output explicitly says so and does not substitute an asymptotic estimate.

45.2 Algebra and numerical illustrations

Exact rational polynomial arithmetic checks the two Airy-density integrals mentioned by the source, reducing derivatives with $\text{Ai}''(s) = s \text{Ai}(s)$. The densities are

$$\begin{aligned} Q_1(s, u) &= s(u^2 - s)/2, \\ Q_2(s, u) &= -u^5/30 + s^2u^4/8 - s^3u^2/4 + s^2u/4 + s^4/8. \end{aligned}$$

Under $u^j \mapsto (-1)^j \text{Ai}^{(j)}(s)$, their integrals are respectively zero and $(4s \text{Ai}(s) + s^2 \text{Ai}'(s))/30$. Finite multiplicity vectors also check (43.6) and the exact tilt (43.7) at rational q . These tests verify algebra and conventions, not Li's analytic uniform remainder.

The illustration output uses ordinary floating point and explicitly recorded approximate Airy constants. It evaluates B, D, K_F, K, α and selected finite normalized log residuals

$$\log A(n) - B\sqrt{n} + Dn^{1/6} + \frac{11}{12} \log n - \log K.$$

No fitting, empirical extrapolation, or certified rounding claim is made. Slowly changing finite residuals do not establish a monotone approach or locate an eventual crossover.

45.3 Bounds and local release contract

All public computational integer inputs reject booleans, nonintegers and out-of-range values. CLI integers use canonical nonnegative ASCII decimal notation. Counts are capped at 400, the independent formal series at 100, direct enumeration at 32, and requested exact threshold values at 10^{200} . The verification command uses fixed smaller workloads. Tests run both normally and with Python

optimization so checks cannot disappear with `-O`. The companion emits JSON to standard output, has no file-writing API, and performs no network I/O.

The release includes the editable LaTeX source, deterministic PDF builder, companion, exact and approximate JSON outputs, attribution record, tests, and a sorted SHA-256 manifest. A fixed allowlist controls the archive; internal reviews, source-paper PDFs, temporary renders, build logs and previous reports are excluded. Two clean PDF builds, two archives and exact JSON replays are compared byte for byte in the recorded toolchain. PDF pages are rendered and visually inspected. The README gives the exact commands and the platform qualification.

[write] 6 October 2026. The package is shipped with the prefix `162-prefactor-`: its companion README at the report root, the companion, its tests, the PDF builder, archive packer and release tools under `code/`, and the four result files and the provenance record under `data/` (the PDF, README and SHA256SUMS are not shipped). At the write, on a Windows copy of the arrival archive, the 14 tests passed and `counts` regenerated `data/counts.json`.

Release outputs are exclusive: existing files, links, directories and special targets are refused. Parent traversal and symlink parents are rejected with descriptor-relative POSIX operations. Input reads and archive payloads have explicit size bounds. These tools assume trusted, nonadversarial source trees and output parents; they are not a sandbox against malicious same-user path replacement or modified TeX. Shell escape is disabled, but arbitrary TeX is not made safe by that alone. There is no upload, publication, external repository change or OEIS edit.

46 Source status and limits of the result

The analytic constants in this report are derived from the exact normalizations (40.1), (40.3) and (40.6). The preprint theorem is a genuine explicit dependency, even though its needed normalization, contour orientation, error order and zero-indexing argument have been independently checked in preparing this work. Such checking is neither formal proof verification nor peer review. The report supplies its own full coefficient transfer rather than treating real radial asymptotics as sufficient.

The 2025 partition paper supplies exact generating functions and identifies the superdiagonal asymptotics as an open direction in that source. The 2017 paper supplies the entire-function product; Li's July 2026 preprint concerns the Ramanujan function itself. A bounded primary-source search on 3 October 2026 did not find the same full partition prefactor and inverse in the inspected sources. That is limited evidence for an application, not exhaustive novelty or priority certification. Attribution and verified source hashes are retained in `SOURCE_PROVENANCE.json` without redistributing the full papers.

The conclusions stop at (39.5), the radial correction (39.4), and the threshold precision (39.7). They do not give a pointwise first relative correction, numerical constants in an eventual relative-error inequality, a certified finite crossover, an all-orders expansion, or complex- q turning-point asymptotics. The central and local probability limits proved here are for the Boltzmann size variable used in coefficient extraction; no unmentioned fluctuation theorem for a uniformly sampled fixed-size partition is inferred.

[write] 6 October 2026. Conditional status: note after Corollary 39.2's statement. The open items of this section, and an unconditional proof of the amplitude, are collected in Section 47, items 5–9.

47 Further questions and research (all Parts)

[write] 6 October 2026. This section is the write's. Under Vladimir's standing rule of 4 October 2026 it gathers every question left open by the five manuscripts, and every non-claim that marks a gap, with its source, the argument offered and what is missing. The manuscripts' own question and limits sections (Sections 9, 19, 29, 38 and 46) stay as delivered. No claim of any manuscript was found false, so nothing is refuted here.

Answered inside this report

- Part I, Section 9, “The superdiagonal exponential constant”: the limit of $\log A(n)/\sqrt{n}$ exists and equals B (Part III, Theorem 20.1), with the matching upper bound $\log A(n) \leq B\sqrt{n} + O(1)$. The “leading amplitude, correction series, and precise inverse” it names are answered *in part*: the next term $-Dn^{1/6}$ and the inverse to $o(x^{2/3})$ unconditionally (Part IV), the amplitude K , the power $n^{-11/12}$ and the inverse to $o(\sqrt{x})$ conditionally on [13] (Part V); the correction series stays open (item 7).
- Part I, Section 9, “The subdiagonal defect”: answered at $r = 0$. By Part II the defect is $\frac{\beta}{4} + (\frac{7}{16} - \frac{3}{2\pi^2})\beta^2 + O(\beta^3)$ (the write's (9.1)), so the scale is $n^{-1/2}$, not $1/\log n$, and $s_0(n)$ has an expansion to every order. Re-scoped in item 1.
- Part III, Section 29: “No optimality claim is made for the error $O(n^{1/3})$ ”. By Part IV the true lower error is $B\sqrt{n} - \log A(n) = Dn^{1/6}(1 + o(1))$, so the order $n^{1/3}$ is not optimal, $n^{1/6}$ is, and no bound $\log A(n) \geq B\sqrt{n} - o(n^{1/6})$ holds (dated note in Section 20).

Open

1. *The defect at fixed $r \geq 1$ and at growing r* (Part I, “The subdiagonal defect” and “Sharper shifts”; re-scoped). Part I proves $0 \leq \mathbb{P}_n(R \leq r) - s_r(n)/p(n) = O(1/\log n)$ uniformly in $r \geq 0$; Part II's endpoint reduction is carried out only for the barrier $\lambda_i \leq i$. Missing: the analogue of Part II's localization and mixed bijection for the shifted barrier $\lambda_i \leq i + r$ (first failure patterns at both ends with the shift), and, for growing r , a reduction uniform in r . Part II evaluates rank atoms only at fixed ranks; the rank asymptotics it cites are uniform in a growing range (Liu–Zhou's Theorem 1.3 for $m = o(n^{3/4})$, Zhou's Theorem 4.1 uniformly in m with $m^2\beta^3 = o(1)$), but no reduction uniform in r has been written. Part I also asks for an error sensitive to $r/\sqrt{n}$, smaller where the logistic tail is small.
2. *Explicit onsets and certified thresholds* (all Parts: Part I, “Explicit numerical onset”; Part II, constants C_J, y_J not explicit; Part III, unspecified $K_0, \dots, K_3$; Part IV, no rate in the o -terms; Part V, “no effective crossover”). Every asymptotic statement here has unspecified constants and onset, so no finite search interval or rounding rule is certified. Missing: explicit Rademacher and rank remainders (Parts I, II), explicit spectral, Chernoff and Mallein-type constants (Parts III, IV), an effective version of Li's theorem (Part V). The companions' threshold commands search actual counts instead.
3. *The subdiagonal series* (Part II, Section 19). Convergence of $\sum c_j\beta^j$, uniformity at growing order, efficient computation, and the coefficients beyond c_3 (the algorithm is proved at every order; the companion stops at order three). Also whether the ceilings in Theorem 10.2 can be

removed: the integer threshold may stay at distance of order one from the smooth centre, so two adjacent candidates can remain.

4. *Fluctuations and microscopic structure of superdiagonal partitions* (Part IV, Sections 37 and 38). Theorem 30.3 gives laws of large numbers with exponential deviation bounds. Open: a central limit theorem or covariance process for the profile, the diagonal fluctuation scale, the number of exact contacts $\lambda_i = i$ (the theorem is about the limiting row curve only), the centring and law of the largest part, row control at the divergent endpoint $y = 2b$, and optimal deviation constants. Part V's central and local limits concern the Boltzmann size variable, not a uniform partition of fixed size.
5. *An unconditional amplitude* (Part V, and the intake record). Part V's equivalent (39.5) rests on Li's unrefereed preprint [13], used for the normalized turning-point expansion (40.7) on compact s -sets with an $O(t)$ remainder and for the global indexing of the first two zeros (40.8). Open: a refereed or independent proof of these two inputs, or a different route to K ; meanwhile the unconditional statements are Part III's and Part IV's. Part IV states that it proves no amplitude and no power prefactor; Part V supplies both, conditionally.
6. *A pointwise relative correction* (Part V). Part V's radial expansion has the relative correction $\alpha t^{1/3}$, but its local limit has only an $o(1)$ error, so no pointwise correction $A(n) = Kn^{-11/12}e^{\cdots}(1 + \alpha'n^{-1/6} + \cdots)$ is proved. The intake's pointwise residual (0.155–0.158 for $n = 3200$ –8000, and least-squares $n^{-1/6}$ coefficients 1.68 and 1.52 in the fits described in the front matter) suggests such a term; it proves nothing. Missing: a local limit theorem with a rate, or an Edgeworth correction, uniform in the tilt.
7. *All orders for $A(n)$* (Parts III–V). No all-orders expansion, radial or pointwise, is proved; each of the three Parts says so. A radial expansion would need Li's turning-point expansion (40.7) to higher order, which Part V does not quote, and a pointwise one would need item 6 at every order.
8. *Finer superdiagonal inverses* (Part V, remark at (44.4)). The first omitted deterministic cross term $-\frac{5D^2}{6B}x^{-1/6}$ suggests an inverse term of order $x^{1/3}$, but the available $o(1)$ error in $\log A(n)$ allows an inverse error as large as $o(\sqrt{x})$, so no such term is certified. Exact rounding is open for all five inverses.
9. *Complex- q turning points* (Part V, Section 46). Part V uses Li's theorem for real t only and avoids complex- q asymptotics by its real local-limit argument. A complex- q version, which would allow a saddle-point proof and possibly the corrections of items 6–7, is open.
10. *Literature and priority* (every Part). Each manuscript made a bounded search and disclaims exhaustive priority; Part II calls its expansion an application of Zhou's published theorem; Part V calls its result “limited evidence for an application”. A full literature comparison, in particular for the subdiagonal expansion and for the superdiagonal rate, was not made at the write either.

[write] **Independent check of the write, 6 October 2026.** An adversarial check made by the intake after the write (commit 0ca2804be) re-read the proofs of Theorems 1.1 and 1.2, Corollary 4.1 and Theorem 5.1 step by step and found no gap; it recomputed b , B_0 , $B_* = 2.0617406$, $B = 2.2829105$, $B - \pi/\sqrt{3} = 0.46911$ and $B_* - \pi/\sqrt{3} = 0.24794$. *Counts.* Its own dynamic programs for $A(n)$ and $s(n)$, cross-checked by enumeration for $n \leq 20$, agree with the OEIS data (61 terms of A238873, 57 of A238875, 49 of A387118 as $p(n) - A(n)$);

$A(200) = 25548856857$ and $s(200) = 1930845918142$. The ratio $q(n)/A(n)$ decreases strictly for $16 \leq n \leq 400$ (0.0021 at $n = 400$, where $\log(A/q)/\sqrt{n} = 0.309$, approaching $B - \pi/\sqrt{3}$ slowly, as the term $-Dn^{1/6}$ predicts). *Proposition 1.3.* An independent bipartite matching on the intervals $\{1, \dots, y_i\}$, with the parts in decreasing and in shuffled order, agrees with superdiagonality partition by partition for every $n \leq 24$. The OEIS records were fetched again: A387112 is revision #19 of 7 September 2026, and Clément Garrot’s comment, signed 31 August 2026, gives the same proof and names A238873, A387118 and A388711, so the credit, date and revision are exactly as printed; A238873 (#36), A238875 (#33) and A387118 (#13) are quoted faithfully. *The defect.* The check re-derived $\frac{a}{2} - c_2 = \frac{7}{16} - \frac{3}{2\pi^2} = 0.285518224536$, reproduced the six numbers of the note after (9.1), and extended them: at $n = 2000$ the defect divided by β is 0.258505 and $(\text{defect} - \beta/4)/\beta^2 = 0.296561$; the cubic residual $(\text{defect} - \frac{\beta}{4} - (\frac{7}{16} - \frac{3}{2\pi^2})\beta^2)/\beta^3$ is 0.553, 0.526, 0.420, 0.397, 0.385 at $n = 100, 400, 1000, 1500, 2000$, drifting towards the next coefficient $\frac{1}{2}B(0) - c_3 = 0.3249$ (from (11.9) and (10.2)), which corroborates the β^2 and β^3 coefficients jointly. $s(n)/p(n) < \frac{1}{2}$ for every $10 \leq n \leq 2000$ (exact to 400). *Part V.* Its conditional status is stated wherever its results appear (title, abstract, Guide, status table, provenance, Part head, both statement headings, the dated notes in Parts I and III–V, Section 47, README), and no sentence states its equivalent, K or its inverse unconditionally; the extraneous factor q in Štampach–Štovíček’s Example 19 (arXiv:1510.01454v1) was confirmed. *The inverses.* The instance of the Lambert core for x_0 was re-derived, and Part II’s $b_0 = \frac{7}{24} + \frac{3}{\pi^2}$ and $b_1 = \frac{23}{32} + \frac{45}{2\pi^4}$ of (10.7) were re-derived from the volume’s coefficient recursion at orders one and two; at the exact thresholds $y = s(n)$, $(n - x_0 - b_0)/\beta_0 = 1.198, 1.065, 1.006, 0.9915$ at $n = 100, 400, 1200, 2000$ (against $b_1 = 0.949735$), and $n(n - X_2)$ stays bounded (3.20 to 2.40). The non-instance classifications were confirmed. No mathematical error was found. One wording is corrected, with a dated note keeping the first wording (the front matter’s “decays like” for an upper bound); three dated additions sharpen the note on Dousse–Mertens (the literal $m = 0$ reading is false and unused), the note after Proposition 1.3 (it also settles the conjectures of A388711 and A387112) and the note on $s(n)/p(n)$ versus $\frac{1}{2}$ (complete below $n = 10$, extended to $n = 2000$).

References

- [1] M. Archibald, A. Blecher, S. Elizalde and A. Knopfmacher, *Subdiagonal and superdiagonal partitions*, Afrika Matematika **36**, article 77 (2025). doi:10.1007/s13370-025-01282-0. *Part I:* The final article’s Section 5 and Lemma 5.4 are the direct targets and injection antecedent; Theorem 3.3 defines shifted subdiagonal partitions and gives their generating function. *Part II:* Section 5, especially Conjecture 5.3 and Lemma 5.4. *Part III:* 17 pp.; published 7 April 2025. *Part IV:* The increasing class and refined generating functions are in Section 2; the asymptotic questions are in Section 5. *Part V:* Theorem 2.1 and equation (6), with the empty term retained. [write] Received 12 November 2024, accepted 27 February 2025; open access under CC BY 4.0. The conjectures are on p. 15.
- [2] J. Dousse and M. H. Mertens, *Asymptotic formulae for partition ranks*, Acta Arithmetica **168** (2015), 83–100. doi:10.4064/aa168-1-5. *Part I:* Theorem 1.2 in the author-hosted manuscript (arXiv:1406.6848), used here only at positive ranks. [write] Statement of Theorem 1.2 read in arXiv:1406.6848, p. 2.
- [3] NIST Digital Library of Mathematical Functions, Section 27.14, Unrestricted Partitions. *Part I:* especially Section 27.14(iii), Asymptotic Formulas. Classical Hardy–Ramanujan and Rademacher estimates; in particular equation 27.14.9 supports the refined estimate used in this report. *Part II:* Section 27.14(iii), Unrestricted Partitions: Asymptotic Formulas. Classical Hardy–Ramanujan and Rademacher theory.
- [4] NIST, Digital Library of Mathematical Functions, Table 9.9.1, first ten real zeros of the Airy functions, checked 3 October 2026 (Part IV).

- [5] OEIS Foundation Inc., [A238875, Subdiagonal partitions](#). *Part I*: Numeric prefix and increasing-order convention checked against the [official data mirror](#), entry revision of 14 April 2025, accessed 3 October 2026. *Part II*: Increasing-order definition and numerical prefix checked against the official data mirror, accessed 3 October 2026. **[write]** Revision #33 of 14 April 2025 read on 6 October 2026.
- [6] OEIS Foundation Inc., [A238873, Superdiagonal partitions](#) (Part III: *A238873: Number of superdiagonal partitions*). *Part I*: Numeric prefix and order convention checked against the [official data mirror](#), entry revision of 7 October 2025, accessed 3 October 2026. *Part III*: Definition and numerical prefix checked 3 October 2026. *Part IV*: *The On-Line Encyclopedia of Integer Sequences*, A238873, checked 3 October 2026. *Part V*: Sequence identification; no full entry text is redistributed. **[write]** Revision #36 of 7 October 2025 read on 6 October 2026.
- [7] **[write]** OEIS Foundation Inc., [A387112, Numbers with \(strictly\) choosable initial intervals of prime indices](#), revision #19 of 7 September 2026, read on 6 October 2026; comment of Clément Garrot, 31 August 2026.
- [8] Z.-G. Liu and N. H. Zhou, *Uniform asymptotic formulas for the Fourier coefficients of the inverse of theta functions*, The Ramanujan Journal **57** (2022), 1085–1123. [doi:10.1007/s11139-021-00409-8](#). [arXiv:1903.00835v3](#), Theorem 1.3, including rank zero (Part II). **[write]** Statement of Theorem 1.3 read in [arXiv:1903.00835v3](#), p. 5.
- [9] N. H. Zhou, *Eventual log-concavity of k -rank statistics for integer partitions*, Journal of Number Theory **259** (2024), 242–272. [doi:10.1016/j.jnt.2024.01.003](#). [arXiv:2110.11174v3](#), Proposition 2.2 and Theorem 4.1, specialized to $k = 2$, $m = 0$, and fixed $j = |r|$ (Part II). **[write]** Statement and hypotheses of Theorem 4.1 read in [arXiv:2110.11174v3](#), p. 13.
- [10] S. Elizalde, *Research and publications*, <https://math.dartmouth.edu/~sergi/research.html>. Primary publication list checked 3 October 2026; used only for the bounded source assessment, not as a mathematical input (Part III).
- [11] B. Mallein, *Maximal displacement of a branching random walk in time-inhomogeneous environment*, Stochastic Processes and their Applications **125** (2015), 3958–4019. [doi:10.1016/j.spa.2015.05.011](#). [arXiv:1307.4496v3](#), Lemmas 2.6–2.7, equations (2.15), (2.19). [Author manuscript](#) (Part IV). **[write]** Lemmas 2.6 and 2.7 and assumptions (2.12)–(2.13) read in [arXiv:1307.4496v3](#), pp. 11 and 13.
- [12] F. Štampach and P. Štoviček, *Factorization of the characteristic function of a Jacobi matrix*, Operators and Matrices **11**(1) (2017), 147–169. [doi:10.7153/oam-11-11](#). Theorem 5.2, positive-sequence discussion and Example 5.2 (Part V). **[write]** The write read [arXiv:1510.01454v1](#) (6 October 2015), where the theorem is Theorem 15, the positive case follows it, and the example is Example 19 (p. 16); the journal version was not read.
- [13] Y.-T. Li, *Airy Turning-Point Asymptotics for Ramanujan’s Entire Function $A_q(z)$* , [arXiv:2607.27504v1](#), submitted 29 July 2026. [Versioned primary record](#); [versioned full text](#). Theorem 1 and Corollary 2. Preprint; no journal publication verified (Part V). **[write]** Unrefereed. On 6 October 2026 arXiv lists only version 1 and no journal reference. Part V is conditional on it.

- [14] Y.-T. Li and R. Wong, *Global asymptotics of Stieltjes–Wigert polynomials*, arXiv:1302.5193v2, 11 June 2013. [Primary preprint record](#). The fixed-argument theorem is not used as the turning-point input (Part V).
- [15] *Proofs of two conjectures on subdiagonal and superdiagonal partitions: Quantitative rank coupling and exponential separation*, Report 156 (3 October 2026). *Part II*: preceding report in this series. The subdiagonal result is $s(n)/p(n) = 1/2 + O(1/\log n)$; no all-orders result was asserted there. *Part III*: Earlier report in this series; it proves the lower constant B_* and explicitly leaves the exact superdiagonal rate undetermined. The present proof does not require access to that report. **[write]** Printed in this report as Part I.
- [16] *Exact exponential growth of superdiagonal partitions*, Report 158, 3 October 2026. Antecedent for the exact rate and coarse pointwise lower bound; unchanged by this report (Part IV). **[write]** Printed in this report as Part III; its delivered title reads “Exact exponential growth of *increasing* superdiagonal partitions”.
- [17] **[write]** *Transseries and inversion*, ProveIt volume, Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex:
Definition p0:def:model, Lemma p0:lem:alpha-reduction with
p0:eq:alpha-inverse-transport, Theorems p0:thm:lambert-core (with
p0:eq:core-lambert-minus), p0:thm:lambert-centered (with p0:eq:c-bell),
p0:thm:staircase and plt:thm:lw-template.